

lation/Homeostasis

Even the smallest organisms are complex and require multiple regulatory mechanisms to coordinate internal functions, such as the transport of nutrients, response to stimuli, and coping with environmental stresses. Homeostasis (literally, "steady state") refers to the relatively stable internal environment required to maintain life.

For example, organ systems such as the digestive or circulatory systems perform specific functions like carrying oxygen throughout the body, removing wastes, delivering nutrients to every cell, and cooling the body.

To function properly, cells require appropriate conditions such as proper temperature, pH, and concentrations of diverse chemicals. These conditions may, however, change from one moment to the next. Organisms are able to

maintain homeostatic internal conditions within a narrow range almost constantly, despite environmental changes,

by activation of regulatory mechanisms. For example, many organisms regulate their body temperature in a process

known as thermoregulation. Organisms that live in cold climates, such as the polar bear (Figure 1.5), have body structures that help them withstand low temperatures and conserve body heat. In hot climates, organisms have methods (such as perspiration in humans or panting in dogs) that help them to shed excess body heat.

FIGURE 1.5 Polar bears and other mammals living in ice-covered regions maintain their body temperature by generating heat and reducing heat loss through thick fur and a dense layer of fat under their skin. (credit: "longhorndave"/Flickr)

Energy Processing

All organisms (such as the California condor shown in Figure 1.6) use a source of energy for their metabolic activities. Some organisms capture energy from the Sun and convert it into chemical energy in food; others use chemical energy from molecules they take in.

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FIGURE 1.6 A lot of energy is required for a California condor to fly. Chemical energy derived from food is used to power flight. California condors are an endangered species; scientists have strived to place a wing tag on each bird to help them identify and locate each individual bird. (credit: Pacific Southwest Region U.S. Fish and Wildlife)

Evolution

The diversity of life on Earth is a result of mutations, or random changes in hereditary material over time. These mutations allow the possibility for organisms to adapt to a changing environment. An organism that evolves characteristics fit for the environment will have greater reproductive success, subject to the forces of natural selection.

Levels of Organization of Living Things

Living things are highly organized and structured, following a hierarchy on a scale from small to large. The atom is

the smallest and most fundamental unit of matter that retains the properties of an element. It consists of a nucleus

surrounded by electrons. Atoms form molecules. A molecule is a chemical structure co

h an observation and apply the scientific method to solve the problem. One Monday morning, a student arrives at class and quickly discovers that the classroom is too warm. That is an observation that also describes a problem: the classroom is too warm. The student then asks a question: ?Why is the classroom so warm??

Recall that a hypothesis is a suggested explanation that can be tested. To solve a problem, several hypotheses may be proposed. For example, one hypothesis might be, ?The classroom is warm because no one turned on the air conditioning.? But there could be other responses to the question, and therefore other hypotheses may be proposed. A second hypothesis might be, ?The classroom is warm because there is a power failure, and so the air conditioning doesn?t work.?

Once a hypothesis has been selected, a prediction may be made. A prediction is similar to a hypothesis but it typically has the format ?If . . . then? For example, the prediction for the first hypothesis might be, ?If the student turns on the air conditioning, then the classroom will no longer be too warm.?

A hypothesis must be testable to ensure that it is valid. For example, a hypothesis that depends on what a bear thinks is not testable, because it can never be known what a bear thinks. It should also be falsifiable, meaning that it can be disproven by experimental results. An example of an unfalsifiable hypothesis is ?Botticelli?s Birth of Venus is beautiful.? There is no experiment that might show this statement to be false. To test a hypothesis, a researcher will conduct one or more experiments designed to eliminate one or more of the hypotheses. This is important. A hypothesis can be disproven, or eliminated, but it can never be proven. Science does not deal in proofs like mathematics. If an experiment fails to disprove a hypothesis, then we find support for that explanation, but this is not to say that down the road a better explanation will not be found, or a more carefully designed experiment will be found to falsify the hypothesis.

Each experiment will have one or more variables and one or more controls. A variable is any part of the experiment that can vary or change during the experiment. A control is a part of the experiment that does not change. Look for the variables and controls in the example that follows. As a simple example, an experiment might be conducted to test the hypothesis that phosphate limits the growth of algae in freshwater ponds. A series of artificial ponds are filled with water and half of them are treated by adding phosphate each week, while the other half are treated by adding a salt that is known not to be used by algae. The variable here is the phosphate (or lack of phosphate), the experimental or treatment cases are the ponds with added phosphate and the control ponds are those with something inert added, such as the salt. Just adding something is also a control against the possibility that adding extra matter to the pond has

ic science is knowledge for knowledge's sake, though this does not mean that in the end it may not result in an application.

In contrast, applied science or "technology," aims to use science to solve real-world problems, making it possible, for example, to improve a crop yield, find a cure for a particular disease, or save animals threatened by a natural disaster. In applied science, the problem is usually defined for the researcher.

Some individuals may perceive applied science as "useful" and basic science as "useless." A question these people might pose to a scientist advocating knowledge acquisition would be, "What for?" A careful look at the history of science, however, reveals that basic knowledge has resulted in many remarkable applications of great value. Many scientists think that a basic understanding of science is necessary before an application is developed; therefore, applied science relies on the results generated through basic science. Other scientists think that it is time to move on from basic science and instead to find solutions to actual problems. Both approaches are valid. It is true that there are problems that demand immediate attention; however, few solutions would be found without the help of the knowledge generated through basic science.

One example of how basic and applied science can work together to solve practical problems occurred after the discovery of DNA structure led to an understanding of the molecular mechanisms governing DNA replication. Strands of DNA, unique in every human, are found in our cells, where they provide the instructions necessary for life. During DNA replication, new copies of DNA are made, shortly before a cell divides to form new cells. Understanding the mechanisms of DNA replication enabled scientists to develop laboratory techniques that are now used to identify genetic diseases, pinpoint individuals who were at a crime scene, and determine paternity. Without basic science, it is unlikely that applied science could exist.

Another example of the link between basic and applied research is the Human Genome Project, a study in which each human chromosome was analyzed and mapped to determine the precise sequence of DNA subunits and the exact location of each gene. (The gene is the basic unit of heredity represented by a specific DNA segment that codes for a functional molecule.) Other organisms have also been studied as part of this project to gain a better understanding of human chromosomes. The Human Genome Project (Figure 1.19) relied on basic research carried out with non-human organisms and, later, with the human genome. An important end goal eventually became using the data for applied research seeking cures for genetically related diseases.

FIGURE 1.19 The Human Genome Project was a 13-year collaborative effort among researchers working in several different fields of science. The project was completed in 2003. (credit: the U.S. Department of Energy Genome Programs)

While research ef

forts in both basic science and applied science are usually carefully planned, it is important to note that some discoveries are made by serendipity, that is, by means of a fortunate accident or a lucky surprise.

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Penicillin was discovered when biologist Alexander Fleming accidentally left a petri dish of *Staphylococcus* bacteria

open. An unwanted mold grew, killing the bacteria. The mold turned out to be *Penicillium*, and a new critically important antibiotic was discovered. In a similar manner, Percy Lavon Julian was an established medicinal chemist

working on a way to mass produce compounds with which to manufacture important drugs. He was focused on using soybean oil in the production of progesterone (a hormone important in the menstrual cycle and pregnancy),

but it wasn't until water accidentally leaked into a large soybean oil storage tank that he found his method. Immediately recognizing the resulting substance as stigmasterol, a primary ingredient in progesterone and similar

drugs, he began the process of replicating and industrializing the process in a manner that has helped millions of

people. Even in the highly organized world of science, luck?when combined with an observant, curious mind focused on the types of reasoning discussed above?can lead to unexpected breakthroughs.

Reporting Scientific Work

Whether scientific research is basic science or applied science, scientists must share their findings for other researchers to expand and build upon their discoveries. Communication and collaboration within and between sub

disciplines of science are key to the advancement of knowledge in science. For this reason, an important aspect of a

scientist's work is disseminating results and communicating with peers. Scientists can share results by presenting

them at a scientific meeting or conference, but this approach can reach only the limited few who are present.

Instead, most scientists present their results in peer-reviewed articles that are published in scientific journals.

Peer-reviewed articles are scientific papers that are reviewed, usually anonymously by a scientist's colleagues, or

peers. These colleagues are qualified individuals, often experts in the same research area, who judge whether or not

the scientist's work is suitable for publication. The process of peer review helps to ensure that the research described in a scientific paper or grant proposal is original, significant, logical, and thorough. Grant proposals, which

are requests for research funding, are also subject to peer review. Scientists publish their work so other scientists

can reproduce their experiments under similar or different conditions to expand on the findings.

There are many journals and the popular press that do not use a peer-review system. A large number of online open-access journals, journals with articles available without cost, are now available many of which use rigorous peer-review sy

ic method a method of research with defined

hypothesis-based science a form of science that steps that include experiments and careful begins with a specific explanation that is then tested observation

inductive reasoning a form of logical thinking that scientific theory a thoroughly tested and confirmed uses related observations to arrive at a general explanation for observations or phenomena conclusion tissue a group of similar cells carrying out the same

life science a field of science, such as biology, that function studies living things variable a part of an experiment that can vary or

macromolecule a large molecule typically formed by change

Chapter Summary

1.1 Themes and Concepts of Biology turn, are grouped as populations, communities,

ecosystems, and the biosphere. Evolution is the source

Biology is the science of life. All living organisms share of the tremendous biological diversity on Earth today. A several key properties such as order, sensitivity or diagram called a phylogenetic tree can be used to response to stimuli, reproduction, adaptation, growth show evolutionary relationships among organisms. and development, regulation, homeostasis, and energy Biology is very broad and includes many branches and processing. Living things are highly organized following sub disciplines. Examples include molecular biology, a hierarchy that includes atoms, molecules, organelles, microbiology, neurobiology, zoology, and botany, cells, tissues, organs, and organ systems. Organisms, in among others.

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1.2 The Process of Science certain circumstances. Two types of logical reasoning are used in science. Inductive reasoning uses results to

Biology is the science that studies living organisms and produce general scientific principles. Deductive their interactions with one another and their reasoning is a form of logical thinking that predicts environments. Science attempts to describe and results by applying general principles. The common understand the nature of the universe in whole or in thread throughout scientific research is the use of the part. Science has many fields; those fields related to scientific method. Scientists present their results in the physical world and its phenomena are considered peer-reviewed scientific papers published in scientific natural sciences. journals.

A hypothesis is a tentative explanation for an Science can be basic or applied. The main goal of basic observation. A generally accepted scientific theory is science is to expand knowledge without any thoroughly tested and confirmed explanation for a set expectation of short-term practical application of that of observations or phenomena. A scientific law is a knowledge. The primary goal of applied research, description, often in the form of a mathematical however, is to solve practical problems. formula, of the behavior of an aspect of nature under

Visual Connection Questions

1. Figure 1.8 Wh

associated with the oxygen atom than they do with hydrogen atoms. There is no overall charge to a water molecule, but there is a slight positive charge on each hydrogen atom and a slight negative charge on the oxygen atom. Because of these charges, the slightly positive hydrogen atoms repel each other and form the unique shape seen in Figure 2.7. Each water molecule attracts other water molecules because of the positive and negative charges in the different parts of the molecule. Water also attracts other polar molecules (such as sugars), forming hydrogen bonds. When a substance readily forms hydrogen bonds with water, it can dissolve in water and is referred to as hydrophilic (?water-loving?). Hydrogen bonds are not readily formed with nonpolar substances like oils and fats (Figure 2.8). These nonpolar compounds are hydrophobic (?water-fearing?) and will not dissolve in water.

FIGURE 2.8 As this macroscopic image of oil and water show, oil is a nonpolar compound and, hence, will not dissolve in water. Oil and water do not mix. (credit: Gautam Dogra)

Water Stabilizes Temperature

The hydrogen bonds in water allow it to absorb and release heat energy more slowly than many other substances.

Temperature is a measure of the motion (kinetic energy) of molecules. As the motion increases, energy is higher and thus temperature is higher. Water absorbs a great deal of energy before its temperature rises. Increased energy disrupts the hydrogen bonds between water molecules. Because these bonds can be created and disrupted rapidly, water absorbs an increase in energy and temperature changes only minimally. This means that water moderates temperature changes within organisms and in their environments. As energy input continues, the balance between hydrogen-bond formation and destruction swings toward the destruction side. More bonds are broken than are formed. This process results in the release of individual water molecules at the surface of the liquid (such as a body of water, the leaves of a plant, or the skin of an organism) in a process called evaporation. Evaporation of sweat, which is 90 percent water, allows for cooling of an organism, because breaking hydrogen bonds requires an input of energy and takes heat away from the body.

Conversely, as molecular motion decreases and temperatures drop, less energy is present to break the hydrogen bonds between water molecules. These bonds remain intact and begin to form a rigid, lattice-like structure (e.g., ice) (Figure 2.9a). When frozen, ice is less dense than liquid water (the molecules are farther apart). This means that ice floats on the surface of a body of water (Figure 2.9b). In lakes, ponds, and oceans, ice will form on the surface of the water, creating an insulating barrier to protect the animal and plant life beneath from freezing in the water. If this did not happen, plants and animals living in water would freeze in a block of ice and could not move freely, making life in cold temperatures di

rt to the cell. Wood and paper are mostly cellulosic in nature. Cellulose is made up of glucose monomers that are linked by bonds between particular carbon atoms in the glucose molecule.

Every other glucose monomer in cellulose is flipped over and packed tightly as extended long chains. This gives cellulose its rigidity and high tensile strength, which is so important to plant cells. Cellulose passing through our digestive system is called dietary fiber. While the glucose-glucose bonds in cellulose cannot be broken down by human digestive enzymes, herbivores such as cows, buffalos, and horses are able to digest grass that is rich in cellulose and use it as a food source. In these animals, certain species of bacteria reside in the digestive system of

herbivores and secrete the enzyme cellulase. The appendix also contains bacteria that break down cellulose, giving

it an important role in the digestive systems of some ruminants. Cellulases can break down cellulose into glucose

monomers that can be used as an energy source by the animal.

Carbohydrates serve other functions in different animals. Arthropods, such as insects, spiders, and crabs, have an

outer skeleton, called the exoskeleton, which protects their internal body parts. This exoskeleton is made of the

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biological macromolecule chitin, which is a nitrogenous carbohydrate. It is made of repeating units of a modified sugar containing nitrogen.

Thus, through differences in molecular structure, carbohydrates are able to serve the very different functions of energy storage (starch and glycogen) and structural support and protection (cellulose and chitin) (Figure 2.16).

FIGURE 2.16 Although their structures and functions differ, all polysaccharide carbohydrates are made up of monosaccharides and have the chemical formula $(CH_2O)_n$.

CAREER CONNECTION

Registered Dietitian

Obesity is a worldwide health concern, and many diseases, such as diabetes and heart disease, are becoming more

prevalent because of obesity. This is one of the reasons why registered dietitians are increasingly sought after for

advice. Registered dietitians help plan food and nutrition programs for individuals in various settings. They often work with patients in health-care facilities, designing nutrition plans to prevent and treat diseases. For example, dietitians may teach a patient with diabetes how to manage blood-sugar levels by eating the correct types and amounts of carbohydrates. Dietitians may also work in nursing homes, schools, and private practices.

To become a registered dietitian, one needs to earn at least a bachelor's degree in dietetics, nutrition, food technology, or a related field. In addition, registered dietitians must complete a supervised internship program and

pass a national exam. Those who pursue careers in dietetics take courses in nutrition, chemistry, biochemistry, biology, microbiology, and human physiology. Dietitians must become experts in the chemistry and function

together, have a distinct shape, and have a unique function.

EVOLUTION CONNECTION

The Evolutionary Significance of Cytochrome c

Cytochrome c is an important component of the molecular machinery that harvests energy from glucose.

Because

this protein's role in producing cellular energy is crucial, it has changed very little over millions of years. Protein sequencing has shown that there is a considerable amount of sequence similarity among cytochrome c molecules of

different species; evolutionary relationships can be assessed by measuring the similarities or differences among various species' protein sequences.

For example, scientists have determined that human cytochrome c contains 104 amino acids. For each cytochrome

c molecule that has been sequenced to date from different organisms, 37 of these amino acids appear in the same

position in each cytochrome c. This indicates that all of these organisms are descended from a common ancestor.

On comparing the human and chimpanzee protein sequences, no sequence difference was found. When human and

rhesus monkey sequences were compared, a single difference was found in one amino acid. In contrast, human-to-

yeast comparisons show a difference in 44 amino acids, suggesting that humans and chimpanzees have a more

recent common ancestor than humans and the rhesus monkey, or humans and yeast.

Protein Structure

As discussed earlier, the shape of a protein is critical to its function. To understand how the protein gets its final shape or conformation, we need to understand the four levels of protein structure: primary, secondary, tertiary, and

quaternary (Figure 2.21).

The unique sequence and number of amino acids in a polypeptide chain is its primary structure. The unique sequence for every protein is ultimately determined by the gene that encodes the protein. Any change in the gene

sequence may lead to a different amino acid being added to the polypeptide chain, causing a change in protein structure and function. William Warrick Cardozo showed that sickle-cell anemia is caused by a change in protein

structure as a result of gene encoding, meaning that it is an inherited disorder. In sickle cell anemia, the hemoglobin

α chain has a single amino acid substitution, causing a change in both the structure and function of the protein.

What is most remarkable to consider is that a hemoglobin molecule is made up of two alpha chains and two beta

chains that each consist of about 150 amino acids. The molecule, therefore, has about 600 amino acids. The structural difference between a normal hemoglobin molecule and a sickle cell molecule—that dramatically decreases life expectancy in the affected individuals—is a single amino acid of the 600.

Because of this change of one amino acid in the chain, the normally biconcave, or disc-shaped, red blood cells assume a crescent or 'sickle' shape, which clogs arteries. This can lead to a myriad of serious health problems,

such as breathlessness, dizziness, headaches, and a

rotations in an atom protein or steroid, secreted by an endocrine gland or
base a substance that absorbs hydrogen ions and group of endocrine cells; acts to control or regulate

therefore raises pH specific physiological processes

buffer a solution that resists a change in pH by hydrogen bond a weak bond between partially

absorbing or releasing hydrogen or hydroxide ions positively charged hydrogen atoms and partially
carbohydrate a biological macromolecule in which negatively charged elements or molecules

the ratio of carbon to hydrogen to oxygen is 1:2:1; hydrophilic describes a substance that dissolves in
carbohydrates serve as energy sources and water; water-loving
structural support in cells hydrophobic describes a substance that does not

cation a positive ion formed by losing electrons dissolve in water; water-fearing

cellulose a polysaccharide that makes up the cell wall an atom or compound that does not contain equal

walls of plants and provides structural support to numbers of protons and electrons, and therefore
the cell has a net charge

chemical bond an interaction between two or more ionic bond a chemical bond that forms between ions
of the same or different elements that results in the of opposite charges
formation of molecules isotope one or more forms of an element that have

chitin a type of carbohydrate that forms the outer different numbers of neutrons

skeleton of arthropods, such as insects and lipids a class of macromolecules that are nonpolar
crustaceans, and the cell walls of fungi and insoluble in water

cohesion the intermolecular forces between water litmus paper filter paper that has been treated with a
molecules caused by the polar nature of water; natural water-soluble dye so it can be used as a pH
creates surface tension indicator

covalent bond a type of strong bond between two or macromolecule a large molecule, often formed by
more of the same or different elements; forms when polymerization of smaller monomers
electrons are shared between elements mass number the number of protons plus neutrons in

denaturation the loss of shape in a protein as a result an atom
of changes in temperature, pH, or exposure to matter anything that has mass and occupies space
chemicals monosaccharide a single unit or monomer of

deoxyribonucleic acid (DNA) a double-stranded carbohydrates

polymer of nucleotides that carries the hereditary neutron a particle with no charge that resides in the
information of the cell nucleus of an atom; has a mass of 1

disaccharide two sugar monomers that are linked nonpolar covalent bond a type of covalent bond that
together by a glycosidic bond forms between atoms when electrons are shared

electron a negatively charged particle that resides equally between atoms, resulting in no regions with
outside of the nucleus in the electron orbital; lacks partial charges as in polar covalent bonds
functional mass and has a charge of -1 nucleic acid a biological macromolecule that carries

electron transfer the

y positively charged or slightly negatively charged
emits particles or energy to form a more stable atoms
element

Chapter Summary

2.1 The Building Blocks of Molecules remain stable, although increased energy is added to

the system. Water's cohesive forces allow for the

Matter is anything that occupies space and has mass. Its property of surface tension. All of these unique
is made up of atoms of different elements. All of the 92 properties of water are important in the chemistry of
elements that occur naturally have unique qualities living organisms.
that allow them to combine in various ways to create

compounds or molecules. Atoms, which consist of The pH of a solution is a measure of the concentration
protons, neutrons, and electrons, are the smallest units of hydrogen ions in the solution. A solution with a high
of an element that retain all of the properties of that number of hydrogen ions is acidic and has a low pH
element. Electrons can be donated or shared between value. A solution with a high number of hydroxide ions
atoms to create bonds, including ionic, covalent, and is basic and has a high pH value. The pH scale ranges
hydrogen bonds, as well as van der Waals interactions. from 0 to 14, with a pH of 7 being neutral. Buffers are

solutions that moderate pH changes when an acid or

2.2 Water base is added to the buffer system. Buffers are

Water has many properties that are critical to important in biological systems because of their ability
maintaining life. It is polar, allowing for the formation of to maintain constant pH conditions.

hydrogen bonds, which allow ions and other polar 2.3 Biological Molecules

molecules to dissolve in water. Therefore, water is an

excellent solvent. The hydrogen bonds between water Living things are carbon-based because carbon plays
molecules give water the ability to hold heat better such a prominent role in the chemistry of living things.
than many other substances. As the temperature rises, The four covalent bonding positions of the carbon atom
the hydrogen bonds between water continually break can give rise to a wide diversity of compounds with
and reform, allowing for the overall temperature to many functions, accounting for the importance of

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carbon in living things. Carbohydrates are a group of perform a diverse range of functions for the cell. They
macromolecules that are a vital energy source for the help in metabolism by providing structural support and
cell, provide structural support to many organisms, and by acting as enzymes, carriers or as hormones. The
can be found on the surface of the cell as receptors or building blocks of proteins are amino acids. Proteins
for cell recognition. Carbohydrates are classified as are organized at four levels: primary, secondary,
monosaccharides, disaccharides, and polysaccharides, tertiary, and quaternary. Protein shape and function are
depending on the number of monomers in the intrica

termining whether a cell is prokaryotic or eukaryotic, let us first examine how biologists study cells.

Microscopy

Cells vary in size. With few exceptions, individual cells are too small to be seen with the naked eye, so scientists use microscopes to study them. A microscope is an instrument that magnifies an object. Most images of cells are taken with a microscope and are called micrographs.

Light Microscopes

To give you a sense of the size of a cell, a typical human red blood cell is about eight millionths of a meter or eight micrometers (abbreviated as μm) in diameter; the head of a pin is about two thousandths of a meter (millimeters, or mm) in diameter. That means that approximately 250 red blood cells could fit on the head of a pin.

The optics of the lenses of a light microscope changes the orientation of the image. A specimen that is right-side up and facing right on the microscope slide will appear upside-down and facing left when viewed through a microscope, and vice versa. Similarly, if the slide is moved left while looking through the microscope, it will appear to move right, and if moved down, it will seem to move up. This occurs because microscopes use two sets of lenses to magnify the image. Due to the manner in which light travels through the lenses, this system of lenses produces an inverted image (binoculars and a dissecting microscope work in a similar manner, but include an additional magnification system that makes the final image appear to be upright).

Most student microscopes are classified as light microscopes (Figure 3.2a). Visible light both passes through and is bent by the lens system to enable the user to see the specimen. Light microscopes are advantageous for viewing living organisms, but since individual cells are generally transparent, their components are not distinguishable unless they are colored with special stains. Staining, however, usually kills the cells.

Light microscopes commonly used in the undergraduate college laboratory magnify up to approximately 400 times. Two parameters that are important in microscopy are magnification and resolving power. Magnification is the degree of enlargement of an object. Resolving power is the ability of a microscope to allow the eye to distinguish two adjacent structures as separate; the higher the resolution, the closer those two objects can be, and the better the clarity and detail of the image. When oil immersion lenses are used, magnification is usually increased to 1,000 times for the study of smaller cells, like most prokaryotic cells. Because light entering a specimen from below is focused onto the eye of an observer, the specimen can be viewed using light microscopy. For this reason, for light to pass through a specimen, the sample must be thin or translucent.

LINK TO LEARNING

For another perspective on cell size, try the HowBig (http://openstax.org//cell_sizes2) interactive.

A second type of microscope used in laboratories is the dissecting microscope (Fi

ll between the plasma membrane and the nuclear envelope (a structure to be discussed shortly). It is made up of organelles suspended in the gel-like cytosol, the cytoskeleton, and various chemicals (Figure 3.7). Even though the cytoplasm consists of 70 to 80 percent water, it has a semi-solid consistency, which comes from the proteins within it. However, proteins are not the only organic molecules found in the cytoplasm. Glucose and other simple sugars, polysaccharides, amino acids, nucleic acids, fatty acids, and derivatives of glycerol are found there too. Ions of sodium, potassium, calcium, and many other elements are also dissolved in the cytoplasm. Many metabolic reactions, including protein synthesis, take place in the cytoplasm.

The Cytoskeleton

If you were to remove all the organelles from a cell, would the plasma membrane and the cytoplasm be the only components left? No. Within the cytoplasm, there would still be ions and organic molecules, plus a network of protein fibers that helps to maintain the shape of the cell, secures certain organelles in specific positions, allows cytoplasm and vesicles to move within the cell, and enables unicellular organisms to move independently. Collectively, this network of protein fibers is known as the cytoskeleton. There are three types of fibers within the cytoskeleton: microfilaments, also known as actin filaments, intermediate filaments, and microtubules (Figure 3.9).

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FIGURE 3.9 Microfilaments, intermediate filaments, and microtubules compose a cell's cytoskeleton.

Microfilaments are the thinnest of the cytoskeletal fibers and function in moving cellular components, for example, during cell division. They also maintain the structure of microvilli, the extensive folding of the plasma membrane found in cells dedicated to absorption. These components are also common in muscle cells and are responsible for muscle cell contraction. Intermediate filaments are of intermediate diameter and have structural functions, such as maintaining the shape of the cell and anchoring organelles. Keratin, the compound that strengthens hair and nails, forms one type of intermediate filament. Microtubules are the thickest of the cytoskeletal fibers. These are hollow tubes that can dissolve and reform quickly. Microtubules guide organelle movement and are the structures that pull chromosomes to their poles during cell division. They are also the structural components of flagella and cilia. In cilia and flagella, the microtubules are organized as a circle of nine double microtubules on the outside and two microtubules in the center.

The centrosome is a region near the nucleus of animal cells that functions as a microtubule-organizing center. It contains a pair of centrioles, two structures that lie perpendicular to each other. Each centriole is a cylinder of nine triplets of microtubules.

The centrosome replicates itself before a cell divides, and the centrioles play a role in pul

release materials into the extracellular space. The primary components of these materials are glycoproteins and the protein collagen. Collectively, these materials are called the extracellular matrix (Figure 3.16). Not only does the extracellular matrix hold the cells together to form a tissue, but it also allows the cells within the tissue to communicate with each other.

FIGURE 3.16 The extracellular matrix consists of a network of substances secreted by cells.

Blood clotting provides an example of the role of the extracellular matrix in cell communication. When the cells lining a blood vessel are damaged, they display a protein receptor called tissue factor. When tissue factor binds with another factor in the extracellular matrix, it causes platelets to adhere to the wall of the damaged blood vessel, stimulates adjacent smooth muscle cells in the blood vessel to contract (thus constricting the blood vessel), and initiates a series of steps that stimulate the platelets to produce clotting factors.

Intercellular Junctions

Cells can also communicate with each other by direct contact, referred to as intercellular junctions. There are some differences in the ways that plant and animal cells do this. Plasmodesmata (singular = plasmodesma) are junctions between plant cells, whereas animal cell contacts include tight and gap junctions, and desmosomes.

In general, long stretches of the plasma membranes of neighboring plant cells cannot touch one another because they are separated by the cell walls surrounding each cell. Plasmodesmata are numerous channels that pass between the cell walls of adjacent plant cells, connecting their cytoplasm and enabling signal molecules and nutrients to be transported from cell to cell (Figure 3.17a).

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FIGURE 3.17 There are four kinds of connections between cells. (a) A plasmodesma is a channel between the cell walls of two adjacent plant cells. (b) Tight junctions join adjacent animal cells. (c) Desmosomes join two animal cells together. (d) Gap junctions act as channels between animal cells. (credit b, c, d: modification of work by Mariana Ruiz Villareal)

A tight junction is a watertight seal between two adjacent animal cells (Figure 3.17b). Proteins hold the cells tightly against each other. This tight adhesion prevents materials from leaking between the cells. Tight junctions are typically found in the epithelial tissue that lines internal organs and cavities, and composes most of the skin. For example, the tight junctions of the epithelial cells lining the urinary bladder prevent urine from leaking into the extracellular space.

Also found only in animal cells are desmosomes, which act like spot welds between adjacent epithelial cells (Figure 3.17c). They keep cells together in a sheet-like formation in organs and tissues that stretch, like the skin, heart, and muscles.

Gap junctions in animal cells are like plasmodesmata in plant cells in that they are channels between adja

The greater the difference in concentration, the more rapid the diffusion.

The closer the distribution of the material gets to equilibrium, the slower the rate of diffusion becomes.

? Mass of the molecules diffusing: More massive molecules move more slowly, because it is more difficult for them to move between the molecules of the substance they are moving through; therefore, they diffuse more slowly.

? Temperature: Higher temperatures increase the energy and therefore the movement of the molecules, increasing the rate of diffusion.

? Solvent density: As the density of the solvent increases, the rate of diffusion decreases. The molecules slow down because they have a more difficult time getting through the denser medium.

LINK TO LEARNING

For an animation of the diffusion process in action, view this short video (http://openstax.org//passive_trnsprt) on cell membrane transport.

Facilitated transport

In facilitated transport, also called facilitated diffusion, material moves across the plasma membrane with the assistance of transmembrane proteins down a concentration gradient (from high to low concentration) without the expenditure of cellular energy. However, the substances that undergo facilitated transport would otherwise not diffuse easily or quickly across the plasma membrane. The solution to moving polar substances and other substances across the plasma membrane rests in the proteins that span its surface. The material being transported is first attached to protein or glycoprotein receptors on the exterior surface of the plasma membrane. This allows the material that is needed by the cell to be removed from the extracellular fluid. The substances are then passed to specific integral proteins that facilitate their passage, because they form channels or pores that allow certain substances to pass through the membrane. The integral proteins involved in facilitated transport are collectively referred to as transport proteins, and they function as either channels for the material or carriers.

Osmosis

Osmosis is the movement of free water molecules through a semipermeable membrane according to the water's concentration gradient across the membrane, which is inversely proportional to the solutes' concentration.

Whereas

diffusion transports material across membranes and within cells, osmosis transports only water across a membrane

and the membrane limits the diffusion of solutes in the water. Osmosis is a special case of diffusion. Water, like other substances, moves from an area of high concentration of free water molecules to one of low free water molecule concentration. Imagine a beaker with a semipermeable membrane, separating the two sides or halves (Figure 3.21). On both sides of the membrane, the water level is the same, but there are different concentrations on

each side of a dissolved substance, or solute, that cannot cross the membrane. If the volume of the water is the same, but the concentrations of solute are different, then ther

r fluid in which they are bathed; at the same time, cells have higher concentrations of potassium (K^+) and lower concentrations of sodium (Na^+) than does the extracellular fluid. Thus, in a living cell, the concentration gradient and electrical gradient of Na^+ promotes diffusion of the ion into the cell, and the electrical gradient of Na^+

(a positive ion) tends to drive it inward to the negatively charged interior. The situation is more complex, however, for other elements such as potassium. The electrical gradient of K^+ promotes diffusion of the ion into the cell, but the concentration gradient of K^+ promotes diffusion out of the cell (Figure 3.24). The combined gradient that affects an ion is called its electrochemical gradient, and it is especially important to muscle and nerve cells.

FIGURE 3.24 Electrochemical gradients arise from the combined effects of concentration gradients and electrical gradients. Na^+ ions are at higher concentration outside the cell, and K^+ ions are at higher concentration inside of the cell, and yet the inside of the cell has negative net charge compared to the other side of the membrane. This is due to the presence of K^+ binding proteins and other negatively charged molecules. The difference in electrical charges attracts the positively charged Na ions toward the inside of the cell, the electrical gradient, while the K ions tend to flow through K channels toward the outside of the cell due to the concentration difference, the concentration gradient. (credit: modification of work by ?Synaptitude?/Wikimedia Commons)

Moving Against a Gradient

To move substances against a concentration or an electrochemical gradient, the cell must use energy. This energy is harvested from ATP that is generated through cellular metabolism. Active transport mechanisms, collectively called pumps or carrier proteins, work against electrochemical gradients. With the exception of ions, small substances constantly pass through plasma membranes. Active transport maintains concentrations of ions and other substances needed by living cells in the face of these passive changes. Much of a cell's supply of metabolic energy may be spent maintaining these processes. Because active transport mechanisms depend on cellular metabolism for energy, they are sensitive to many metabolic poisons that interfere with the supply of ATP.

Two mechanisms exist for the transport of small-molecular weight material and macromolecules. Primary active transport moves ions across a membrane and creates a difference in charge across that membrane. The primary active transport system uses ATP to move a substance, such as an ion, into the cell, and often at the same time, a

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second substance is moved out of the cell. The sodium-potassium pump, an important pump in animal cells, expends energy to move potassium ions into the cell and a different number of sodium ions out of the cell (Figure 3.25). The action of this pump results in a

sulting in a fluid rather than static

concentration gradient an area of high concentration character

across from an area of low concentration gap junction a channel between two adjacent animal

cytoplasm the entire region between the plasma cells that allows ions, nutrients, and other low-membrane and the nuclear envelope, consisting of molecular weight substances to pass between the organelles suspended in the gel-like cytosol, the cells, enabling the cells to communicate cytoskeleton, and various chemicals Golgi apparatus a eukaryotic organelle made up of a

cytoskeleton the network of protein fibers that series of stacked membranes that sorts, tags, and collectively maintains the shape of the cell, secures packages lipids and proteins for distribution some organelles in specific positions, allows hypertonic describes a solution in which extracellular cytoplasm and vesicles to move within the cell, and fluid has higher osmolarity than the fluid inside the enables unicellular organisms to move cell

cytosol the gel-like material of the cytoplasm in hypotonic describes a solution in which extracellular which cell structures are suspended fluid has lower osmolarity than the fluid inside the

desmosome a linkage between adjacent epithelial cell

cells that forms when cadherins in the plasma isotonic describes a solution in which the membrane attach to intermediate filaments extracellular fluid has the same osmolarity as the

diffusion a passive process of transport of low- fluid inside the cell

molecular weight material down its concentration lysosome an organelle in an animal cell that functions gradient as the cell's digestive component; it breaks down

electrochemical gradient a gradient produced by the proteins, polysaccharides, lipids, nucleic acids, and combined forces of the electrical gradient and the even worn-out organelles

chemical gradient microscope the instrument that magnifies an object

endocytosis a type of active transport that moves mitochondria (singular: mitochondrion) the cellular substances, including fluids and particles, into a cell organelles responsible for carrying out cellular

endomembrane system the group of organelles and respiration, resulting in the production of ATP, the membranes in eukaryotic cells that work together to cell's main energy-carrying molecule modify, package, and transport lipids and proteins nuclear envelope the double-membrane structure

endoplasmic reticulum (ER) a series of that constitutes the outermost portion of the interconnected membranous structures within nucleus

eukaryotic cells that collectively modify proteins nucleolus the darkly staining body within the nucleus and synthesize lipids that is responsible for assembling ribosomal

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subunits proteins in the plasma membrane for specific

nucleus the cell organelle that houses the cell's DNA molecules or particles

and directs the synthesis of ribosomes and proteins ribosome a cell

ular structure that carries out protein

organelle a membrane-bound compartment or sac synthesis

within a cell rough endoplasmic reticulum (RER) the region of

osmolarity the total amount of substances dissolved the endoplasmic reticulum that is studded with

in a specific amount of solution ribosomes and engages in protein modification

osmosis the transport of water through a selectively permeable the characteristic of a

semipermeable membrane from an area of high membrane that allows some substances through

water concentration to an area of low water but not others (also known as semipermeable)

concentration across a membrane smooth endoplasmic reticulum (SER) the region of

passive transport a method of transporting material the endoplasmic reticulum that has few or no

that does not require energy ribosomes on its cytoplasmic surface and

peroxisome a small, round organelle that contains synthesizes carbohydrates, lipids, and steroid

hydrogen peroxide, oxidizes fatty acids and amino hormones; detoxifies chemicals like pesticides,

acids, and detoxifies many poisons preservatives, medications, and environmental

phagocytosis a process that takes particulate matter pollutants, and stores calcium ions

like macromolecules, cells, or cell fragments that solute a substance dissolved in another to form a

the cell needs from the extracellular fluid; a solution

variation of endocytosis tight junction a firm seal between two adjacent

pinocytosis a process that takes solutes that the cell animal cells created by protein adherence

needs from the extracellular fluid; a variation of tonicity the amount of solute in a solution.

endocytosis unified cell theory the biological concept that states

plasma membrane a phospholipid bilayer with that all organisms are composed of one or more

embedded (integral) or attached (peripheral) cells, the cell is the basic unit of life, and new cells

proteins that separates the internal contents of the arise from existing cells

cell from its surrounding environment vacuole a membrane-bound sac, somewhat larger

plasmodesma (plural: plasmodesmata) a channel than a vesicle, that functions in cellular storage and

that passes between the cell walls of adjacent plant transport

cells, connects their cytoplasm, and allows vesicle a small, membrane-bound sac that functions

materials to be transported from cell to cell in cellular storage and transport; its membrane is

prokaryotic cell a unicellular organism that lacks a capable of fusing with the plasma membrane and

nucleus or any other membrane-bound organelle the membranes of the endoplasmic reticulum and

receptor-mediated endocytosis a variant of Golgi apparatus

endocytosis that involves the use of specific binding

Chapter Summary

3.1 How Cells Are Studied have plasma membranes, cytoplasm, ribosomes, a cell

wall, DNA, and lack membrane-bound organelles.

A cell is the smallest unit of life. Most cells are so small Many also have polysaccharide capsules. Prokaryotic that they c

cannot be viewed with the naked eye. cells range in diameter from 0.1-5.0 μm .

Therefore, scientists must use microscopes to study

cells. Electron microscopes provide higher magnification, higher resolution, and more detail than light microscopes. Like a prokaryotic cell, a eukaryotic cell has a plasma membrane, cytoplasm, and ribosomes, but a eukaryotic cell is typically larger than a prokaryotic cell, organisms are composed of one or more cells, the cell has a true nucleus (meaning its DNA is surrounded by a nuclear envelope), and has other membrane-bound organelles that allow for compartmentalization of

functions. Eukaryotic cells tend to be 10 to 100 times

3.2 Comparing Prokaryotic and Eukaryotic the size of prokaryotic cells.

Cells

3.3 Eukaryotic Cells

Prokaryotes are predominantly single-celled organisms

of the domains Bacteria and Archaea. All prokaryotes Like a prokaryotic cell, a eukaryotic cell has a plasma

membrane, cytoplasm, and ribosomes, but a

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eukaryotic cell is typically larger than a prokaryotic cell, membrane is composed of a bilayer of phospholipids, has a true nucleus (meaning its DNA is surrounded by a nuclear envelope), and has other membrane-bound organelles that allow for compartmentalization of

functions. The plasma membrane is a phospholipid bilayer embedded with proteins. Some of these proteins serve to transport materials

into or out of the cell. Carbohydrates are attached to the surface of the membrane. These form complexes that

function to identify the cell to other cells. The fluid mosaic model of the membrane owes itself to the discovery of the phospholipid bilayer and the presence of proteins and lipids on the outward-facing surface of the membrane. These form complexes that

function to identify the cell to other cells. The fluid mosaic model of the membrane owes itself to the

discovery of the phospholipid bilayer and the presence of proteins and lipids on the outward-facing surface of the membrane. These form complexes that

function to identify the cell to other cells. The fluid mosaic model of the membrane owes itself to the discovery of the phospholipid bilayer and the presence of proteins and lipids on the outward-facing surface of the membrane. These form complexes that

of cells, but rather than being a static bag, they are

Animal cells also have a centrosome and lysosomes.

dynamic and constantly in flux.

The centrosome has two bodies, the centrioles, with an

unknown role in cell division. Lysosomes are the 3.5 Passive Transport

digestive organelles of animal cells.

The passive forms of transport, diffusion and osmosis

Plant cells have a cell wall, chloroplasts, and a central vacuole. The plant cell wall, whose primary component is cellulose, protects the cell, provides structural support, and gives shape to the cell. Photosynthesis takes place in chloroplasts. The central vacuole contains solutions of more than one substance, each type of which expands, enlarging the cell without the need for more cytoplasm. The endomembrane system includes the nuclear envelope, the endoplasmic reticulum, Golgi apparatus, lysosomes, vesicles, as well as the plasma membrane. These cellular components work together to modify, package, tag, and transport membrane lipids and proteins. In living systems, diffusion of substances into and out of cells is mediated by the plasma membrane. Some materials diffuse readily through the membrane, but others are hindered, and their passage is only made possible by protein channels and carriers. The cytoskeleton has three different types of elements. Microfilaments provide rigidity and shape to the cell, and facilitate cellular movements. Intermediate filaments bear tension and anchor the nucleus and other organelles in place. Microtubules help the cell resist compression, serve as tracks for some substances, and help move vesicles through the cell, and pull replicated chromosomes to opposite ends of a dividing cell. They are also the structural elements of centrioles, flagella, and cilia. The combined concentration gradient that affects an ion includes its

concentration gradient and its electrical gradient.

Animal cells communicate through their extracellular matrices and are connected to each other by tight junctions, desmosomes, and gap junctions. Plant cells are connected and communicate with each other by plasmodesmata. Living cells need certain substances in concentrations greater than they exist in the extracellular space. Moving substances up their electrochemical gradients requires energy from the cell. Active transport uses energy stored in ATP to fuel the transport. Active transport uses integral

proteins in the cell membrane to move the

The modern understanding of the plasma membrane is referred to as the fluid mosaic model. The plasma membrane is material?these proteins are analogous to pumps. Some pumps, which carry out pri

erse reaction to photosynthesis. In this reaction, oxygen is consumed and carbon dioxide is released as a waste product. The reaction is summarized as:

Both of these reactions involve many steps.

The processes of making and breaking down sugar molecules illustrate two examples of metabolic pathways. A metabolic pathway is a series of chemical reactions that takes a starting molecule and modifies it, step-by-step, through a series of metabolic intermediates, eventually yielding a final product. In the example of sugar metabolism, the first metabolic pathway synthesized sugar from smaller molecules, and the other pathway broke sugar down into smaller molecules. These two opposite processes—the first requiring energy and the second producing energy—are referred to as anabolic pathways (building polymers) and catabolic pathways (breaking down polymers into their monomers), respectively. Consequently, metabolism is composed of synthesis (anabolism) and degradation (catabolism) (Figure 4.3).

It is important to know that the chemical reactions of metabolic pathways do not take place on their own. Each reaction step is facilitated, or catalyzed, by a protein called an enzyme. Enzymes are important for catalyzing all types of biological reactions—those that require energy as well as those that release energy.

FIGURE 4.3 Catabolic pathways are those that generate energy by breaking down larger molecules. Anabolic pathways are those that require energy to synthesize larger molecules. Both types of pathways are required for maintaining the cell's energy balance.

Energy

Thermodynamics refers to the study of energy and energy transfer involving physical matter. The matter relevant to

a particular case of energy transfer is called a system, and everything outside of that matter is called the surroundings. For instance, when heating a pot of water on the stove, the system includes the stove, the pot, and the

water. Energy is transferred within the system (between the stove, pot, and water). There are two types of systems:

open and closed. In an open system, energy can be exchanged with its surroundings. The stovetop system is open

because heat can be lost to the air. A closed system cannot exchange energy with its surroundings.

Biological organisms are open systems. Energy is exchanged between them and their surroundings as they use energy from the sun to perform photosynthesis or consume energy-storing molecules and release energy to the environment by doing work and releasing heat. Like all things in the physical world, energy is subject to physical laws. The laws of thermodynamics govern the transfer of energy in and among all systems in the universe.

In general, energy is defined as the ability to do work, or to create some kind of change. Energy exists in different

forms. For example, electrical energy, light energy, and heat energy are all different types of energy. To appreciate

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the way energy flows into and out o

in physical systems is that of order and disorder. The more energy that is lost by a system to its surroundings, the less ordered and more random the system is. Scientists refer to the measure of randomness or disorder within a system as entropy. High entropy means high disorder and low energy. Molecules and chemical reactions have varying entropy as well. For example, entropy increases as molecules at a high concentration in one place diffuse and spread out. The second law of thermodynamics says that energy will always be lost as heat in energy transfers or transformations.

Living things are highly ordered, requiring constant energy input to be maintained in a state of low entropy.

Potential and Kinetic Energy

When an object is in motion, there is energy associated with that object. Think of a wrecking ball. Even a slow-moving wrecking ball can do a great deal of damage to other objects. Energy associated with objects in motion is called kinetic energy (Figure 4.5). A speeding bullet, a walking person, and the rapid movement of molecules in the air (which produces heat) all have kinetic energy.

Now what if that same motionless wrecking ball is lifted two stories above ground with a crane? If the wrecking ball is unmoving, is there energy associated with it? The answer is yes. The energy that was required to lift the wrecking ball did not disappear, but is now stored in the wrecking ball by virtue of its position and the force of gravity acting on it. This type of energy is called potential energy (Figure 4.5). If the ball were to fall, the potential energy would be transformed into kinetic energy until all of the potential energy was exhausted when the ball rested on the ground. Wrecking balls also swing like a pendulum; through the swing, there is a constant change of potential energy (highest at the top of the swing) to kinetic energy (highest at the bottom of the swing). Other examples of potential energy include the energy of water held behind a dam or a person about to skydive out of an airplane.

FIGURE 4.5 Still water has potential energy; moving water, such as in a waterfall or a rapidly flowing river, has kinetic energy. (credit "dam": modification of work by "Pascal"/Flickr; credit "waterfall": modification of work by Frank Gualtieri)

Potential energy is not only associated with the location of matter, but also with the structure of matter. Even a spring on the ground has potential energy if it is compressed; so does a rubber band that is pulled taut. On a molecular level, the bonds that hold the atoms of molecules together exist in a particular structure that has potential energy. Remember that anabolic cellular pathways require energy to synthesize complex molecules from simpler ones and catabolic pathways release energy when complex molecules are broken down. The fact that

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energy can be released by the breakdown of certain chemical bonds implies that those bonds have potential energy

FIGURE 4.8 The induced-fit model is an adjustment to the lock-and-key model and explains how enzymes and substrates undergo dynamic modifications during the transition state to increase the affinity of the substrate for the active site.

It would seem ideal to have a scenario in which all of an organism's enzymes existed in abundant supply and functioned optimally under all cellular conditions, in all cells, at all times. However, a variety of mechanisms ensures that this does not happen. Cellular needs and conditions constantly vary from cell to cell, and change within individual cells over time. The required enzymes of stomach cells differ from those of fat storage cells, skin cells, blood cells, and nerve cells. Furthermore, a digestive organ cell works much harder to process and break down nutrients during the time that closely follows a meal compared with many hours after a meal. As these cellular demands and conditions vary, so must the amounts and functionality of different enzymes.

Since the rates of biochemical reactions are controlled by activation energy, and enzymes lower and determine activation energies for chemical reactions, the relative amounts and functioning of the variety of enzymes within a cell ultimately determine which reactions will proceed and at what rates. This determination is tightly controlled in cells. In certain cellular environments, enzyme activity is partly controlled by environmental factors like pH, temperature, salt concentration, and, in some cases, cofactors or coenzymes.

Enzymes can also be regulated in ways that either promote or reduce enzyme activity. There are many kinds of molecules that inhibit or promote enzyme function, and various mechanisms by which they do so. In some cases of enzyme inhibition, an inhibitor molecule is similar enough to a substrate that it can bind to the active site and simply block the substrate from binding. When this happens, the enzyme is inhibited through competitive inhibition, because an inhibitor molecule competes with the substrate for binding to the active site.

On the other hand, in noncompetitive inhibition, an inhibitor molecule binds to the enzyme in a location other than the active site, called an allosteric site, but still manages to prevent substrate binding to the active site. Some inhibitor molecules bind to enzymes in a location where their binding induces a conformational change that reduces the enzyme activity as it no longer effectively catalyzes the conversion of the substrate to product. This type of inhibition is called allosteric inhibition (Figure 4.9). Most allosterically regulated enzymes are made up of more than one polypeptide, meaning that they have more than one protein subunit. When an allosteric inhibitor binds to a region on an enzyme, all active sites on the protein subunits are changed slightly such that they bind their substrates with less efficiency. There are allosteric activators as well as inhibitors. Allosteric activators bind to

n atoms come into the citric acid cycle from each acetyl group. Two carbon dioxide molecules are released on each turn of the cycle; however, these do not contain the same carbon atoms contributed by the acetyl group on that turn of the pathway. The two acetyl-carbon atoms will eventually be released on later turns of the cycle; in this way, all six carbon atoms from the original glucose molecule will be eventually released as carbon dioxide. It takes two turns of the cycle to process the equivalent of one glucose molecule. Each turn of the cycle forms three high-energy NADH molecules and one high-energy FADH₂ molecule. These high-energy carriers will connect with the last portion of aerobic respiration to produce ATP molecules. One ATP (or an equivalent) is also made in each cycle. Several of the intermediate compounds in the citric acid cycle can be used in synthesizing non-essential amino acids; therefore, the cycle is both anabolic and catabolic.

Oxidative Phosphorylation

You have just read about two pathways in glucose catabolism—glycolysis and the citric acid cycle—that generate ATP. Most of the ATP generated during the aerobic catabolism of glucose, however, is not generated directly from these pathways. Rather, it derives from a process that begins with passing electrons through a series of chemical reactions to a final electron acceptor, oxygen. These reactions take place in specialized protein complexes located in the inner membrane of the mitochondria of eukaryotic organisms and on the inner part of the cell membrane of prokaryotic organisms. The energy of the electrons is harvested and used to generate an electrochemical gradient across the inner mitochondrial membrane. The potential energy of this gradient is used to generate ATP. The entirety of this process is called oxidative phosphorylation.

The electron transport chain (Figure 4.15a) is the last component of aerobic respiration and is the only part of metabolism that uses atmospheric oxygen. Oxygen continuously diffuses into plants for this purpose. In animals, oxygen enters the body through the respiratory system. Electron transport is a series of chemical reactions that resembles a bucket brigade in that electrons are passed rapidly from one component to the next, to the endpoint of the chain where oxygen is the final electron acceptor and water is produced. There are four complexes composed of proteins, labeled I through IV in Figure 4.15c, and the aggregation of these four complexes, together with associated mobile, accessory electron carriers, is called the electron transport chain. The electron transport chain is present in multiple copies in the inner mitochondrial membrane of eukaryotes and in the plasma membrane of prokaryotes. In each transfer of an electron through the electron transport chain, the electron loses energy, but with some transfers, the energy is stored as potential energy by using it to pump hydrogen ions across the inner

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ut the catabolism of glucose, which provides energy to living cells. But living things consume more than just glucose for food. How does a turkey sandwich, which contains protein, provide energy to your cells?

This happens because all of the catabolic pathways for carbohydrates, proteins, and lipids eventually connect into

glycolysis and the citric acid cycle pathways (Figure 4.20). Metabolic pathways should be thought of as porous?that

is, substances enter from other pathways, and other substances leave for other pathways. These pathways are not

closed systems. Many of the products in a particular pathway are reactants in other pathways.

Connections of Other Sugars to Glucose Metabolism

Glycogen, a polymer of glucose, is a short-term energy storage molecule in animals. When there is adequate ATP

present, excess glucose is converted into glycogen for storage. Glycogen is made and stored in the liver and muscle.

Glycogen will be taken out of storage if blood sugar levels drop. The presence of glycogen in muscle cells as a source

of glucose allows ATP to be produced for a longer time during exercise.

Sucrose is a disaccharide made from glucose and fructose bonded together. Sucrose is broken down in the small

intestine, and the glucose and fructose are absorbed separately. Fructose is one of the three dietary monosaccharides, along with glucose and galactose (which is part of milk sugar, the disaccharide lactose), that are

absorbed directly into the bloodstream during digestion. The catabolism of both fructose and galactose produces

the same number of ATP molecules as glucose.

Connections of Proteins to Glucose Metabolism

Proteins are broken down by a variety of enzymes in cells. Most of the time, amino acids are recycled into new proteins. If there are excess amino acids, however, or if the body is in a state of famine, some amino acids will be

shunted into pathways of glucose catabolism. Each amino acid must have its amino group removed prior to entry

into these pathways. The amino group is converted into ammonia. In mammals, the liver synthesizes urea from two

ammonia molecules and a carbon dioxide molecule. Thus, urea is the principal waste product in mammals from the

nitrogen originating in amino acids, and it leaves the body in urine.

Connections of Lipids to Glucose Metabolism

The lipids that are connected to the glucose pathways are cholesterol and triglycerides. Cholesterol is a lipid that

contributes to cell membrane flexibility and is a precursor of steroid hormones. The synthesis of cholesterol starts

with acetyl CoA and proceeds in only one direction. The process cannot be reversed, and ATP is not produced.

Triglycerides are a form of long-term energy storage in animals. Triglycerides store about twice as much energy as

carbohydrates. Triglycerides are made of glycerol and three fatty acids. Animals can make most of the fatty acids

they need. Triglycerides can be both made and broken down through parts of the glucose catabolism pat

rgy

conformation change in the active site, preventing feedback inhibition a mechanism of enzyme activity binding with the substrate regulation in which the product of a reaction or the

anabolic describes the pathway that requires a net final product of a series of sequential reactions energy input to synthesize complex molecules from simpler ones series

anaerobic cellular respiration the use of an electron fermentation the steps that follow the partial acceptor other than oxygen to complete metabolism oxidation of glucose via glycolysis to regenerate using electron transport-based chemiosmosis NAD^+ ; occurs in the absence of oxygen and uses an

ATP (also, adenosine triphosphate) the cell's energy organic compound as the final electron acceptor currency glycolysis the process of breaking glucose into two

ATP synthase a membrane-embedded protein three-carbon molecules with the production of ATP complex that regenerates ATP from ADP with energy and NADH from protons diffusing through it heat energy the energy transferred from one system

bioenergetics the concept of energy flow through to another that is not work living systems kinetic energy the type of energy associated with

catabolic describes the pathway in which complex objects in motion molecules are broken down into simpler ones, metabolism all the chemical reactions that take yielding energy as an additional product of the place inside cells, including those that use energy reaction and those that release energy

chemiosmosis the movement of hydrogen ions down noncompetitive inhibition a general mechanism of their electrochemical gradient across a membrane enzyme activity regulation in which a regulatory through ATP synthase to generate ATP molecule binds to a site other than the active site

citric acid cycle a series of enzyme-catalyzed and prevents the active site from binding the chemical reactions of central importance in all living substrate; thus, the inhibitor molecule does not cells that harvests the energy in carbon-carbon compete with the substrate for the active site; bonds of sugar molecules to generate ATP; the citric allosteric inhibition is a form of noncompetitive acid cycle is an aerobic metabolic pathway because inhibition it requires oxygen in later reactions to proceed oxidative phosphorylation the production of ATP by

competitive inhibition a general mechanism of the transfer of electrons down the electron enzyme activity regulation in which a molecule transport chain to create a proton gradient that is other than the enzyme's substrate is able to bind used by ATP synthase to add phosphate groups to the active site and prevent the substrate itself from ADP molecules binding, thus inhibiting the overall rate of reaction potential energy the type of energy that refers to the for the enzyme potential to do work

electron transport chain a series of four large, multi- substrate a molecule on which the enzyme acts p

protein complexes embedded in the inner mitochondrial membrane that accept electrons between heat, energy, and work from donor compounds and harvests energy from a

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Chapter Summary

4.1 Energy and Metabolism

atoms and attaches them to NAD⁺. Two ATP molecules

are invested in the first half and four ATP molecules are produced during the second half. This produces a net gain of two ATP molecules per molecule of glucose for the combination of chemical reactions that take place within the cell. Catabolic reactions break down complex chemicals into simpler ones and are associated with energy release. Anabolic processes build complex molecules out of simpler ones and require energy.

The citric acid cycle is a series of chemical reactions in which the electron transport chain to generate ATP. One molecule of ATP (or an equivalent) is produced per turn of the cycle. The laws of thermodynamics. The first law states that the total amount of energy in the universe is constant. The electron transport chain is the portion of aerobic respiration that uses free oxygen as the final electron acceptor for electrons removed from the intermediate unusable form, such as heat energy. Energy comes in compounds in glucose catabolism. The electrons are different forms: kinetic, potential, and free. The change passed through a series of chemical reactions, with a negative free energy of a reaction can be negative (releases small amount of free energy used at three points to energy, exergonic) or positive (consumes energy, transport hydrogen ions across the membrane. This is endergonic). All reactions require an initial input of energy to proceed, called the activation energy. The electrons are passed from NADH or FADH₂ down

the electron transport chain, they lose energy. The products of the electron transport chain are water and chemical reactions by lowering their activation energy. A number of intermediate compounds can be diverted into the anabolism of other biochemical environment that fits particular chemical reactants for molecules, such as nucleic acids, non-essential amino acids, that enzyme, called substrates. Enzymes and acids, sugars, and lipids. These same molecules, substrates are thought to bind according to

organelle where photosynthesis stroma the fluid-filled space surrounding the grana
takes place inside a chloroplast where the Calvin cycle reactions

electromagnetic spectrum the range of all possible of photosynthesis take place
frequencies of radiation thylakoid a disc-shaped membranous structure

granum a stack of thylakoids located inside a inside a chloroplast where the light-dependent
chloroplast reactions of photosynthesis take place using

heterotroph an organism that consumes other chlorophyll embedded in the membranes
organisms for food wavelength the distance between consecutive points

light-dependent reaction the first stage of of a wave
photosynthesis where visible light is absorbed to

Chapter Summary

5.1 Overview of Photosynthesis sunlight. The most common and abundant pigment is

chlorophyll a. A photon strikes photosystem II to

The process of photosynthesis transformed life on initiate photosynthesis. Energy travels through the
earth. By harnessing energy from the sun, electron transport chain, which pumps hydrogen ions
photosynthesis allowed living things to access into the thylakoid space. This forms an electrochemical
enormous amounts of energy. Because of gradient. The ions flow through ATP synthase from the
photosynthesis, living things gained access to sufficient thylakoid space into the stroma in a process called
energy, allowing them to evolve new structures and chemiosmosis to form molecules of ATP, which are
achieve the biodiversity that is evident today. used for the formation of sugar molecules in the
Only certain organisms, called autotrophs, can perform second stage of photosynthesis. Photosystem I
photosynthesis; they require the presence of absorbs a second photon, which results in the
chlorophyll, a specialized pigment that can absorb light formation of an NADPH molecule, another energy
and convert light energy into chemical energy. carrier for the Calvin cycle reactions.

Photosynthesis uses carbon dioxide and water to

assemble carbohydrate molecules (usually glucose) 5.3 The Calvin Cycle

and releases oxygen into the air. Eukaryotic autotrophs, Using the energy carriers formed in the first stage of
such as plants and algae, have organelles called photosynthesis, the Calvin cycle reactions fix CO₂ from
chloroplasts in which photosynthesis takes place. the environment to build carbohydrate molecules. An

enzyme, RuBisCO, catalyzes the fixation reaction, by

5.2 The Light-Dependent Reactions of combining CO₂ with RuBP. The resulting six-carbon

Photosynthesis compound is broken down into two three-carbon

In the first part of photosynthesis, the light-dependent compounds, and the energy in ATP and NADPH is used
reaction, pigment molecules absorb energy from to convert these molecules into G3P. One of the three-

carbon molecules of G3P leaves the cycle to become a

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part of a carbohydrate molecule. The remaining G3P process of cellular respiration. Plants are capable of
molecules

ea. Chromosomes become more condensed and visually discrete. Each sister chromatid attaches to spindle microtubules at the centromere via a protein complex called the kinetochore.

During metaphase, all of the chromosomes are aligned in a plane called the metaphase plate, or the equatorial plane, midway between the two poles of the cell. The sister chromatids are still tightly attached to each other. At this time, the chromosomes are maximally condensed.

During anaphase, the sister chromatids at the equatorial plane are split apart at the centromere. Each chromatid, now called a chromosome, is pulled rapidly toward the centrosome to which its microtubule was attached. The cell becomes visibly elongated as the non-kinetochore microtubules slide against each other at the metaphase plate where they overlap.

During telophase, all of the events that set up the duplicated chromosomes for mitosis during the first three phases are reversed. The chromosomes reach the opposite poles and begin to decondense (unravel). The mitotic spindles are broken down into monomers that will be used to assemble cytoskeleton components for each daughter cell. Nuclear envelopes form around chromosomes.

LINK TO LEARNING

This page of movies (http://openstax.org//divisn_newtcell) illustrates different aspects of mitosis. Watch the movie entitled "DIC microscopy of cell division in a newt lung cell" and identify the phases of mitosis.

Cytokinesis

Cytokinesis is the second part of the mitotic phase during which cell division is completed by the physical separation of the cytoplasmic components into two daughter cells. Although the stages of mitosis are similar for most eukaryotes, the process of cytokinesis is quite different for eukaryotes that have cell walls, such as plant cells.

In cells such as animal cells that lack cell walls, cytokinesis begins following the onset of anaphase. A contractile ring composed of actin filaments forms just inside the plasma membrane at the former metaphase plate. The actin filaments pull the equator of the cell inward, forming a fissure. This fissure, or "crack," is called the cleavage furrow. The furrow deepens as the actin ring contracts, and eventually the membrane and cell are cleaved in two (Figure 6.5).

In plant cells, a cleavage furrow is not possible because of the rigid cell walls surrounding the plasma membrane. A new cell wall must form between the daughter cells. During interphase, the Golgi apparatus accumulates enzymes, structural proteins, and glucose molecules prior to breaking up into vesicles and dispersing throughout the dividing cell. During telophase, these Golgi vesicles move on microtubules to collect at the metaphase plate. There, the vesicles fuse from the center toward the cell walls; this structure is called a cell plate. As more vesicles fuse, the cell plate enlarges until it merges with the cell wall at the periphery of the cell. Enzymes use the glucose that has accumulated between the membrane layers to bu

ore readily. Over and over, small, uncorrected errors are passed from parent cell to daughter cells and accumulate as each generation of cells produces more non-functional proteins from uncorrected DNA damage. Eventually, the pace of the cell cycle speeds up as the effectiveness of the control and repair mechanisms decreases. Uncontrolled growth of the mutated cells outpaces the growth of normal cells in the area, and a tumor can result.

Proto-oncogenes

The genes that code for the positive cell-cycle regulators are called proto-oncogenes. Proto-oncogenes are normal genes that, when mutated, become oncogenes?genes that cause a cell to become cancerous. Consider what might happen to the cell cycle in a cell with a recently acquired oncogene. In most instances, the alteration of the DNA sequence will result in a less functional (or non-functional) protein. The result is detrimental to the cell and will likely prevent the cell from completing the cell cycle; however, the organism is not harmed because the mutation will not be carried forward. If a cell cannot reproduce, the mutation is not propagated and the damage is minimal. Occasionally, however, a gene mutation causes a change that increases the activity of a positive regulator. For example, a mutation that allows Cdk, a protein involved in cell-cycle regulation, to be activated before it should be could push the cell cycle past a checkpoint before all of the required conditions are met. If the resulting daughter cells are too damaged to undertake further cell divisions, the mutation would not be propagated and no harm comes to the organism. However, if the atypical daughter cells are able to divide further, the subsequent generation of cells will likely accumulate even more mutations, some possibly in additional genes that regulate the cell cycle.

The Cdk example is only one of many genes that are considered proto-oncogenes. In addition to the cell-cycle regulatory proteins, any protein that influences the cycle can be altered in such a way as to override cell-cycle checkpoints. Once a proto-oncogene has been altered such that there is an increase in the rate of the cell cycle, it is then called an oncogene.

Tumor Suppressor Genes

Like proto-oncogenes, many of the negative cell-cycle regulatory proteins were discovered in cells that had become cancerous. Tumor suppressor genes are genes that code for the negative regulator proteins, the type of regulator that?when activated?can prevent the cell from undergoing uncontrolled division. The collective function of the best-understood tumor suppressor gene proteins, retinoblastoma protein (RB1), p53, and p21, is to put up a roadblock to cell-cycle progress until certain events are completed. A cell that carries a mutated form of a negative regulator might not be able to halt the cell cycle if there is a problem.

n a chromosome

preparedness of a eukaryotic cell to advance metaphase the stage of mitosis during which through the various cell cycle stages chromosomes are lined up at the metaphase plate

cell plate a structure formed during plant-cell metaphase plate the equatorial plane midway cytokinesis by Golgi vesicles fusing at the between two poles of a cell where the metaphase plate; will ultimately lead to formation of chromosomes align during metaphase a cell wall to separate the two daughter cells mitosis the period of the cell cycle at which the

centriole a paired rod-like structure constructed of duplicated chromosomes are separated into microtubules at the center of each animal cell identical nuclei; includes prophase, prometaphase, centrosome metaphase, anaphase, and telophase

cleavage furrow a constriction formed by the actin mitotic phase the period of the cell cycle when ring during animal-cell cytokinesis that leads to duplicated chromosomes are distributed into two cytoplasmic division nuclei and the cytoplasmic contents are divided;

cytokinesis the division of the cytoplasm following includes mitosis and cytokinesis mitosis to form two daughter cells mitotic spindle the microtubule apparatus that

diploid describes a cell, nucleus, or organism orchestrates the movement of chromosomes during containing two sets of chromosomes ($2n$) mitosis

FtsZ a tubulin-like protein component of the oncogene a mutated version of a proto-oncogene, prokaryotic cytoskeleton that is important in which allows for uncontrolled progression of the cell prokaryotic cytokinesis (name origin: Filamenting cycle, or uncontrolled cell reproduction temperature-sensitive mutant Z) origin the region of the prokaryotic chromosome at

G0 phase a cell-cycle phase distinct from the G1 which replication begins phase of interphase; a cell in G0 is not preparing to prometaphase the stage of mitosis during which divide mitotic spindle fibers attach to kinetochores

G1 phase (also, first gap) a cell-cycle phase; first prophase the stage of mitosis during which phase of interphase centered on cell growth during chromosomes condense and the mitotic spindle mitosis begins to form

G2 phase (also, second gap) a cell-cycle phase; third proto-oncogene a normal gene that controls cell phase of interphase where the cell undergoes the division by regulating the cell cycle that becomes an final preparations for mitosis oncogene if it is mutated

gamete a haploid reproductive cell or sex cell (sperm quiescent describes a cell that is performing normal or egg) cell functions and has not initiated preparations for

gene the physical and functional unit of heredity; a cell division sequence of DNA that codes for a specific peptide or S phase the second, or synthesis phase, of RNA molecule interphase during which DNA replication occurs

genome the entire genetic complement (DNA) of an septum a wall formed between bacterial daughter organism cells as a precursor to cell separation

haploid des

ast majority of eukaryotic organisms can or must employ some form of meiosis and fertilization to reproduce.

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7.1 Sexual Reproduction

LEARNING OBJECTIVES

By the end of this section, you will be able to:

? Explain that variation among offspring is a potential evolutionary advantage resulting from sexual reproduction

? Describe the three different life-cycle strategies among sexual multicellular organisms and their commonalities

Sexual reproduction was an early evolutionary innovation after the appearance of eukaryotic cells. The fact that most eukaryotes reproduce sexually is evidence of its evolutionary success. In many animals, it is the only mode of reproduction. And yet, scientists recognize some real disadvantages to sexual reproduction. On the surface, offspring that are genetically identical to the parent may appear to be more advantageous. If the parent organism is successfully occupying a habitat, offspring with the same traits would be similarly successful. There is also the obvious benefit to an organism that can produce offspring by asexual budding, fragmentation, or asexual eggs. These methods of reproduction do not require another organism of the opposite sex. There is no need to expend energy finding or attracting a mate. That energy can be spent on producing more offspring. Indeed, some organisms that lead a solitary lifestyle have retained the ability to reproduce asexually. In addition, asexual populations only have female individuals, so every individual is capable of reproduction. In contrast, the males in sexual populations (half the population) are not producing offspring themselves. Because of this, an asexual population can grow twice as fast as a sexual population in theory. This means that in competition, the asexual population would have the advantage. All of these advantages to asexual reproduction, which are also disadvantages to sexual reproduction, should mean that the number of species with asexual reproduction should be more common.

However, multicellular organisms that exclusively depend on asexual reproduction are exceedingly rare. Why is sexual reproduction so common? This is one of the important questions in biology and has been the focus of much research from the latter half of the twentieth century until now. A likely explanation is that the variation that sexual reproduction creates among offspring is very important to the survival and reproduction of those offspring. The only source of variation in asexual organisms is mutation. This is the ultimate source of variation in sexual organisms. In addition, those different mutations are continually reshuffled from one generation to the next when different parents combine their unique genomes, and the genes are mixed into different combinations by the process of meiosis. Meiosis is the division of the contents of the nucleus that divides the chromosomes among gametes. Variation is introduced during meiosis, as well

ation among the products of meiosis

Sexual reproduction requires fertilization, a union of two cells from two individual organisms. If those two cells each contain one set of chromosomes, then the resulting cell contains two sets of chromosomes. The number of sets of chromosomes in a cell is called its ploidy level. Haploid cells contain one set of chromosomes. Cells containing two sets of chromosomes are called diploid. If the reproductive cycle is to continue, the diploid cell must somehow reduce its number of chromosome sets before fertilization can occur again, or there will be a continual doubling in the number of chromosome sets in every generation. So, in addition to fertilization, sexual reproduction includes a nuclear division, known as meiosis, that reduces the number of chromosome sets.

Most animals and plants are diploid, containing two sets of chromosomes; in each somatic cell (the nonreproductive cells of a multicellular organism), the nucleus contains two copies of each chromosome that are referred to as homologous chromosomes. Somatic cells are sometimes referred to as "body" cells.

Homologous

chromosomes are matched pairs containing genes for the same traits in identical locations along their length. Diploid organisms inherit one copy of each homologous chromosome from each parent; all together, they are considered a full set of chromosomes. In animals, haploid cells containing a single copy of each homologous chromosome are found only within gametes. Gametes fuse with another haploid gamete to produce a diploid cell.

The nuclear division that forms haploid cells, which is called meiosis, is related to mitosis. As you have learned, mitosis is part of a cell reproduction cycle that results in identical daughter nuclei that are also genetically identical

to the original parent nucleus. In mitosis, both the parent and the daughter nuclei contain the same number of chromosome sets—diploid for most plants and animals. Meiosis employs many of the same mechanisms as mitosis.

However, the starting nucleus is always diploid and the nuclei that result at the end of a meiotic cell division are haploid. To achieve the reduction in chromosome number, meiosis consists of one round of chromosome duplication

and two rounds of nuclear division. Because the events that occur during each of the division stages are analogous

to the events of mitosis, the same stage names are assigned. However, because there are two rounds of division, the

stages are designated with a "I" or "II." Thus, meiosis I is the first round of meiotic division and consists of prophase I, prometaphase I, and so on. Meiosis I reduces the number of chromosome sets from two to one.

The

genetic information is also mixed during this division to create unique recombinant chromosomes. Meiosis II, in which the second round of meiotic division takes place in a way that is similar to mitosis, includes prophase II, prometaphase II, and so on.

Interphase

Meiosis is preceded by an interphase

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FIGURE 7.5 In prometaphase I, microtubules attach to the fused kinetochores of homologous chromosomes. In anaphase I, the homologous chromosomes are separated. In prometaphase II, microtubules attach to individual kinetochores of sister chromatids. In anaphase II, the sister chromatids are separated.

In telophase II, the chromosomes arrive at opposite poles and begin to decondense. Nuclear envelopes form around the chromosomes. Cytokinesis separates the two cells into four genetically unique haploid cells. At this point, the nuclei in the newly produced cells are both haploid and have only one copy of the single set of chromosomes. The cells produced are genetically unique because of the random assortment of paternal and maternal homologs and because of the recombination of maternal and paternal segments of chromosomes?with their sets of genes?that occurs during crossover.

Comparing Meiosis and Mitosis

Mitosis and meiosis, which are both forms of division of the nucleus in eukaryotic cells, share some similarities, but also exhibit distinct differences that lead to their very different outcomes. Mitosis is a single nuclear division that results in two nuclei, usually partitioned into two new cells. The nuclei resulting from a mitotic division are genetically identical to the original. They have the same number of sets of chromosomes: one in the case of haploid cells, and two in the case of diploid cells. On the other hand, meiosis is two nuclear divisions that result in four nuclei, usually partitioned into four new cells. The nuclei resulting from meiosis are never genetically identical, and they contain one chromosome set only?this is half the number of the original cell, which was diploid (Figure 7.6).

The differences in the outcomes of meiosis and mitosis occur because of differences in the behavior of the chromosomes during each process. Most of these differences in the processes occur in meiosis I, which is a very different nuclear division than mitosis. In meiosis I, the homologous chromosome pairs become associated with each other, are bound together, experience chiasmata and crossover between sister chromatids, and line up along the metaphase plate in tetrads with spindle fibers from opposite spindle poles attached to each kinetochore of a homolog in a tetrad. All of these events occur only in meiosis I, never in mitosis.

Homologous chromosomes move to opposite poles during meiosis I so the number of sets of chromosomes in each nucleus-to-be is reduced from two to one. For this reason, meiosis I is referred to as a reduction division. There is no such reduction in ploidy level in mitosis.

Meiosis II is much more analogous to a mitotic division. In this case, duplicated chromosomes (only one set of them) line up at the center of the cell with divided kinetochores attached to spindle fibers from opposite poles. During anaphase II, as in mitotic anaphase, the kinetochores divide and one sister chromatid is pulled

me dissociates and reattaches to a
banding patterns, and centromere position different, nonhomologous chromosome

life cycle the sequence of events in the development trisomy an otherwise diploid genotype in which one
of an organism and the production of cells that entire chromosome is duplicated
produce offspring X inactivation the condensation of X chromosomes

meiosis a nuclear division process that results in four into Barr bodies during embryonic development in
haploid cells females to compensate for the double genetic dose

Chapter Summary

7.1 Sexual Reproduction Fertilization, the fusion of haploid gametes from two

individuals, restores the diploid condition. Thus,

Nearly all eukaryotes undergo sexual reproduction. The sexually reproducing organisms alternate between
variation introduced into the reproductive cells by haploid and diploid stages. However, the ways in which
meiosis appears to be one of the advantages of sexual reproductive cells are produced and the timing
reproduction that has made it so successful. Meiosis between meiosis and fertilization vary greatly. There
and fertilization alternate in sexual life cycles. There are three main categories of life cycles: diploid-
process of meiosis produces genetically unique dominant, demonstrated by most animals; haploid-
reproductive cells called gametes, which have half the dominant, demonstrated by all fungi and some algae;
number of chromosomes as the parent cell. and alternation of generations, demonstrated by plants

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and some algae. occur in the first division of meiosis. The homologous
chromosomes separate into different nuclei during

7.2 Meiosis meiosis I causing a reduction of ploidy level. The

Sexual reproduction requires that diploid organisms second division of meiosis is much more similar to a
produce haploid cells that can fuse during fertilization mitotic division.

to form diploid offspring. The process that results in

haploid cells is called meiosis. Meiosis is a series of 7.3 Variations in Meiosis

events that arrange and separate chromosomes into The number, size, shape, and banding pattern of
daughter cells. During the interphase of meiosis, each chromosomes make them easily identifiable in a
chromosome is duplicated. In meiosis, there are two karyogram and allow for the assessment of many
rounds of nuclear division resulting in four nuclei and chromosomal abnormalities. Disorders in chromosome
usually four haploid daughter cells, each with half the number, or aneuploidies, are typically lethal to the
number of chromosomes as the parent cell. During embryo, although a few trisomic genotypes are viable.
meiosis, variation in the daughter nuclei is introduced Because of X inactivation, aberrations in sex
because of crossover in prophase I and random chromosomes typically have milder effects on an
alignment at metaphase I. The cells that are produced individual. Aneuploidies also include instances in which
by meiosis are genetically

shed his work, *Experiments in Plant Hybridization*,¹ in the proceedings of the Natural History Society of Brunn. As stated earlier, in genetics, "parent" is often used to describe the individual organism(s) that contribute genetic material to an offspring, usually in the form of gamete cells.

Mendel's work went virtually unnoticed by the scientific community, which incorrectly believed that the process of inheritance involved a blending of parental traits that produced an intermediate physical appearance in offspring. This hypothetical process appeared to be correct because of what we know now as continuous variation. Continuous variation is the range of small differences we see among individuals in a characteristic like human height. It does appear that offspring are a "blend" of their parents' traits when we look at characteristics that exhibit continuous variation. Mendel worked instead with traits that show discontinuous variation. Discontinuous variation is the variation seen among individuals when each individual shows one of two or a very few easily distinguishable traits, such as violet or white flowers. Mendel's choice of these kinds of traits allowed him to see experimentally that the traits were not blended in the offspring as would have been expected at the time, but that they were inherited as distinct traits. In 1868, Mendel became abbot of the monastery and exchanged his scientific pursuits for his

1 Johann Gregor Mendel, "Versuche über Pflanzenhybriden." *Verhandlungen des naturforschenden Vereines in Brunn*, Bd. IV für das Jahr, 1865 *Abhandlungen* (1866):3-47. [for English translation, see <http://www.mendelweb.org/Mendel.plain.html>]

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pastoral duties. He was not recognized for his extraordinary scientific contributions during his lifetime; in fact, it was not until 1900 that his work was rediscovered, reproduced, and revitalized by scientists on the brink of discovering the chromosomal basis of heredity.

Mendel's Crosses

Mendel's seminal work was accomplished using the garden pea, *Pisum sativum*, to study inheritance. This species naturally self-fertilizes, meaning that pollen encounters ova within the same flower. Because every pea plant has both male reproductive organs and female reproductive organs, each plant produces both types of gametes required for reproduction—both pollen and ova. In plants, just as in animals, reproductive organs are classified by the size of the gametes produced. The organs producing the smaller pollen are called male reproductive organs, while the organs producing the larger ova are called female reproductive organs.

In garden peas, the flower petals remain sealed tightly until pollination is completed to prevent the pollination of other plants. The result is highly inbred, or "true-breeding," pea plants. These are plants that always produce offspring that look like the parent. By experimenting with true-breeding pea plants, Mendel avo

Once these validations were complete, Mendel applied the pollen from a plant with violet flowers to the stigma of a plant with white flowers. After gathering and sowing the seeds that resulted from this cross, Mendel found that 100 percent of the F1 hybrid generation had violet flowers. Conventional wisdom at that time would have predicted the hybrid flowers to be pale violet or for hybrid plants to have equal numbers of white and violet flowers. In other words, the contrasting parental traits were expected to blend in the offspring. Instead, Mendel's results demonstrated that the white flower trait had completely disappeared in the F1 generation.

Importantly, Mendel did not stop his experimentation there. He allowed the F1 plants to self-fertilize and found that 705 plants in the F2 generation had violet flowers and 224 had white flowers. This was a ratio of 3.15 violet flowers to one white flower, or approximately 3:1. Mendel performed an additional experiment to ascertain differences in inheritance of traits carried in the pollen versus the ovum. When Mendel transferred pollen from a plant with violet flowers to fertilize the ova of a plant with white flowers and vice versa, he obtained approximately the same ratio irrespective of which gamete contributed which trait. This is called a reciprocal cross—a paired cross in which the respective traits of the male and female in one cross become the respective traits of the female and male in the other cross. For the other six characteristics that Mendel examined, the F1 and F2 generations behaved in the same way that they behaved for flower color. One of the two traits would disappear completely from the F1 generation, only to reappear in the F2 generation at a ratio of roughly 3:1 (Figure 8.4).

FIGURE 8.4 Mendel identified seven pea plant characteristics.

Upon compiling his results for many thousands of plants, Mendel concluded that the characteristics could be divided into expressed and latent traits. He called these dominant and recessive traits, respectively. Dominant traits are those that are inherited unchanged in a hybridization. Recessive traits become latent, or disappear in the offspring of a hybridization. The recessive trait does, however, reappear in the progeny of the hybrid offspring. An example of a dominant trait is the violet-colored flower trait. For this same characteristic (flower color), white-colored flowers are a recessive trait. The fact that the recessive trait reappeared in the F2 generation meant that the traits remained separate (and were not blended) in the plants of the F1 generation. Mendel proposed that this was because the plants possessed two copies of the trait for the flower-color characteristic, and that each parent transmitted one of their two copies to their offspring, where they came together. Moreover, the physical observation of a dominant trait could mean that the genetic composition of the organism included two dominant versions of the character.

the top (for one parent) and side (for the other parent) of a grid. The combinations of egg and sperm gametes are then made in the boxes in the table on the basis of which alleles are combining. Each box then represents the diploid genotype of a zygote, or fertilized egg. Because each possibility is equally likely, genotypic ratios can be determined from a Punnett square. If the pattern of inheritance (dominant and recessive) is known, the phenotypic ratios can be inferred as well. For a monohybrid cross of two true-breeding parents, each parent contributes one type of allele. In this case, only one genotype is possible in the F1 offspring. All offspring are Yy and have yellow seeds.

When the F1 offspring are crossed with each other, each has an equal probability of contributing either a Y or a y to the F2 offspring. The result is a 1 in 4 (25 percent) probability of both parents contributing a Y, resulting in an offspring with a yellow phenotype; a 25 percent probability of parent A contributing a Y and parent B a y, resulting in an offspring with a yellow phenotype; a 25 percent probability of parent A contributing a y and parent B a Y, also resulting in a yellow phenotype; and a (25 percent) probability of both parents contributing a y, resulting in a green

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phenotype. When counting all four possible outcomes, there is a 3 in 4 probability of offspring having the yellow phenotype and a 1 in 4 probability of offspring having the green phenotype. This explains why the results of Mendel's F2 generation occurred in a 3:1 phenotypic ratio. Using large numbers of crosses, Mendel was able to calculate probabilities, found that they fit the model of inheritance, and use these to predict the outcomes of other crosses.

Law of Segregation

Observing that true-breeding pea plants with contrasting traits gave rise to F1 generations that all expressed the dominant trait and F2 generations that expressed the dominant and recessive traits in a 3:1 ratio, Mendel proposed the law of segregation. This law states that paired unit factors (genes) must segregate equally into gametes such that offspring have an equal likelihood of inheriting either factor. For the F2 generation of a monohybrid cross, the following three possible combinations of genotypes result: homozygous dominant, heterozygous, or homozygous recessive. Because heterozygotes could arise from two different pathways (receiving one dominant and one recessive allele from either parent), and because heterozygotes and homozygous dominant individuals are phenotypically identical, the law supports Mendel's observed 3:1 phenotypic ratio. The equal segregation of alleles is the reason we can apply the Punnett square to accurately predict the offspring of parents with known genotypes. The physical basis of Mendel's law of segregation is the first division of meiosis in which the homologous chromosomes with their different versions of each gene are

the heterozygote exhibits a

the F1 generation are described as recessive. When the phenotype that is intermediate between the F1 plants in Mendel's experiment were self-crossed, homozygous phenotypes. Codominance describes the the F2 offspring exhibited the dominant trait or the simultaneous expression of both of the alleles in the recessive trait in a 3:1 ratio, confirming that the heterozygote. Although diploid organisms can only recessive trait had been transmitted faithfully from the have two alleles for any given gene, it is common for original P parent. Reciprocal crosses generated more than two alleles for a gene to exist in a identical F1 and F2 offspring ratios. By examining population. In humans, as in many animals and some sample sizes, Mendel showed that traits were inherited plants, females have two X chromosomes and males as independent events. have one X and one Y chromosome. Genes that are

present on the X but not the Y chromosome are said to

8.2 Laws of Inheritance be X-linked, such that males only inherit one allele for

When true-breeding, or homozygous, individuals that the gene, and females inherit two.

differ for a certain trait are crossed, all of the offspring According to Mendel's law of independent assortment, will be heterozygous for that trait. If the traits are genes sort independently of each other into gametes inherited as dominant and recessive, the F1 offspring during meiosis. This occurs because chromosomes, on will all exhibit the same phenotype as the parent which the genes reside, assort independently during homozygous for the dominant trait. If these meiosis and crossovers cause most genes on the same heterozygous offspring are self-crossed, the resulting chromosomes to also behave independently. When F2 offspring will be equally likely to inherit gametes genes are located in close proximity on the same carrying the dominant or recessive trait, giving rise to chromosome, their alleles tend to be inherited offspring of which one quarter are homozygous together. This results in offspring ratios that violate dominant, half are heterozygous, and one quarter are Mendel's law of independent assortment. However, homozygous recessive. Because homozygous dominant recombination serves to exchange genetic material on and heterozygous individuals are phenotypically homologous chromosomes such that maternal and identical, the observed traits in the F2 offspring will paternal alleles may be recombined on the same exhibit a ratio of three dominant to one recessive. chromosome. This is why alleles on a given Mendel postulated that genes (characteristics) are chromosome are not always inherited together. inherited as pairs of alleles (traits) that behave in a Recombination is a random event occurring anywhere dominant and recessive pattern. Alleles segregate into on a chromosome. Therefore, genes that are far apart gametes such that each gamete is equally likely to on the same chromosome are likely to still assort receive ei

ven the different numbers of ?letters? in the mRNA and protein ?alphabets,? combinations of nucleotides corresponded to single amino acids. Using a three-nucleotide code means that there are a total of 64 ($4 \times 4 \times 4$) possible combinations; therefore, a given amino acid is encoded by more than one nucleotide triplet (Figure 9.20).

FIGURE 9.20 This figure shows the genetic code for translating each nucleotide triplet, or codon, in mRNA into an amino acid or a termination signal in a nascent protein. (credit: modification of work by NIH)

Three of the 64 codons terminate protein synthesis and release the polypeptide from the translation machinery. These triplets are called stop codons. Another codon, AUG, also has a special function. In addition to specifying the amino acid methionine, it also serves as the start codon to initiate translation. The reading frame for translation is set by the AUG start codon near the 5' end of the mRNA. The genetic code is universal. With a few exceptions, virtually all species use the same genetic code for protein synthesis, which is powerful evidence that all life on Earth shares a common origin.

The Mechanism of Protein Synthesis

Just as with mRNA synthesis, protein synthesis can be divided into three phases: initiation, elongation, and termination. The process of translation is similar in prokaryotes and eukaryotes. Here we will explore how translation occurs in *E. coli*, a representative prokaryote, and specify any differences between prokaryotic and eukaryotic translation.

Protein synthesis begins with the formation of an initiation complex. In *E. coli*, this complex involves the small ribosome subunit, the mRNA template, three initiation factors, and a special initiator tRNA. The initiator tRNA interacts with the AUG start codon, and links to a special form of the amino acid methionine that is typically removed from the polypeptide after translation is complete.

In prokaryotes and eukaryotes, the basics of polypeptide elongation are the same, so we will review elongation from the perspective of *E. coli*. The large ribosomal subunit of *E. coli* consists of three compartments: the A site binds incoming charged tRNAs (tRNAs with their attached specific amino acids). The P site binds charged tRNAs carrying amino acids that have formed bonds with the growing polypeptide chain but have not yet dissociated from their corresponding tRNA. The E site releases dissociated tRNAs so they can be recharged with free amino acids. The

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ribosome shifts one codon at a time, catalyzing each process that occurs in the three sites. With each step, a charged tRNA enters the complex, the polypeptide becomes one amino acid longer, and an uncharged tRNA departs.

The energy for each bond between amino acids is derived from GTP, a molecule similar to ATP (Figure 9.21). Amazingly, the *E. coli* translation apparatus takes only 0.05 seconds to add each amino acid, meaning that a 200-amino acid polypeptide could be

ecial conditions.

The control of gene expression is extremely complex. Malfunctions in this process are detrimental to the cell and can lead to the development of many diseases, including cancer.

Prokaryotic versus Eukaryotic Gene Expression

To understand how gene expression is regulated, we must first understand how a gene becomes a functional protein in a cell. The process occurs in both prokaryotic and eukaryotic cells, just in slightly different fashions.

Because prokaryotic organisms lack a cell nucleus, the processes of transcription and translation occur almost simultaneously. When the protein is no longer needed, transcription stops. As a result, the primary method to control what type and how much protein is expressed in a prokaryotic cell is through the regulation of DNA transcription into RNA. All the subsequent steps happen automatically. When more protein is required, more transcription occurs. Therefore, in prokaryotic cells, the control of gene expression is almost entirely at the transcriptional level.

The first example of such control was discovered using *E. coli* in the 1950s and 1960s by French researchers and is called the lac operon. The lac operon is a stretch of DNA with three adjacent genes that code for proteins that participate in the absorption and metabolism of lactose, a food source for *E. coli*. When lactose is not present in the bacterium's environment, the lac genes are transcribed in small amounts. When lactose is present, the genes are transcribed and the bacterium is able to use the lactose as a food source. The operon also contains a promoter sequence to which the RNA polymerase binds to begin transcription; between the promoter and the three genes is a region called the operator. When there is no lactose present, a protein known as a repressor binds to the operator and prevents RNA polymerase from binding to the promoter, except in rare cases. Thus very little of the protein products of the three genes is made. When lactose is present, an end product of lactose metabolism binds to the repressor protein and prevents it from binding to the operator. This allows RNA polymerase to bind to the promoter and freely transcribe the three genes, allowing the organism to metabolize the lactose.

Eukaryotic cells, in contrast, have intracellular organelles and are much more complex. Recall that in eukaryotic cells, the DNA is contained inside the cell's nucleus and it is transcribed into mRNA there. The newly synthesized mRNA is then transported out of the nucleus into the cytoplasm, where ribosomes translate the mRNA into protein.

The processes of transcription and translation are physically separated by the nuclear membrane; transcription occurs only within the nucleus, and translation only occurs outside the nucleus in the cytoplasm. The regulation of

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gene expression can occur at all stages of the process (Figure 9.22). Regulation may occur when the DNA is uncoiled and loosened from nucl

re likely to lead to their failure. Such "mistakes" would more than likely produce a nonfunctional protein. Indeed, the cause of many genetic diseases is alternative splicing rather than mutations in a sequence. However, alternative splicing would create a protein variant without the loss of the original protein, opening up possibilities for adaptation of the new variant to new functions. Gene duplication has played an important role in the evolution of new functions in a similar way by providing genes that may evolve without eliminating the original functional protein.

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Key Terms

alternative RNA splicing a post-transcriptional gene nontemplate strand the strand of DNA that is not

regulation mechanism in eukaryotes in which used to transcribe mRNA; this strand is identical to multiple protein products are produced by a single the mRNA except that T nucleotides in the DNA are gene through alternative splicing combinations of replaced by U nucleotides in the mRNA the RNA transcript nucleotide excision repair a form of DNA repair in

codon three consecutive nucleotides in mRNA that which the DNA molecule is unwound and separated specify the addition of a specific amino acid or the in the region of the nucleotide damage, the release of a polypeptide chain during translation damaged nucleotides are removed and replaced

deoxyribose a five-carbon sugar molecule with a with new nucleotides using the complementary hydrogen atom rather than a hydroxyl group in the strand, and the DNA strand is resealed and allowed 2' position; the sugar component of DNA to rejoin its complement nucleotides Okazaki fragments the DNA fragments that are

DNA ligase the enzyme that catalyzes the joining of synthesized in short stretches on the lagging strand DNA fragments together phosphate group a molecular group consisting of a

DNA polymerase an enzyme that synthesizes a new central phosphorus atom bound to four oxygen strand of DNA complementary to a template strand atoms

double helix the molecular shape of DNA in which post-transcriptional control of gene expression after two strands of nucleotides wind around each other the RNA molecule has been created but before it is in a spiral shape translated into protein

epigenetic describing non-genetic regulatory factors, post-translational control of gene expression after a such as changes in modifications to histone proteins protein has been created and DNA that control accessibility to genes in primer a short stretch of RNA nucleotides that is chromosomes required to initiate replication and allow DNA

exon a sequence present in protein-coding mRNA polymerase to bind and begin replication after completion of pre-mRNA splicing promoter a sequence on DNA to which RNA

gene expression processes that control whether a polymerase and associated factors bind and initiate gene is expressed transcription

genetic code the amino acids that correspond to replication fork the Y-s

haped structure formed

three-nucleotide codons of mRNA during the initiation of replication

helicase an enzyme that helps to open up the DNA RNA polymerase an enzyme that synthesizes an RNA helix during DNA replication by breaking the strand from a DNA template strand

hydrogen bonds rRNA ribosomal RNA; molecules of RNA that combine

intron non-protein-coding intervening sequences to form part of the ribosome

that are spliced from mRNA during processing semiconservative replication the method used to

lagging strand during replication of the 3' to 5' replicate DNA in which the double-stranded strand, the strand that is replicated in short molecule is separated and each strand acts as a fragments and away from the replication fork template for a new strand to be synthesized, so the

leading strand the strand that is synthesized resulting DNA molecules are composed of one new continuously in the 5' to 3' direction that is strand of nucleotides and one old strand of synthesized in the direction of the replication fork nucleotides

mismatch repair a form of DNA repair in which non-splicing the process of removing introns and complementary nucleotides are recognized, reconnecting exons in a pre-mRNA excised, and replaced with correct nucleotides start codon the AUG (or, rarely GUG) on an mRNA

mRNA messenger RNA; a form of RNA that carries from which translation begins; always specifies the nucleotide sequence code for a protein methionine

sequence that is translated into a polypeptide stop codon one of the three mRNA codons that sequence specifies termination of translation

mutation a permanent variation in the nucleotide telomerase an enzyme that contains a catalytic part sequence of a genome and an inbuilt RNA template; it functions to maintain

nitrogenous base a nitrogen-containing molecule telomeres at chromosome ends

that acts as a base; often referring to one of the telomere the DNA at the end of linear chromosomes purine or pyrimidine components of nucleic acids template strand the strand of DNA that specifies the

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complementary mRNA molecule specific three-nucleotide anticodon sequence to

transcription bubble the region of locally unwound pair with the mRNA codon and also binds to a

DNA that allows for transcription of mRNA specific amino acid

tRNA transfer RNA; an RNA molecule that contains a

Chapter Summary

9.1 The Structure of DNA a single origin of replication. The DNA is opened with

enzymes, resulting in the formation of the replication

The model of the double-helix structure of DNA was fork. Primase synthesizes an RNA primer to initiate proposed by Watson and Crick. The DNA molecule is a synthesis by DNA polymerase, which can add polymer of nucleotides. Each nucleotide is composed nucleotides in only one direction. One strand is of a nitrogenous base, a five-carbon sugar synthesized continuously in the direction of the (deoxyribose), and a phosphate group. There are four replication fork; this is called

the leading strand. The

nitrogenous bases in DNA, two purines (adenine and other strand is synthesized in a direction away from the guanine) and two pyrimidines (cytosine and thymine). replication fork, in short stretches of DNA known as A DNA molecule is composed of two strands. Each Okazaki fragments. This strand is known as the lagging strand is composed of nucleotides bonded together strand. Once replication is completed, the RNA primers covalently between the phosphate group of one and are replaced by DNA nucleotides and the DNA is sealed the deoxyribose sugar of the next. From this backbone with DNA ligase.

extend the bases. The bases of one strand bond to the

bases of the second strand with hydrogen bonds. The ends of eukaryotic chromosomes pose a problem, Adenine always bonds with thymine, and cytosine as polymerase is unable to extend them without a always bonds with guanine. The bonding causes the primer. Telomerase, an enzyme with an inbuilt RNA two strands to spiral around each other in a shape template, extends the ends by copying the RNA called a double helix. Ribonucleic acid (RNA) is a template and extending one end of the chromosome. second nucleic acid found in cells. RNA is a single- DNA polymerase can then extend the DNA using the stranded polymer of nucleotides. It also differs from primer. In this way, the ends of the chromosomes are DNA in that it contains the sugar ribose, rather than protected. Cells have mechanisms for repairing DNA deoxyribose, and the nucleotide uracil rather than when it becomes damaged or errors are made in thymine. Various RNA molecules function in the replication. These mechanisms include mismatch process of forming proteins from the genetic code in repair to replace nucleotides that are paired with a DNA. non-complementary base and nucleotide excision

repair, which removes bases that are damaged such as

Prokaryotes contain a single, double-stranded circular thymine dimers.

chromosome. Eukaryotes contain double-stranded

linear DNA molecules packaged into chromosomes. 9.3 Transcription

The DNA helix is wrapped around proteins to form

In prokaryotes, mRNA synthesis is initiated at a

nucleosomes. The protein coils are further coiled, and

promoter sequence on the DNA template. Elongation

during mitosis and meiosis, the chromosomes become

synthesizes new mRNA. Termination liberates the

even more greatly coiled to facilitate their movement.

mRNA and occurs by mechanisms that stall the RNA

Chromosomes have two distinct regions which can be

polymerase and cause it to fall off the DNA template.

distinguished by staining, reflecting different degrees

Newly transcribed eukaryotic mRNAs are modified with

of packaging and determined by whether the DNA in a

a cap and a poly-A tail. These structures protect the

region is being expressed (euchromatin) or not

mature mRNA from degradation and help export it from

(heterochromatin).

the nucleus. Eukaryotic mRNAs also undergo splicing,

9.2 DNA Replication in

which introns are removed and exons are

reconnected with single-nucleotide accuracy. Only

DNA replicates by a semi-conservative method in which each of the two parental DNA strands act as a template for new DNA to be synthesized. After

replication, each DNA has one parental or "old" strand, and one daughter or "new" strand.

The central dogma describes the flow of genetic

information. Replication in eukaryotes starts at multiple origins of information in the cell from genes to mRNA to proteins. Replication, while replication in prokaryotes starts from a single origin. Genes are used to make mRNA by the process of

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transcription; mRNA is used to synthesize proteins by dissociates the components and frees the new protein. the process of translation. The genetic code is the

correspondence between the three-nucleotide mRNA codon and an amino acid. The genetic code is

While all somatic cells within an organism contain the "translated" by the tRNA molecules, which associate a same DNA, not all cells within that organism express specific codon with a specific amino acid. The genetic code is degenerate because 64 triplet codons in mRNA encode in every cell, but not specify only 20 amino acids and three stop codons. necessarily all at the same time. Proteins are This means that more than one codon corresponds to expressed only when they are needed. Eukaryotic an amino acid. Almost every species on the planet uses organisms express a subset of the DNA that is encoded

the same genetic code. in any given cell. In each cell type, the type and amount

The players in translation include the mRNA template, of protein is regulated by controlling gene expression. ribosomes, tRNAs, and various enzymatic factors. The To express a protein, the DNA is first transcribed into small ribosomal subunit binds to the mRNA template. RNA, which is then translated into proteins. In Translation begins at the initiating AUG on the mRNA. prokaryotic cells, these processes occur almost The formation of bonds occurs between sequential simultaneously. In eukaryotic cells, transcription amino acids specified by the mRNA template according occurs in the nucleus and is separate from the to the genetic code. The ribosome accepts charged translation that occurs in the cytoplasm. Gene tRNAs, and as it steps along the mRNA, it catalyzes expression in prokaryotes is regulated only at the bonding between the new amino acid and the end of transcriptional level, whereas in eukaryotic cells, gene the growing polypeptide. The entire mRNA is translated expression is regulated at the epigenetic, in three-nucleotide "steps" of the ribosome. When a transcriptional, post-transcriptional, translational, and stop codon is encountered, a release factor binds and post-translational levels.

ive cloning is a method used to make a clone or an identical copy of an entire multicellular organism. Most multicellular organisms undergo reproduction by sexual means, which involves the contribution of DNA from two individuals (parents), making it impossible to generate an identical copy or a clone of either parent. Recent Access for free at openstax.org

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advances in biotechnology have made it possible to reproductively clone mammals in the laboratory.

Natural sexual reproduction involves the union, during fertilization, of a sperm and an egg. Each of these gametes is haploid, meaning they contain one set of chromosomes in their nuclei. The resulting cell, or zygote, is then diploid and contains two sets of chromosomes. This cell divides mitotically to produce a multicellular organism. However, the union of just any two cells cannot produce a viable zygote; there are components in the cytoplasm of the egg cell that are essential for the early development of the embryo during its first few cell divisions. Without these provisions, there would be no subsequent development. Therefore, to produce a new individual, both a diploid genetic complement and an egg cytoplasm are required. The approach to producing an artificially cloned individual is to take the egg cell of one individual and to remove the haploid nucleus. Then a diploid nucleus from a body cell of a second individual, the donor, is put into the egg cell. The egg is then stimulated to divide so that development proceeds. This sounds simple, but in fact it takes many attempts before each of the steps is completed successfully.

The first cloned agricultural animal was Dolly, a sheep who was born in 1996. The success rate of reproductive cloning at the time was very low. Dolly lived for six years and died of a lung tumor (Figure 10.7). There was speculation that because the cell DNA that gave rise to Dolly came from an older individual, the age of the DNA may have affected her life expectancy. Since Dolly, several species of animals (such as horses, bulls, and goats) have been successfully cloned.

There have been attempts at producing cloned human embryos as sources of embryonic stem cells. In the procedure, the DNA from an adult human is introduced into a human egg cell, which is then stimulated to divide. The technology is similar to the technology that was used to produce Dolly, but the embryo is never implanted into a surrogate carrier. The cells produced are called embryonic stem cells because they have the capacity to develop into many different kinds of cells, such as muscle or nerve cells. The stem cells could be used to research and ultimately provide therapeutic applications, such as replacing damaged tissues. The benefit of cloning in this instance is that the cells used to regenerate new tissues would be a perfect match to the donor of the original DNA. For example, a leukemia patient would not require a sibling with a tissue mat

projects, particularly the Human Genome

Project, has expanded the applicability of DNA sequence information. Genomics is now being used in a wide variety

of fields, such as metagenomics, pharmacogenomics, and mitochondrial genomics. The most commonly known application of genomics is to understand and find cures for diseases.

Predicting Disease Risk at the Individual Level

Predicting the risk of disease involves screening and identifying currently healthy individuals by genome analysis at

the individual level. Intervention with lifestyle changes and drugs can be recommended before disease onset.

However, this approach is most applicable when the problem arises from a single gene mutation. Such defects only

account for about 5 percent of diseases found in developed countries. Most of the common diseases, such as heart

disease, are multifactorial or polygenic, which refers to a phenotypic characteristic that is determined by two or more genes, and also environmental factors such as diet. In April 2010, scientists at Stanford University published

the genome analysis of a healthy individual (Stephen Quake, a scientist at Stanford University, who had his genome

sequenced); the analysis predicted his propensity to acquire various diseases. A risk assessment was done to analyze Quake's percentage of risk for 55 different medical conditions. A rare genetic mutation was found that showed him to be at risk for sudden heart attack. He was also predicted to have a 23 percent risk of developing prostate cancer and a 1.4 percent risk of developing Alzheimer's disease. The scientists used databases and several

publications to analyze the genomic data. Even though genomic sequencing is becoming more affordable and analytical tools are becoming more reliable, ethical issues surrounding genomic analysis at a population level remain to be addressed. For example, could such data be legitimately used to charge more or less for insurance or

to affect credit ratings?

Gene Editing

For thousands of years, humans have engaged in some level of control over genes and heredity regarding the plants

and animals we rely on. The technology now exists to exert that control more directly by precisely altering the DNA

of organisms. The technique is usually referred to as CRISPR, for the portions of DNA it targets: "Clustered Regularly

Interspaced Short Palindromic Repeats." In essence, DNA contains repetitive sequences with "spacers" between

them. CRISPR-associated nucleases (known as "Cas") are enzymes that can identify, attach to, and cut the strand at

precise locations. In 2012, Jennifer Doudna and Emmanuelle Charpentier developed a method to combine the Cas

nuclease with a synthetically produced "guide RNA" that leads the nuclease to selected locations on the DNA strand. The discovery revolutionized gene editing. Researchers around the world have used CRISPR to manipulate

the actual DNA of plants, animals, laboratory cell lines, and (in trials) human patients. Doudna and Charpe

agnosed individuals, and the particular genes identified can be targets for research into the molecular pathway of the disease and potential therapies. An offshoot of the discovery of gene associations with disease has been the formation of companies that provide so-called "personal genomics" that will identify risk levels for various diseases based on an individual's SNP complement. The science behind these services is controversial.

Because GWAS looks for associations between genes and disease, these studies provide data for other research into causes, rather than answering specific questions themselves. An association between a gene difference and a disease does not necessarily mean there is a cause-and-effect relationship. However, some studies have provided useful information about the genetic causes of diseases. For example, three different studies in 2005 identified a gene for a protein involved in regulating inflammation in the body that is associated with a disease-causing blindness called age-related macular degeneration. This opened up new possibilities for research into the cause of this disease. A large number of genes have been identified to be associated with Crohn's disease using GWAS, and some of these have suggested new hypothetical mechanisms for the cause of the disease.

Pharmacogenomics

Pharmacogenomics involves evaluating the effectiveness and safety of drugs on the basis of information from an individual's genomic sequence. Personal genome sequence information can be used to prescribe medications that will be most effective and least toxic on the basis of the individual patient's genotype. Studying changes in gene expression could provide information about the gene transcription profile in the presence of the drug, which can be used as an early indicator of the potential for toxic effects. For example, genes involved in cellular growth and controlled cell death, when disturbed, could lead to the growth of cancerous cells. Genome-wide studies can also help to find new genes involved in drug toxicity. The gene signatures may not be completely accurate, but can be tested further before pathologic symptoms arise.

Metagenomics

Traditionally, microbiology has been taught with the view that microorganisms are best studied under pure culture conditions, which involves isolating a single type of cell and culturing it in the laboratory. Because microorganisms can go through several generations in a matter of hours, their gene expression profiles adapt to the new laboratory environment very quickly. On the other hand, many species resist being cultured in isolation. Most microorganisms do not live as isolated entities, but in microbial communities known as biofilms. For all of these reasons, pure culture is not always the best way to study microorganisms. Metagenomics is the study of the collective genomes of multiple species that grow and interact in an environmental niche. Metagenomics can be used to identify new species more rap

polymerase chain reaction (PCR) a technique used through a semisolid gel in response to an electric current to make multiple copies of DNA
current protein signature a set of over- or under-expressed

gene therapy the technique used to cure heritable diseases by replacing mutant genes with good diseased tissue
genes proteomics study of the function of proteomes

genetic engineering alteration of the genetic makeup recombinant DNA a combination of DNA fragments of an organism using the molecular methods of generated by molecular cloning that does not exist in nature

genetic map an outline of genes and their location on a chromosome that is based on recombination frequencies between markers recombinant protein a protein that is expressed from recombinant DNA molecules

genetic testing identifying gene variants in an individual that may lead to a genetic disease in that specific nucleotide sequence in DNA and cuts the individual DNA double strand at that recognition site, often

genetically modified organism (GMO) an organism with a staggered cut leaving short single strands or whose genome has been artificially changed ?sticky? ends

genomics the study of entire genomes, including the reverse genetics a form of genetic analysis that complete set of genes, their nucleotide sequence manipulates DNA to disrupt or affect the product of and organization, and their interactions within a gene to analyze the gene's function
species and with other species transgenic describing an organism that receives DNA

metagenomics the study of the collective genomes of from a different species
multiple species that grow and interact in an whole genome sequencing a process that environmental niche determines the nucleotide sequence of an entire

model organism a species that is studied and used as genome

Chapter Summary

10.1 Cloning and Genetic Engineering modifying them to achieve desirable traits.

Nucleic acids can be isolated from cells for the Cloning may involve cloning small DNA fragments purposes of further analysis by breaking open the cells (molecular cloning), or cloning entire organisms and enzymatically destroying all other major (reproductive cloning). In molecular cloning with macromolecules. Fragmented or whole chromosomes bacteria, a desired DNA fragment is inserted into a can be separated on the basis of size by gel bacterial plasmid using restriction enzymes and the electrophoresis. Short stretches of DNA can be plasmid is taken up by a bacterium, which will then amplified by PCR. DNA can be cut (and subsequently express the foreign DNA. Using other techniques, re-spliced together) using restriction enzymes. The foreign genes can be inserted into eukaryotic molecular and cellular techniques of biotechnology organisms. In each case, the organisms are called allow researchers to genetic

ally engineer organisms, transgenic organisms. In reproductive cloning, a donor

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nucleus is put into an enucleated egg cell, which is provide an outline for the location of genes within a then stimulated to divide and develop into an genome, and they estimate the distance between organism. genes and genetic markers on the basis of the

recombination frequency during meiosis. Physical
In reverse genetics methods, a gene is mutated or

maps provide detailed information about the physical
removed in some way to identify its effect on the

distance between the genes. The most detailed
phenotype of the whole organism as a way to

information is available through sequence mapping.
determine its function.

Information from all mapping and sequencing sources

10.2 Biotechnology in Medicine and is combined to study an entire genome.

Agriculture Whole genome sequencing is the latest available

Genetic testing is performed to identify disease- resource to treat genetic diseases. Some doctors are causing genes, and can be used to benefit affected using whole genome sequencing to save lives.

individuals and their relatives who have not developed Genomics has many industrial applications, including disease symptoms yet. Gene therapy?by which biofuel development, agriculture, pharmaceuticals, and functioning genes are incorporated into the genomes of pollution control.

individuals with a non-functioning mutant gene?has Imagination is the only barrier to the applicability of the potential to cure heritable diseases. Transgenic genomics. Genomics is being applied to most fields of organisms possess DNA from a different species, biology; it can be used for personalized medicine, usually generated by molecular cloning techniques. prediction of disease risks at an individual level, the Vaccines, antibiotics, and hormones are examples of study of drug interactions before the conduction of products obtained by recombinant DNA technology. clinical trials, and the study of microorganisms in the Transgenic animals have been created for experimental environment as opposed to the laboratory. It is also purposes and some are used to produce some human being applied to the generation of new biofuels, proteins. genealogical assessment using mitochondria,

Genes are inserted into plants, using plasmids in the advances in forensic science, and improvements in bacterium *Agrobacterium tumefaciens*, which infects agriculture.

plants. Transgenic plants have been created to improve Proteomics is the study of the entire set of proteins the characteristics of crop plants?for example, by expressed by a given type of cell under certain giving them insect resistance by inserting a gene for a environmental conditions. In a multicellular organism, bacterial toxin. different cell types will have different proteomes, and

10.3 Genomics and Proteomics these will vary with changes in the environment. Unlike

a genome, a proteome is dynamic and under constant

s that is still occurring. (credit

?wolf?: modification of work by Gary Kramer, USFWS; credit ?coral?: modification of work by William Harrigan, NOAA;

credit ?river?: modification of work by Vojtěch Dostál; credit ?protozoa?: modification of work by Sharon Franklin,

Stephen Ausmus, USDA ARS; credit ?fish? modification of work by Christian Mehlführer; credit ?mushroom?, ?bee?:

modification of work by Cory Zanker; credit ?tree?: modification of work by Joseph Kranak)

CHAPTER OUTLINE

11.1 Discovering How Populations Change

11.2 Mechanisms of Evolution

11.3 Evidence of Evolution

11.4 Speciation

11.5 Common Misconceptions about Evolution

INTRODUCTION All species of living organisms—from the bacteria on our skin, to the trees in our yards, to the birds outside—evolved at some point from a different species. Although it may seem that living things today stay much the same from generation to generation, that is not the case: evolution is ongoing. Evolution is the process through which the characteristics of species change and through which new species arise.

248 11 ? Evolution and Its Processes

The theory of evolution is the unifying theory of biology, meaning it is the framework within which biologists ask questions about the living world. Its power is that it provides direction for predictions about living things that are borne out in experiment after experiment. The Ukrainian-born American geneticist Theodosius Dobzhansky famously wrote that “nothing makes sense in biology except in the light of evolution.”¹ He meant that the principle that all life has evolved and diversified from a common ancestor is the foundation from which we understand all other questions in biology. This chapter will explain some of the mechanisms for evolutionary change and the kinds of questions that biologists can and have answered using evolutionary theory.

11.1 Discovering How Populations Change

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Explain how Darwin’s theory of evolution differed from the current view at the time
- ? Describe how the present-day theory of evolution was developed
- ? Describe how population genetics is used to study the evolution of populations

The theory of evolution by natural selection describes a mechanism for species change over time. That species change had been suggested and debated well before Darwin. The view that species were static and unchanging was grounded in the writings of Plato, yet there were also ancient Greeks that expressed evolutionary ideas.

In the eighteenth century, ideas about the evolution of animals were reintroduced by the naturalist Georges-Louis Leclerc, Comte de Buffon and even by Charles Darwin’s grandfather, Erasmus Darwin. During this time, it was also accepted that there were extinct species. At the same time, James Hutton, the Scottish naturalist, proposed that geological change occurred gradually by the accumulation of small changes from processes (over long periods of time) jus

t

like those happening today. This contrasted with the predominant view that the geology of the planet was a consequence of catastrophic events occurring during a relatively brief past. Hutton's view was later popularized by the geologist Charles Lyell in the nineteenth century. Lyell became a friend to Darwin and his ideas were very influential on Darwin's thinking. Lyell argued that the greater age of Earth gave more time for gradual change in species, and the process provided an analogy for gradual change in species.

In the early nineteenth century, Jean-Baptiste Lamarck published a book that detailed a mechanism for evolutionary change that is now referred to as inheritance of acquired characteristics. In Lamarck's theory, modifications in an individual caused by its environment, or the use or disuse of a structure during its lifetime, could be inherited by its offspring and, thus, bring about change in a species. While this mechanism for evolutionary change as described by Lamarck was discredited, Lamarck's ideas were an important influence on evolutionary thought. The inscription on the statue of Lamarck that stands at the gates of the Jardin des Plantes in Paris describes him as the "founder of the doctrine of evolution."

1 Theodosius Dobzhansky. "Biology, Molecular and Organismic." *American Zoologist* 4, no. 4 (1964): 449.

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Charles Darwin and Natural Selection

The actual mechanism for evolution was independently conceived of and described by two naturalists, Charles Darwin and Alfred Russell Wallace, in the mid-nineteenth century. Importantly, each spent time exploring the natural world on expeditions to the tropics. From 1831 to 1836, Darwin traveled around the world on H.M.S. Beagle,

visiting South America, Australia, and the southern tip of Africa. Wallace traveled to Brazil to collect insects in the

Amazon rainforest from 1848 to 1852 and to the Malay Archipelago from 1854 to 1862. Darwin's journey, like Wallace's later journeys in the Malay Archipelago, included stops at several island chains, the last being the Galápagos Islands (west of Ecuador). On these islands, Darwin observed species of organisms on different islands

that were clearly similar, yet had distinct differences. For example, the ground finches inhabiting the Galápagos Islands comprised several species that each had a unique beak shape (Figure 11.2). He observed both that these

finches closely resembled another finch species on the mainland of South America and that the group of species in

the Galápagos formed a graded series of beak sizes and shapes, with very small differences between the most similar. Darwin imagined that the island species might be all species modified from one original mainland species. In

1860, he wrote, "Seeing this gradation and diversity of structure in one small, intimately related group of birds, one

might really fancy that from an original paucity of birds i

on may affect the phenotype of the organism in a way that gives it reduced fitness?lower likelihood of survival, resulting in fewer offspring.

? A mutation may produce a phenotype with a beneficial effect on fitness.

? Many mutations, called neutral mutations, will have no effect on fitness.

Mutations may also have a whole range of effect sizes on the fitness of the organism that expresses them in their

phenotype, from a small effect to a great effect. Sexual reproduction and crossing over in meiosis also lead to genetic diversity: when two parents reproduce, unique combinations of alleles assemble to produce unique genotypes and, thus, phenotypes in each of the offspring.

A heritable trait that aids the survival and reproduction of an organism in its present environment is called an adaptation. An adaptation is a ?match? of the organism to the environment. Adaptation to an environment comes

about when a change in the range of genetic variation occurs over time that increases or maintains the match of the

population with its environment. The variations in finch beaks shifted from generation to generation providing adaptation to food availability.

Whether or not a trait is favorable depends on the environment at the time. The same traits do not always have the

same relative benefit or disadvantage because environmental conditions can change. For example, finches with large bills were benefited in one climate, while small bills were a disadvantage; in a different climate, the relationship reversed.

Patterns of Evolution

The evolution of species has resulted in enormous variation in form and function. When two species evolve in different directions from a common point, it is called divergent evolution. Such divergent evolution can be seen in

the forms of the reproductive organs of flowering plants, which share the same basic anatomies; however, they can

look very different as a result of selection in different physical environments, and adaptation to different kinds of pollinators (Figure 11.5).

FIGURE 11.5 Flowering plants evolved from a common ancestor. Notice that the (a) dense blazing star and (b) purple coneflower vary in

appearance, yet both share a similar basic morphology. (credit a, b: modification of work by Cory Zanker)

In other cases, similar phenotypes evolve independently in distantly related species. For example, flight has evolved

in both bats and insects, and they both have structures we refer to as wings, which are adaptations to flight. The wings of bats and insects, however, evolved from very different original structures. When similar structures arise through evolution independently in different species it is called convergent evolution. The wings of bats and insects

are called analogous structures; they are similar in function and appearance, but do not share an origin in a common ancestor. Instead they evolved independently in the two lineages. The wings of a hummingbird and an

strich are homologous structures, meaning they share similarities (despite their differences resulting from evolutionary divergence). The wings of hummingbirds and ostriches did not evolve independently in the hummingbird lineage and the ostrich lineage—they descended from a common ancestor with wings.

The Modern Synthesis

The mechanisms of inheritance, genetics, were not understood at the time Darwin and Wallace were developing

their idea of natural selection. This lack of understanding was a stumbling block to comprehending many aspects of

evolution. In fact, blending inheritance was the predominant (and incorrect) genetic theory of the time, which made

it difficult to understand how natural selection might operate. Darwin and Wallace were unaware of the genetics work by Austrian monk Gregor Mendel, which was published in 1866, not long after publication of *On the Origin of*

Species. Mendel's work was rediscovered in the early twentieth century at which time geneticists were rapidly coming to an understanding of the basics of inheritance. Initially, the newly discovered particulate nature of genes

made it difficult for biologists to understand how gradual evolution could occur. But over the next few decades genetics and evolution were integrated in what became known as the modern synthesis—the coherent understanding of the relationship between natural selection and genetics that took shape by the 1940s and is generally accepted today. In sum, the modern synthesis describes how evolutionary pressures, such as natural selection, can affect a population's genetic makeup, and, in turn, how this can result in the gradual evolution of populations and species. The theory also connects this gradual change of a population over time, called microevolution, with the processes that gave rise to new species and higher taxonomic groups with widely divergent characters, called macroevolution.

Population Genetics

Recall that a gene for a particular character may have several variants, or alleles, that code for different traits associated with that character. For example, in the ABO blood type system in humans, three alleles determine the

particular blood-type carbohydrate on the surface of red blood cells. Each individual in a population of diploid organisms can only carry two alleles for a particular gene, but more than two may be present in the individuals that

make up the population. Mendel followed alleles as they were inherited from parent to offspring. In the early twentieth century, biologists began to study what happens to all the alleles in a population in a field of study known

as population genetics.

Until now, we have defined evolution as a change in the characteristics of a population of organisms, but behind that

phenotypic change is genetic change. In population genetic terms, evolution is defined as a change in the frequency

of an allele in a population. Using the ABO system as an example, the frequency of one of the alleles, I^A , is the number

ratio of copies of that allele divided by all the copies of the ABO gene in the population. For example, a study in Jordan found a frequency of I^A to be 26.1 percent.³ The I^B , I^O alleles made up 13.4 percent and 60.5 percent of the alleles respectively, and all of the frequencies add up to 100 percent. A change in this frequency over time would constitute evolution in the population.

There are several ways the allele frequencies of a population can change. One of those ways is natural selection. If a given allele confers a phenotype that allows an individual to have more offspring that survive and reproduce, that allele, by virtue of being inherited by those offspring, will be in greater frequency in the next generation. Since allele frequencies always add up to 100 percent, an increase in the frequency of one allele always means a corresponding decrease in one or more of the other alleles. Highly beneficial alleles may, over a very few generations, become "fixed" in this way, meaning that every individual of the population will carry the allele. Similarly, detrimental alleles may be swiftly eliminated from the gene pool, the sum of all the alleles in a population. Part of the study of population genetics is tracking how selective forces change the allele frequencies in a population over time, which can give scientists clues regarding the selective forces that may be operating on a given population. The studies of changes in wing coloration in the peppered moth from mottled white to dark in response to soot-covered tree trunks and then back to mottled white when factories stopped producing so much soot is a classic example of studying evolution in natural populations (Figure 11.6).

3 Sahar S. Hanania, Dhia S. Hassawi, and Nidal M. Irshaid, "Allele Frequency and Molecular Genotypes of ABO Blood Group System in a Jordanian Population," *Journal of Medical Sciences* 7 (2007): 51-58, doi:10.3923/jms.2007.51.58

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FIGURE 11.6 As the Industrial Revolution caused trees to darken from soot, darker colored peppered moths were better camouflaged than the lighter colored ones, which caused there to be more of the darker colored moths in the population.

In the early twentieth century, English mathematician Godfrey Hardy and German physician Wilhelm Weinberg independently provided an explanation for a somewhat counterintuitive concept. Hardy's original explanation was in response to a misunderstanding as to why a "dominant" allele, one that masks a recessive allele, should not increase in frequency in a population until it eliminated all the other alleles. The question resulted from a common confusion about what "dominant" means, but it forced Hardy, who was not even a biologist, to point out that if there are no factors that affect an allele frequency those frequencies will remain constant from one generation to the next. This principle is now known as the Hardy-Weinberg equilibrium

. The theory states that a population's allele and genotype frequencies are inherently stable unless some kind of evolutionary force is acting on the population, the population would carry the same alleles in the same proportions generation after generation. Individuals would, as a whole, look essentially the same and this would be unrelated to whether the alleles were dominant or recessive. The four most important evolutionary forces, which will disrupt the equilibrium, are natural selection, mutation, genetic drift, and migration into or out of a population. A fifth factor, nonrandom mating, will also disrupt the Hardy-Weinberg equilibrium but only by shifting genotype frequencies, not allele frequencies (unless the allele contributes toward increased or decreased reproductive potential). In nonrandom mating, individuals are more likely to mate with individuals with specific phenotypes rather than at random. Since nonrandom mating does not change allele frequencies, it does not cause evolution directly. Natural selection has been described. Mutation creates one allele out of another one and changes an allele's frequency by a small, but continuous amount each generation. Each allele is generated by a low, constant mutation rate that will slowly increase the allele's frequency in a population if no other forces act on the allele. If natural selection acts against the allele, it will be removed from the population at a low rate leading to a frequency that results from a balance between selection and mutation. This is one reason that genetic diseases remain in the human population at very low frequencies. If the allele is favored by selection, it will increase in frequency. Genetic drift causes random changes in allele frequencies when populations are small. Genetic drift can often be important in evolution, as discussed in the next section. Finally, if two populations of a species have different allele frequencies, migration of individuals between them will cause frequency changes in both populations. As it happens, there is no population in which one or more of these processes are not operating, so populations are always evolving, and the Hardy-Weinberg equilibrium will never be exactly observed. However, the Hardy-Weinberg principle gives scientists a baseline expectation for allele frequencies in a non-evolving population to which they can compare evolving populations and thereby infer what evolutionary forces might be at play. The population is evolving if the frequencies of alleles or genotypes deviate from the value expected from the Hardy-Weinberg principle.

Darwin identified a special case of natural selection that he called sexual selection. Sexual selection affects an individual's ability to mate and thus produce offspring, and it leads to the evolution of dramatic traits that often appear maladaptive in terms of survival but persist because they give their owners greater reproductive success. Sexual selection occurs in two ways: through

h intrasexual selection, as male?male or female?female competition for mates, and through intersexual selection, as female or male selection of mates. Male?male competition takes the form of conflicts between males, which are often ritualized, but may also pose significant threats to a male?s survival. Sometimes the competition is for territory, with females more likely to mate with males with higher quality territories. Female choice occurs when females choose a male based on a particular trait, such as feather colors, the performance of a mating dance, or the building of an elaborate structure. In some cases male?male competition and female choice combine in the mating process. In each of these cases, the traits selected for, such as fighting ability or feather color and length, become enhanced in the males. In general, it is thought that sexual selection can

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proceed to a point at which natural selection against a character?s further enhancement prevents its further evolution because it negatively impacts the male?s ability to survive. For example, colorful feathers or an elaborate display make the male more obvious to predators.

11.2 Mechanisms of Evolution

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe the four basic causes of evolution: natural selection, mutation, genetic drift, and gene flow
- ? Explain how each evolutionary force can influence the allele frequencies of a population

The Hardy-Weinberg equilibrium principle says that allele frequencies in a population will remain constant in the absence of the four factors that could change them. Those factors are natural selection, mutation, genetic drift, and migration (gene flow). In fact, we know they are probably always affecting populations.

Natural Selection

Natural selection has already been discussed. Alleles are expressed in a phenotype. Depending on the environmental conditions, the phenotype confers an advantage or disadvantage to the individual with the phenotype relative to the other phenotypes in the population. If it is an advantage, then that individual will likely have more offspring than individuals with the other phenotypes, and this will mean that the allele behind the phenotype will have greater representation in the next generation. If conditions remain the same, those offspring, which are carrying the same allele, will also benefit. Over time, the allele will increase in frequency in the population.

Mutation

Mutation is a source of new alleles in a population. Mutation is a change in the DNA sequence of the gene. A mutation can change one allele into another, but the net effect is a change in frequency. The change in frequency resulting from mutation is small, so its effect on evolution is small unless it interacts with one of the other factors, such as selection. A mutation may produce an allele that is selected against, selected for, or selectively neutral. Harmful mutations are removed fr

occur when an individual travels from one geographic location to another and joins a different population of the species. In the example shown here, the brown allele is introduced into the green population.

11.3 Evidence of Evolution

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Explain sources of evidence for evolution
- ? Define homologous and vestigial structures

The evidence for evolution is compelling and extensive. Looking at every level of organization in living systems, biologists see the signature of past and present evolution. Darwin dedicated a large portion of his book, *On the Origin of Species*, identifying patterns in nature that were consistent with evolution and since Darwin our understanding has become clearer and broader.

Fossils

Fossils provide solid evidence that organisms from the past are not the same as those found today; fossils show the gradual evolutionary changes over time. Scientists determine the age of fossils and categorize them all over the world to determine when the organisms lived relative to each other. The resulting fossil record tells the story of the past, and shows the evolution of form over millions of years (Figure 11.10). For example, highly detailed fossil 4 A. J. Tipping et al., ?Molecular and Genealogical Evidence for a Founder Effect in Fanconi Anemia Families of the Afrikaner Population of South Africa,? PNAS 98, no. 10 (2001): 5734-5739, doi: 10.1073/pnas.091402398.

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records have been recovered for sequences of species in the evolution of whales and modern horses. The fossil record of horses in North America is especially rich and many contain transition fossils: those showing intermediate anatomy between earlier and later forms. The fossil record extends back to a dog-like ancestor some 55 million years ago that gave rise to the first horse-like species 55 to 42 million years ago in the genus *Eohippus*. The series of fossils tracks the change in anatomy resulting from a gradual drying trend that changed the landscape from a forested one to a prairie. Successive fossils show the evolution of teeth shapes and foot and leg anatomy to a grazing habit, with adaptations for escaping predators, for example in species of *Mesohippus* found from 40 to 30 million years ago. Later species showed gains in size, such as those of *Hipparion*, which existed from about 23 to 2 million years ago. The fossil record shows several adaptive radiations in the horse lineage, which is now much reduced to only one genus, *Equus*, with several species.

FIGURE 11.10 This illustration shows an artist's renderings of these species derived from fossils of the evolutionary history of the horse and its ancestors. The species depicted are only four from a very diverse lineage that contains many branches, dead ends, and adaptive radiations. One of the trends, depicted here is the evolutionary tracking of a drying climate and increase in prairie

ay to illustrate the relationships of species. (b) Modern elephants evolved from the Palaeomastodon, a species that lived in Egypt 35?50 million years ago.

Speciation through Geographic Separation

A geographically continuous population has a gene pool that is relatively homogeneous. Gene flow, the movement of alleles across the range of the species, is relatively free because individuals can move and then mate with individuals in their new location. Thus, the frequency of an allele at one end of a distribution will be similar to the frequency of the allele at the other end. When populations become geographically discontinuous that free-flow of alleles is prevented. When that separation lasts for a period of time, the two populations are able to evolve along different trajectories. Thus, their allele frequencies at numerous genetic loci gradually become more and more different as new alleles independently arise by mutation in each population. Typically, environmental conditions, such as climate, resources, predators, and competitors, for the two populations will differ causing natural selection to favor divergent adaptations in each group. Different histories of genetic drift, enhanced because the populations are smaller than the parent population, will also lead to divergence.

Given enough time, the genetic and phenotypic divergence between populations will likely affect characters that influence reproduction enough that were individuals of the two populations brought together, mating would be less likely, or if a mating occurred, offspring would be non-viable or infertile. Many types of diverging characters may affect the reproductive isolation (inability to interbreed) of the two populations. These mechanisms of reproductive isolation can be divided into prezygotic mechanisms (those that operate before fertilization) and postzygotic mechanisms (those that operate after fertilization). Prezygotic mechanisms include traits that allow the individuals to find each other, such as the timing of mating, sensitivity to pheromones, or choice of mating sites. If individuals

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are able to encounter each other, character divergence may prevent courtship rituals from leading to a mating either because female preferences have changed or male behaviors have changed. Physiological changes may interfere with successful fertilization if mating is able to occur. Postzygotic mechanisms include genetic incompatibilities that prevent proper development of the offspring, or if the offspring live, they may be unable to produce viable gametes themselves as in the example of the mule, the infertile offspring of a female horse and a male donkey.

If the two isolated populations are brought back together and the hybrid offspring that formed from matings between individuals of the two populations have lower survivorship or reduced fertility, then selection will favor individuals that are able to discriminate between

n potential mates of their own population and the other population.
This selection will enhance the reproductive isolation.

Isolation of populations leading to allopatric speciation can occur in a variety of ways: from a river forming a new branch, erosion forming a new valley, or a group of organisms traveling to a new location without the ability to return, such as seeds floating over the ocean to an island. The nature of the geographic separation necessary to

isolate populations depends entirely on the biology of the organism and its potential for dispersal. If two flying insect populations took up residence in separate nearby valleys, chances are that individuals from each population

would fly back and forth, continuing gene flow. However, if two rodent populations became divided by the formation

of a new lake, continued gene flow would be unlikely; therefore, speciation would be more likely.

Biologists group allopatric processes into two categories. If a few members of a species move to a new geographical

area, this is called dispersal. If a natural situation arises to physically divide organisms, this is called vicariance.

Scientists have documented numerous cases of allopatric speciation taking place. For example, along the west coast of the United States, two separate subspecies of spotted owls exist. The northern spotted owl has genetic and

phenotypic differences from its close relative, the Mexican spotted owl, which lives in the south (Figure 11.15). The

cause of their initial separation is not clear, but it may have been caused by the glaciers of the ice age dividing an

initial population into two.⁵

FIGURE 11.15 The northern spotted owl and the Mexican spotted owl inhabit geographically separate locations with different climates and

ecosystems. The owl is an example of incipient speciation. (credit "northern spotted owl": modification of work by John and Karen

Hollingsworth, USFWS; credit "Mexican spotted owl": modification of work by Bill Radke, USFWS)

⁵ Courtney, S.P., et al, "Scientific Evaluation of the Status of the Northern Spotted Owl," Sustainable Ecosystems Institute (2004), Portland, OR.

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Additionally, scientists have found that the further the distance between two groups that once were the same species, the more likely for speciation to take place. This seems logical because as the distance increases, the various environmental factors would likely have less in common than locations in close proximity. Consider the two

owls; in the north, the climate is cooler than in the south; the other types of organisms in each ecosystem differ, as

do their behaviors and habits; also, the hunting habits and prey choices of the owls in the south vary from the northern ones. These variances can lead to evolved differences in the owls, and over time speciation will likely occur

unless gene flow between the populations is restored.

In some cases, a population of one species disperses throughout a

that are also diploid ($2n = 6$, when they should be $n = 3$), the gametes now have twice as many chromosomes as they should have. These new gametes will be incompatible with the normal gametes produced by this plant species. But they could either self-pollinate or reproduce with other autopolyploid plants with gametes having the same diploid number. In this way, sympatric speciation can occur quickly by forming offspring with $4n$ called a tetraploid. These individuals would immediately be able to reproduce only with those of this new kind and not those of the ancestral species. The other form of polyploidy occurs when individuals of two different species reproduce to form a viable offspring called an allopolyploid. The prefix "allo" means "other" (recall from allopatric); therefore, an allopolyploid occurs when gametes from two different species combine. Figure 11.18 illustrates one possible way an allopolyploidy can form. Notice how it takes two generations, or two reproductive acts, before the viable fertile hybrid results.

FIGURE 11.18 Allopolyploidy results when two species mate to produce viable offspring. In the example shown, a normal gamete from one species fuses with a polyploid gamete from another. Two matings are necessary to produce viable offspring.

The cultivated forms of wheat, cotton, and tobacco plants are all allopolyploids. Although polyploidy occurs occasionally in animals, most chromosomal abnormalities in animals are lethal; it takes place most commonly in plants. Scientists have discovered more than 1/2 of all plant species studied relate back to a species evolved through polyploidy.

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Sympatric speciation may also take place in ways other than polyploidy. For example, imagine a species of fish that lived in a lake. As the population grew, competition for food also grew. Under pressure to find food, suppose that a group of these fish had the genetic flexibility to discover and feed off another resource that was unused by the other fish. What if this new food source was found at a different depth of the lake? Over time, those feeding on the second food source would interact more with each other than the other fish; therefore they would breed together as well. Offspring of these fish would likely behave as their parents and feed and live in the same area, keeping them separate from the original population. If this group of fish continued to remain separate from the first population, eventually sympatric speciation might occur as more genetic differences accumulated between them.

This scenario does play out in nature, as do others that lead to reproductive isolation. One such place is Lake Victoria in Africa, famous for its sympatric speciation of cichlid fish. Researchers have found hundreds of sympatric speciation events in these fish, which have not only happened in great number, but also over a short period of time.

Figure 11.19 shows this type of speciation among a cichlid fish populatio

about the living world. As such, a theory in science has survived significant efforts to discredit it by scientists, who are naturally skeptical. While theories can sometimes be overturned or revised, this does not lessen their weight but simply reflects the constantly evolving state of scientific knowledge. In contrast, a "theory" in common vernacular means a guess or suggested explanation for something. This meaning is more akin to the concept of a "hypothesis" used by scientists, which is a tentative explanation for something that is proposed to either be supported or disproved. When critics of evolution say evolution is "just a theory," they are implying that there is little evidence supporting it and that it is still in the process of being rigorously tested. This is a mischaracterization. If this were the case, geneticist Theodosius Dobzhansky would not have said that "nothing in biology makes sense, except in the light of evolution."⁶

Individuals Evolve

An individual is born with the genes it has—these do not change as the individual ages. Therefore, an individual cannot evolve or adapt through natural selection. Evolution is the change in genetic composition of a population over time, specifically over generations, resulting from differential reproduction of individuals with certain alleles. Individuals do change over their lifetime, but this is called development; it involves changes programmed by the set of genes the individual acquired at birth in coordination with the individual's environment. When thinking about the evolution of a characteristic, it is probably best to think about the change of the average value of the characteristic in the population over time. For example, when natural selection leads to bill-size change in medium ground finches in the Galápagos, this does not mean that individual bills on the finches are changing. If one measures the average bill size among all individuals in the population at one time, and then measures the average bill size in the population several years later after there has been a strong selective pressure, this average value may be different as a result of evolution. Although some individuals may survive from the first time to the second, those individuals will still have the same bill size. However, there may be enough new individuals with different bill sizes to change the average bill size.

Evolution Explains the Origin of Life

It is a common misunderstanding that evolution includes an explanation of life's origins. Some of the theory's critics complain that it cannot explain the origin of life. The theory does not try to explain the origin of life. The theory of evolution explains how populations change over time and how life diversifies—the origin of species. It does not shed light on the beginnings of life including the origins of the first cells, which is how life is defined. The mechanisms of the origin of life on Earth are a particularly difficult problem because it occurred a ve

ry long time ago, over a very long time, and presumably just occurred once. Importantly, biologists believe that the presence of life on Earth precludes the possibility that the events that led to life on Earth can be repeated because the intermediate stages would immediately become food for existing living things. The early stages of life included the formation of organic molecules such as carbohydrates, amino acids, or nucleotides. If these were formed from inorganic precursors today, they would simply be broken down by living things. The early stages of life also probably included more complex aggregations of molecules into enclosed structures with an internal environment, a boundary layer of some form, and the external environment. Such structures, if they were formed now, would be quickly consumed or broken down by living organisms.

However, once a mechanism of inheritance was in place in the form of a molecule like DNA or RNA, either within a cell or within a pre-cell, these entities would be subject to the principle of natural selection. More effective reproducers would increase in frequency at the expense of inefficient reproducers. So while evolution does not explain the origin of life, it may have something to say about some of the processes operating once pre-living entities acquired certain properties.

6 Theodosius Dobzhansky. "Biology, Molecular and Organismic." *American Zoologist* 4, no. 4 (1964): 449.

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Organisms Evolve on Purpose

Statements such as "organisms evolve in response to a change in an environment," are quite common. There are two easy misunderstandings possible with such a statement. First of all, the statement must not be understood to mean that individual organisms evolve, as was discussed above. The statement is shorthand for "a population evolves in response to a changing environment." However, a second misunderstanding may arise by interpreting the statement to mean that the evolution is somehow intentional. A changed environment results in some individuals in the population, those with particular phenotypes, benefiting and, therefore, producing proportionately more offspring than other phenotypes. This results in change in the population if the characters are genetically determined.

It is also important to understand that the variation that natural selection works on is already in a population and does not arise in response to an environmental change. For example, applying antibiotics to a population of bacteria will, over time, select for a population of bacteria that are resistant to antibiotics. The resistance, which is caused by a gene, did not arise by mutation because of the application of the antibiotic. The gene for resistance was already present in the gene pool of the bacteria, likely at a low frequency. The antibiotic, which kills the bacterial cells without the resistance gene, strongly selects for individuals that are resistant, since these would be the only on

ing scientists

who reject the theory of evolution, or question its validity and say so, is small. A Pew Research poll in 2009 found

that 97 percent of the 2500 scientists polled believe species evolve.⁷ The support for the theory is reflected in signed statements from many scientific societies such as the American Association for the Advancement of Science,

which includes working scientists as members. Many of the scientists that reject or question the theory of evolution

are non-biologists, such as engineers, physicians, and chemists. There are no experimental results or research programs that contradict the theory. There are no papers published in peer-reviewed scientific journals that appear

to refute the theory. The latter observation might be considered a consequence of suppression of dissent, but it must be remembered that scientists are skeptics and that there is a long history of published reports that challenged scientific orthodoxy in unpopular ways. Examples include the endosymbiotic theory of eukaryotic origins, the theory of group selection, the microbial cause of stomach ulcers, the asteroid-impact theory of the Cretaceous extinction, and the theory of plate tectonics. Research with evidence and ideas with scientific merit are

considered by the scientific community. Research that does not meet these standards is rejected.

Other Theories Should Be Taught

A common argument from some people is that alternative theories to evolution should be taught in public schools.

Critics of evolution use this strategy to create uncertainty about the validity of the theory without offering actual evidence. In fact, there are no viable alternative scientific theories to evolution. The last such theory, proposed by

Lamarck in the nineteenth century, was replaced by the theory of natural selection. A single exception was a research program in the Soviet Union based on Lamarck's theory during the early twentieth century that set that

country's agricultural research back decades. Special creation is not a viable alternative scientific theory because it

is not a scientific theory, since it relies on an untestable explanation. Intelligent design, despite the claims of its proponents, is also not a scientific explanation. This is because intelligent design posits the existence of an unknown designer of living organisms and their systems. Whether the designer is unknown or supernatural, it is a

cause that cannot be measured; therefore, it is not a scientific explanation. There are two reasons not to teach nonscientific theories. First, these explanations for the diversity of life lack scientific usefulness because they do not, and cannot, give rise to research programs that promote our understanding of the natural world.

Experiments

cannot test non-material explanations for natural phenomena. For this reason, teaching these explanations as science in public schools is not in the public interest. Second, in the United States, it is illegal to teach them as s

science because the U.S. Supreme Court and lower courts have ruled that the teaching of religious belief, such as special creation or intelligent design, violates the establishment clause of the First Amendment of the U.S. Constitution, which prohibits government sponsorship of a particular religion.

The theory of evolution and science in general is, by definition, silent on the existence or non-existence of the spiritual world. Science is only able to study and know the material world. Individual biologists have sometimes been vocal atheists, but it is equally true that there are many deeply religious biologists. Nothing in biology precludes the existence of a god or other supreme beings, indeed biology as a science has nothing to say about it.

Individual biologists are free to reconcile their personal and scientific knowledge as they see fit. The Voices for Evolution project (<http://ncse.com/voices>), developed through the National Center for Science Education, works to gather the diversity of perspectives on evolution to advocate it being taught in public schools.

7 Pew Research Center for the People & the Press, Public Praises Science; Scientists Fault Public, Media (Washington, DC, 2009), 37.

268 11 ? Key Terms

Key Terms

adaptation a heritable trait or behavior in an proposed by Lamarck in which traits acquired by

organism that aids in its survival in its present individuals through use or disuse could be passed environment on to their offspring thus leading to evolutionary

adaptive radiation a speciation when one species change in the population radiates out to form several other species macroevolution a broader scale of evolutionary

allopatric speciation a speciation that occurs via a changes seen over paleontological time geographic separation microevolution the changes in a population's genetic

analogous structure a structure that is similar structure (i.e., allele frequency) because of evolution in response to similar selection migration the movement of individuals of a pressures resulting in convergent evolution, not population to a new location; in population genetics similar because of descent from a common ancestor it refers to the movement of individuals and their

bottleneck effect the magnification of genetic drift as alleles from one population to another, potentially a result of natural events or catastrophes changing allele frequencies in both the old and the

convergent evolution an evolution that results in new population similar forms on different species modern synthesis the overarching evolutionary

dispersal an allopatric speciation that occurs when a paradigm that took shape by the 1940s and is few members of a species move to a new generally accepted today geographical area natural selection the greater relative survival and

divergent evolution an evolution that results in reproduction of individuals in a population that have different forms in two species with a common favorable heritable traits, leading to e

voluntary
ancestral change

founder effect a magnification of genetic drift in a population genetics the study of how selective small population that migrates away from a large forces change the allele frequencies in a population parent population carrying with it an over time unrepresentative set of alleles speciation a formation of a new species

gene flow the flow of alleles in and out of a sympatric speciation a speciation that occurs in the population due to the migration of individuals or same geographic space gametes variation the variety of alleles in a population

gene pool all of the alleles carried by all of the vestigial structure a physical structure present in an individuals in the population organism but that has no apparent function and

genetic drift the effect of chance on a population's appears to be from a functional structure in a gene pool distant ancestor

homologous structure a structure that is similar vicariance an allopatric speciation that occurs when because of descent from a common ancestor something in the environment separates organisms

inheritance of acquired characteristics a phrase of the same species into separate groups that describes the mechanism of evolution

Chapter Summary

11.1 Discovering How Populations Change The modern synthesis of evolutionary theory grew out

of the reconciliation of Darwin's, Wallace's, and Evolution by natural selection arises from three Mendel's thoughts on evolution and heredity. conditions: individuals within a species vary, some of Population genetics is a theoretical framework for those variations are heritable, and organisms have describing evolutionary change in populations through more offspring than resources can support. The the change in allele frequencies. Population genetics consequence is that individuals with relatively defines evolution as a change in allele frequency over advantageous variations will be more likely to survive generations. In the absence of evolutionary forces and have higher reproductive rates than those allele frequencies will not change in a population; this individuals with different traits. The advantageous is known as Hardy-Weinberg equilibrium principle. traits will be passed on to offspring in greater However, in all populations, mutation, natural proportion. Thus, the trait will have higher selection, genetic drift, and migration act to change representation in the next and subsequent generations allele frequencies. leading to genetic change in the population.

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11.2 Mechanisms of Evolution form gametes with extra chromosomes, called polyploidy. Autopolyploidy occurs within a single

There are four factors that can change the allele species, whereas allopolyploidy occurs because of a frequencies of a population. Natural selection works by mating between closely related species. Once the selecting for alleles that confer beneficial traits or populations are isolated,

ometimes two segments of DNA in distantly related organisms randomly share a high percentage of bases in the same locations, causing these organisms to appear closely related when they are not. For example, the fruit fly shares 60 percent of its DNA with humans.² In this situation, computer-based statistical algorithms have been developed to help identify the actual relationships, and ultimately, the coupled use of both morphologic and molecular information is more effective in determining phylogeny.

EVOLUTION CONNECTION

Why Does Phylogeny Matter?

In addition to enhancing our understanding of the evolutionary history of species, our own included, phylogenetic analysis has numerous practical applications. Two of those applications include understanding the evolution and transmission of disease and making decisions about conservation efforts. A 2010 study³ of MRSA (methicillin-resistant *Staphylococcus aureus*), an antibiotic resistant pathogenic bacterium, traced the origin and spread of the strain throughout the past 40 years. The study uncovered the timing and patterns in which the resistant strain moved from its point of origin in Europe to centers of infection and evolution in South America, Asia, North America, and Australasia. The study suggested that introductions of the bacteria to new populations occurred very few times, perhaps only once, and then spread from that limited number of individuals. This is in contrast to the possibility that many individuals had carried the bacteria from one place to another. This result suggests that public health officials should concentrate on quickly identifying the contacts of individuals infected with a new strain of bacteria to control its spread.

A second area of usefulness for phylogenetic analysis is in conservation. Biologists have argued that it is important to protect species throughout a phylogenetic tree rather than just those from one branch of the tree. Doing this will preserve more of the variation produced by evolution. For example, conservation efforts should focus on a single species without sister species rather than another species that has a cluster of close sister species that recently evolved. If the single evolutionarily distinct species goes extinct a disproportionate amount of variation from the tree will be lost compared to one species in the cluster of closely related species. A study published in 2007⁴ made recommendations for conservation of mammal species worldwide based on how evolutionarily distinct and at risk of extinction they are. The study found that their recommendations differed from priorities based on simply the level of extinction threat to the species. The study recommended protecting some threatened and valued large mammals such as the orangutans, the giant and lesser pandas, and the African and Asian elephants. But they also found that some much lesser known species should be protected based on how evolutionary distinct they are. These include a number of ro

vents occurred in the simplest, most obvious way. This means that the "best" tree is the one with the fewest number of character reversals, the fewest number of independent character changes, and the fewest number of character changes throughout the tree. Computer programs search through all of the possible trees to find the small number of trees with the simplest evolutionary pathways. Starting with all of the homologous traits in a group of organisms, scientists can determine the order of evolutionary events of which those traits occurred that is the most obvious and simple.

LINK TO LEARNING

Practice Parsimony: Go to this website (<http://openstax.org//parsimony2>) to learn how maximum parsimony is used to create phylogenetic trees (be sure to continue to the second page).

These tools and concepts are only a few of the strategies scientists use to tackle the task of revealing the evolutionary history of life on Earth. Recently, newer technologies have uncovered surprising discoveries with unexpected relationships, such as the fact that people seem to be more closely related to fungi than fungi are to plants. Sound unbelievable? As the information about DNA sequences grows, scientists will become closer to mapping the evolutionary history of all life on Earth.

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12 ? Key Terms 285

Key Terms

analogous structure a character found in two taxa relationships

that looks similar because of convergent evolution, monophyletic group (also, clade) organisms that not because of descent from a common ancestor share a single ancestor

binomial nomenclature a system of two-part order the category in the taxonomic classification scientific names for an organism, which includes system that falls within class and includes families genus and species names phylogenetic tree diagram used to reflect the

branch point a point on a phylogenetic tree where a evolutionary relationships between organisms or single lineage splits to distinct new ones groups of organisms

clade a group of taxa with the same set of shared phylogeny evolutionary history and relationship of an derived characters, including an ancestral species organism or group of organisms and all its descendants phylum the category in the taxonomic classification

cladistics a method used to organize homologous system that falls within kingdom and includes traits to describe phylogenies using common classes
descendent as the primary criterion used to classify rooted describing a phylogenetic tree with a single organisms ancestral lineage to which all organisms

class the category in the taxonomic classification represented in the diagram relate
system that falls within phylum and includes orders shared ancestral character a character on a

domain the highest level category in the classification phylogenetic branch that is shared by a particular system and that includes all taxonomic clade
classifications below it; it is the most inclusive taxon shared derived character a

the late twentieth century, scientists most commonly grouped living things into five kingdoms—animals, plants, fungi, protists, and bacteria—based on several criteria, such as absence or presence of a nucleus and other membrane-bound organelles, absence or presence of cell walls, multicellularity, and mode of nutrition. In the late twentieth century, the pioneering work of Carl Woese and others compared nucleotide sequences of small-subunit ribosomal RNA (SSU rRNA), which resulted in a dramatically different way to group organisms on Earth. Based on differences in the structure of cell membranes and in rRNA, Woese and his colleagues proposed

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that all life on Earth evolved along three lineages, called domains. The three domains are called Bacteria, Archaea, and Eukarya.

Two of the three domains—Bacteria and Archaea—are prokaryotic, meaning that they lack both a nucleus and true membrane-bound organelles. However, they are now considered, on the basis of membrane structure and rRNA, to be as different from each other as they are from the third domain, the Eukarya. Prokaryotes were the first inhabitants on Earth, perhaps appearing approximately 3.9 billion years ago. Today they are ubiquitous—inhabiting the harshest environments on the planet, from boiling hot springs to permanently frozen environments in Antarctica, as well as more benign environments such as compost heaps, soils, ocean waters, and the guts of animals (including humans). The Eukarya include the familiar kingdoms of animals, plants, and fungi. They also include a diverse group of kingdoms formerly grouped together as protists.

13.1 Prokaryotic Diversity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe the evolutionary history of prokaryotes
- ? Describe the basic structure of a typical prokaryote
- ? Identify bacterial diseases that caused historically important plagues and epidemics
- ? Describe the uses of prokaryotes in food processing and bioremediation

Prokaryotes are present everywhere. They cover every imaginable surface where there is sufficient moisture, and they live on and inside of other living things. There are more prokaryotes inside and on the exterior of the human body than there are human cells in the body. Some prokaryotes thrive in environments that are inhospitable for most other living things. Prokaryotes recycle nutrients—essential substances (such as carbon and nitrogen)—and they drive the evolution of new ecosystems, some of which are natural while others are man-made. Prokaryotes have been on Earth since long before multicellular life appeared.

Prokaryotic Diversity

The advent of DNA sequencing provided immense insight into the relationships and origins of prokaryotes that were not possible using traditional methods of classification. A major insight identified two groups of prokaryotes that were found to be as different from each other as they were from eukaryotes. This re

cognition of prokaryotic diversity forced a new understanding of the classification of all life and brought us closer to understanding the fundamental relationships of all living things, including ourselves.

Early Life on Earth

When and where did life begin? What were the conditions on Earth when life began? Prokaryotes were the first forms of life on Earth, and they existed for billions of years before plants and animals appeared. Earth is about 4.54 billion years old. This estimate is based on evidence from the dating of meteorite material, since surface rocks on Earth are not as old as Earth itself. Most rocks available on Earth have undergone geological changes that make them younger than Earth itself. Some meteorites are made of the original material in the solar disk that formed the objects of the solar system, and they have not been altered by the processes that altered rocks on Earth. Thus, the age of meteorites is a good indicator of the age of the formation of Earth. The original estimate of 4.54 billion years was obtained by Clair Patterson in 1956. His meticulous work has since been corroborated by ages determined from other sources, all of which point to an Earth age of about 4.54 billion years.

Early Earth had a very different atmosphere than it does today. Evidence indicates that during the first 2 billion years of Earth's existence, the atmosphere was anoxic, meaning that there was no oxygen. Therefore, only those organisms that can grow without oxygen—anaerobic organisms—were able to live. Organisms that convert solar energy into chemical energy are called

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phototrophs. Phototrophic organisms that required an organic source of carbon appeared within one billion years of the formation of Earth. Then, cyanobacteria, also known as blue-green algae, evolved from these simple phototrophs one billion years later. Cyanobacteria are able to use carbon dioxide as a source of carbon. Cyanobacteria (Figure 13.2) began the oxygenation of the atmosphere. The increase in oxygen concentration allowed the evolution of other life forms.

FIGURE 13.2 This hot spring in Yellowstone National Park flows toward the foreground. Cyanobacteria in the spring are green, and as water flows down the heat gradient, the intensity of the color increases because cell density increases. The water is cooler at the edges of the stream than in the center, causing the edges to appear greener. (credit: Graciela Brelles-Mariño)

Before the atmosphere became oxygenated, the planet was subjected to strong radiation; thus, the first organisms would have flourished where they were more protected, such as in ocean depths or beneath the surface of Earth. At this time, too, strong volcanic activity was common on Earth, so it is likely that these first organisms—the first prokaryotes—were adapted to very high temperatures. These are not the typical temperate environments in which most life flourishes today;

thus, we can conclude that the first organisms that appeared on Earth likely were able to withstand harsh conditions.

Microbial mats may represent the earliest forms of life on Earth, and there is fossil evidence of their presence, starting about 3.5 billion years ago. A microbial mat is a large biofilm, a multi-layered sheet of prokaryotes (Figure 13.3a), including mostly bacteria, but also archaea. Microbial mats are a few centimeters thick, and they typically grow on moist surfaces. Their various types of prokaryotes carry out different metabolic pathways, and for this reason, they reflect various colors. Prokaryotes in a microbial mat are held together by a gummy-like substance that they secrete.

The first microbial mats likely obtained their energy from hydrothermal vents. A hydrothermal vent is a fissure in Earth's surface that releases geothermally heated water. With the evolution of photosynthesis about 3 billion years ago, some prokaryotes in microbial mats came to use a more widely available energy source—sunlight—whereas others were still dependent on chemicals from hydrothermal vents for food.

FIGURE 13.3 (a) This microbial mat grows over a hydrothermal vent in the Pacific Ocean. Chimneys such as the one indicated by the arrow allow gases to escape. (b) This photo shows stromatolites that are nearly 1.5 billion years old, found in Glacier National Park, Montana.

(credit a: modification of work by Dr. Bob Embley, NOAA PMEL; credit b: modification of work by P. Carrara, NPS)

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Fossilized microbial mats represent the earliest record of life on Earth. A stromatolite is a sedimentary structure formed when minerals are precipitated from water by prokaryotes in a microbial mat (Figure 13.3b).

Stromatolites

form layered rocks made of carbonate or silicate. Although most stromatolites are artifacts from the past, there are

places on Earth where stromatolites are still forming. For example, living stromatolites have been found in the Anza-

Borrego Desert State Park in San Diego County, California. Efforts to understand the earliest microbial mats may also

have implications on the search for life elsewhere. Nora Noffke, who identified many of the features used in dating

and categorizing microbially produced sedimentary structures, emphasizes that similar formations could be used to

find evidence of life on Mars. When analyzing photos taken by the NASA Curiosity rover, Noffke showcased the similarities between fossilized microbial mats found on the two planets.

Some prokaryotes are able to thrive and grow under conditions that would kill a plant or animal. Bacteria and archaea that grow under extreme conditions are called extremophiles, meaning "lovers of extremes."

Extremophiles have been found in extreme environments of all kinds, including the depths of the oceans, hot springs, the Arctic and the Antarctic, very dry places, deep inside Earth, harsh chemical environments, and high radiation

aryotes can use different sources of energy to assemble macromolecules from smaller molecules. Phototrophs obtain their energy from sunlight. Chemotrophs obtain their energy from chemical compounds.

Bacterial Diseases in Humans

Devastating pathogen-borne diseases and plagues, both viral and bacterial in nature, have affected and continue to affect humans. It is worth noting that all pathogenic prokaryotes are Bacteria; there are no known pathogenic Archaea in humans or any other organism. Pathogenic organisms evolved alongside humans. In the past, the true cause of these diseases was not understood, and some cultures thought that diseases were a spiritual punishment or were mistaken about material causes. Over time, people came to realize that staying apart from afflicted persons, improving sanitation, and properly disposing of the corpses and personal belongings of victims of illness reduced their own chances of getting sick.

Historical Perspective

There are records of infectious diseases as far back as 3,000 B.C. A number of significant pandemics caused by Bacteria have been documented over several hundred years. Some of the largest pandemics led to the decline of cities and cultures. Many were zoonoses that appeared with the domestication of animals, as in the case of tuberculosis. A zoonosis is a disease that infects animals but can be transmitted from animals to humans.

Infectious diseases remain among the leading causes of death worldwide. Their impact is less significant in many developed countries, but they are important determiners of mortality in developing countries. The development of antibiotics did much to lessen the mortality rates from bacterial infections, but access to antibiotics is not universal, and the overuse of antibiotics has led to the development of resistant strains of bacteria. Public sanitation efforts that dispose of sewage and provide clean drinking water have done as much or more than medical advances to prevent deaths caused by bacterial infections.

In 430 B.C., the plague of Athens killed one-quarter of the Athenian troops that were fighting in the Great Peloponnesian War. The disease killed a quarter of the population of Athens in over 4 years and weakened Athens' dominance and power. The source of the plague may have been identified recently when researchers from the University of Athens were able to analyze DNA from teeth recovered from a mass grave. The scientists identified nucleotide sequences from a pathogenic bacterium that causes typhoid fever.¹

From 541 to 750 A.D., an outbreak called the plague of Justinian (likely a bubonic plague) eliminated, by some estimates, one-quarter to one-half of the human population. The population in Europe declined by 50 percent during this outbreak. Bubonic plague would decimate Europe more than once.

One of the most devastating pandemics was the Black Death (1346 to 1361), which is believed to have been another outbreak of bubonic plague caused by the bacterium *Yersinia pestis*.

s easily treatable with antibiotics. A very dangerous strain, however, has made the news over the past few years (Figure 13.7). This strain, methicillin-resistant *Staphylococcus aureus* (MRSA), is resistant to many commonly used antibiotics, including methicillin, amoxicillin, penicillin, and oxacillin. While MRSA infections have been common among people in healthcare facilities, it is appearing more commonly in healthy people who live or work in dense groups (like military personnel and prisoners). The Journal of the American Medical Association reported that, among MRSA-afflicted persons in healthcare facilities, the average age is 68 years, while people with community-associated MRSA (CA-MRSA) have an average age of 23 years.²

FIGURE 13.7 This scanning electron micrograph shows methicillin-resistant *Staphylococcus aureus* bacteria, commonly known as MRSA.

(credit: modification of work by Janice Haney Carr, CDC; scale-bar data from Matt Russell)

2 Naimi, T. S., LeDell, K. H., Como-Sabetti, K., et al., "Comparison of community- and health care-associated methicillin-resistant *Staphylococcus aureus* infection," *JAMA* 290 (2003): 2976-2984, doi: 10.1001/jama.290.22.2976.

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In summary, society is facing an antibiotic crisis. Some scientists believe that after years of being protected from bacterial infections by antibiotics, we may be returning to a time in which a simple bacterial infection could again devastate the human population. Researchers are working on developing new antibiotics, but few are in the drug development pipeline, and it takes many years to generate an effective and approved drug.

Foodborne Diseases

Prokaryotes are everywhere: They readily colonize the surface of any type of material, and food is not an exception.

Outbreaks of bacterial infection related to food consumption are common. A foodborne disease (colloquially called

food poisoning) is an illness resulting from the consumption of food contaminated with pathogenic bacteria, viruses, or other parasites. Although the United States has one of the safest food supplies in the world, the Center

for Disease Control and Prevention (CDC) has reported that 76 million people get sick, more than 300,000 are hospitalized, and 5,000 Americans die each year from foodborne illness.³

The characteristics of foodborne illnesses have changed over time. In the past, it was relatively common to hear about sporadic cases of botulism, the potentially fatal disease produced by a toxin from the anaerobic bacterium *Clostridium botulinum*. A can, jar, or package created a suitable anaerobic environment where *Clostridium* could grow. Proper sterilization and canning procedures have reduced the incidence of this disease.

Most cases of foodborne illnesses are now linked to produce contaminated by animal waste. For example, there have been serious, produce-related outbreaks associated with raw spinach in the United States and with

To clean up these spills, bioremediation is promoted by adding inorganic nutrients that help bacteria already present in the environment to grow. Hydrocarbon-degrading bacteria feed on the hydrocarbons in the oil droplet, breaking them into inorganic compounds. Some species, such as *Alcanivorax borkumensis*, produce surfactants that solubilize the oil, while other bacteria degrade the oil into carbon dioxide. In the case of oil spills in the ocean, ongoing, natural bioremediation tends to occur, inasmuch as there are oil-consuming bacteria in the ocean prior to the spill. Under ideal conditions, it has been reported that up to 80 percent of the nonvolatile components in oil can be degraded within 1 year of the spill. Other oil fractions containing aromatic and highly branched hydrocarbon chains are more difficult to remove and remain in the environment for longer periods of time. Researchers have genetically engineered other bacteria to consume petroleum products; indeed, the first patent application for a bioremediation application in the U.S. was for a genetically modified oil-eating bacterium.

FIGURE 13.10 (a) Cleaning up oil after the Valdez spill in Alaska, the workers hosed oil from beaches and then used a floating boom to corral the oil, which was finally skimmed from the water surface. Some species of bacteria are able to solubilize and degrade the oil. (b) One of the most catastrophic consequences of oil spills is the damage to fauna. (credit a: modification of work by NOAA; credit b: modification of work by GOLUBENKOV, NGO: Saving Taman)

Prokaryotes in and on the Body

Humans are no exception when it comes to forming symbiotic relationships with prokaryotes. We are accustomed to thinking of ourselves as single organisms, but in reality, we are walking ecosystems. There are 10 to 100 times as many bacterial and archaeal cells inhabiting our bodies as we have cells in our bodies. Some of these are in mutually beneficial relationships with us, in which both the human host and the bacterium benefit, while some of the relationships are classified as commensalism, a type of relationship in which the bacterium benefits and the human host is neither benefited nor harmed.

Human gut flora lives in the large intestine and consists of hundreds of species of bacteria and archaea, with different individuals containing different species mixes. The term "flora," which is usually associated with plants, is traditionally used in this context because bacteria were once classified as plants. The primary functions of these prokaryotes for humans appear to be metabolism of food molecules that we cannot break down, assistance with the

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absorption of ions by the colon, synthesis of vitamin K, training of the infant immune system, maintenance of the adult immune system, maintenance of the epithelium of the large intestine, and formation of a protective barrier against pathogens.

The surface of the skin

is also coated with prokaryotes. The different surfaces of the skin, such as the underarms, the head, and the hands, provide different habitats for different communities of prokaryotes. Unlike with gut flora, the possible beneficial roles of skin flora have not been well studied. However, the few studies conducted so far have identified bacteria that produce antimicrobial compounds as probably responsible for preventing infections by pathogenic bacteria.

As they would in any ecosystem, the organisms in the microbiome are subject to selection and evolution. Simply to survive through changes in the human body requires a degree of adaptability and resistance. Because humans and other animals (such as livestock) ingest more and more antibiotics, the bacteria within our bodies are becoming resistant to those medicines. Abigail A. Salyers, who provided much of the foundational research on the human microbiome, focused much of her early work on a type of bacteria (*Bacteroides*) that could not only become resistant to antibiotics, but pass on that resistance to surrounding bacteria. Salyers was among the first researchers to sound the alarm about the dangers of antibiotic resistance and its potential impact on health. Researchers are actively studying the relationships between various diseases and alterations to the composition of human microbial flora. Some of this work is being carried out by the Human Microbiome Project, funded in the United States by the National Institutes of Health.

13.2 Eukaryotic Origins

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe the endosymbiotic theory
- ? Explain the origin of mitochondria and chloroplasts

The fossil record and genetic evidence suggest that prokaryotic cells were the first organisms on Earth. These cells originated approximately 3.5 billion years ago, which was about 1 billion years after Earth's formation, and were the only life forms on the planet until eukaryotic cells emerged approximately 2.1 billion years ago. During the prokaryotic reign, photosynthetic prokaryotes evolved that were capable of applying the energy from sunlight to synthesize organic materials (like carbohydrates) from carbon dioxide and an electron source (such as hydrogen, hydrogen sulfide, or water).

Photosynthesis using water as an electron donor consumes carbon dioxide and releases molecular oxygen (O_2) as a byproduct. The functioning of photosynthetic bacteria over millions of years progressively saturated Earth's water with oxygen and then oxygenated the atmosphere, which previously contained much greater concentrations of carbon dioxide and much lower concentrations of oxygen. Older anaerobic prokaryotes of the era could not function in their new, aerobic environment. Some species perished, while others survived in the remaining anaerobic environments left on Earth. Still other early prokaryotes evolved mechanisms, such as aerobic respiration, to exploit the oxygenated atmosphere by using ox

as the Chytridiomycota that reside in the rumen of cattle, are obligate anaerobes, meaning that they cannot grow and reproduce in an environment with oxygen. Yeasts are intermediate: They grow best in the presence of oxygen but can use fermentation in the absence of oxygen. The alcohol produced from yeast fermentation is used in wine and beer production, and the carbon dioxide they produce carbonates beer and sparkling wine, and makes bread rise.

The reproductive stage could be sexual or asexual. In both sexual and asexual reproduction, fungi produce spores that disperse from the parent organism by either floating in the wind or hitching a ride on an animal. Fungal spores are smaller and lighter than plant seeds, but they are not usually released as high in the air. The giant puffball mushroom bursts open and releases trillions of spores: The huge number of spores released increases the likelihood of spores landing in an environment that will support growth (Figure 13.23).

FIGURE 13.23 The (a) giant puffball mushroom releases (b) a cloud of spores when it reaches maturity. (credit a: modification of work by Roger Griffith; credit b: modification of work by Pearson Scott Foresman, donated to the Wikimedia Foundation)

How Fungi Obtain Nutrition

Like animals, fungi are heterotrophs: They use complex organic compounds as a source of carbon rather than fixing carbon dioxide from the atmosphere, as some bacteria and most plants do. In addition, fungi do not fix nitrogen from the atmosphere. Like animals, they must obtain it from their diet. However, unlike most animals that ingest food and then digest it internally in specialized organs, fungi perform these steps in the reverse order. Digestion precedes ingestion. First, exoenzymes, enzymes that catalyze reactions on compounds outside of the cell, are transported out of the hyphae where they break down nutrients in the environment. Then, the smaller molecules produced by the external digestion are absorbed through the large surface areas of the mycelium. As with animal cells, the fungal storage polysaccharide is glycogen rather than starch, as found in plants.

Fungi are mostly saprobes, organisms that derive nutrients from decaying organic matter. They obtain their nutrients from dead or decomposing organic matter, mainly plant material. Fungal exoenzymes are able to break down insoluble polysaccharides, such as the cellulose and lignin of dead wood, into readily absorbable glucose molecules. Decomposers are important components of ecosystems, because they return nutrients locked in dead bodies to a form that is usable for other organisms. This role is discussed in more detail later. Because of their varied metabolic pathways, fungi fulfill an important ecological role and are being investigated as potential tools in bioremediation. For example, some species of fungi can be used to break down diesel oil and polycyclic aromatic hydrocarbons. Other species take up heavy metals such as cadmium and lead.

tensive network of hyphae and large surface area in contact with the soil to channel water and minerals from the soil into the plant. In exchange, the plant supplies the products of photosynthesis to fuel the metabolism of the

fungus. Ectomycorrhizae (?outside? mycorrhiza) depend on fungi enveloping the roots in a sheath (called a mantle)

and a net of hyphae that extends into the roots between cells. In a second type, the Glomeromycota fungi form arbuscular mycorrhiza. In these mycorrhiza, the fungi form arbuscles, a specialized highly branched hypha, which

penetrate root cells and are the sites of the metabolic exchanges between the fungus and the host plant.

Orchids

rely on a third type of mycorrhiza. Orchids form small seeds without much storage to sustain germination and growth. Their seeds will not germinate without a mycorrhizal partner (usually Basidiomycota). After nutrients in the

seed are depleted, fungal symbionts support the growth of the orchid by providing necessary carbohydrates and minerals. Some orchids continue to be mycorrhizal throughout their lifecycle.

Lichens blanket many rocks and tree bark, displaying a range of colors and textures. Lichens are important pioneer

organisms that colonize rock surfaces in otherwise lifeless environments such as are created by glacial recession.

The lichen is able to leach nutrients from the rocks and break them down in the first step to creating soil.

Lichens are

also present in mature habitats on rock surfaces or the trunks of trees. They are an important food source for caribou. Lichens are not a single organism, but rather a fungus (usually an Ascomycota or Basidiomycota species)

living in close contact with a photosynthetic organism (an alga or cyanobacterium). The body of a lichen, referred to

as a thallus, is formed of hyphae wrapped around the green partner. The photosynthetic organism provides carbon

and energy in the form of carbohydrates and receives protection from the elements by the thallus of the fungal partner. Some cyanobacteria fix nitrogen from the atmosphere, contributing nitrogenous compounds to the association. In return, the fungus supplies minerals and protection from dryness and excessive light by encasing the

algae in its mycelium. The fungus also attaches the symbiotic organism to the substrate.

Fungi have evolved mutualistic associations with numerous arthropods. The association between species of Basidiomycota and scale insects is one example. The fungal mycelium covers and protects the insect colonies.

The

scale insects foster a flow of nutrients from the parasitized plant to the fungus. In a second example, leaf-cutting ants of Central and South America literally farm fungi. They cut disks of leaves from plants and pile them up in gardens. Fungi are cultivated in these gardens, digesting the cellulose that the ants cannot break down. Once smaller sugar molecules are produced and consumed by the fungi, they in turn become a meal for the ants. The insects also p

atrol their garden, preying on competing fungi. Both ants and fungi benefit from the association. The fungus receives a steady supply of leaves and freedom from competition, while the ants feed on the fungi they cultivate.

Importance to Humans

Although we often think of fungi as organisms that cause diseases and rot food, fungi are important to human life on many levels. As we have seen, they influence the well-being of human populations on a large scale because they help nutrients cycle in ecosystems. They have other ecosystem roles as well. For example, as animal pathogens, fungi help to control the population of damaging pests. These fungi are very specific to the insects they attack and do not infect other animals or plants. The potential to use fungi as microbial insecticides is being investigated, with several species already on the market. For example, the fungus *Beauveria bassiana* is a pesticide that is currently being tested as a possible biological control for the recent spread of emerald ash borer. It has been released in Michigan, Illinois, Indiana, Ohio, West Virginia, and Maryland.

The mycorrhizal relationship between fungi and plant roots is essential for the productivity of farmland. Without the fungal partner in the root systems, 80-90% of trees and grasses would not survive. Mycorrhizal fungal inoculants are available as soil amendments from gardening supply stores and are promoted by supporters of organic agriculture, but there is little evidence as to the effectiveness.

We also eat some types of fungi. Mushrooms figure prominently in the human diet. Morels, shiitake mushrooms, chanterelles, and truffles are considered delicacies (Figure 13.27). The humble meadow mushroom, *Agaricus campestris*, appears in many dishes. Molds of the genus *Penicillium* ripen many cheeses. They originate in the natural environment such as the caves of Roquefort, France, where wheels of sheep milk cheese are stacked to

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capture the molds responsible for the blue veins and pungent taste of the cheese.

FIGURE 13.27 The morel mushroom is an ascomycete that is much appreciated for its delicate taste. (credit: Jason Hollinger)

Fermentation of grains to produce beer, and of fruits to produce wine is an ancient art that humans in most cultures have practiced for millennia. Wild yeasts are acquired from the environment and used to ferment sugars into CO₂ and ethyl alcohol under anaerobic conditions. It is now possible to purchase isolated strains of wild yeasts from different wine-making regions. Pasteur was instrumental in developing a reliable strain of brewer's yeast, *Saccharomyces cerevisiae*, for the French brewing industry in the late 1850s. It was one of the first examples of biotechnology patenting. Yeast is also used to make breads that rise. The carbon dioxide they produce is responsible for the bubbles produced in the dough that become the air pockets of the baked bread.

Many secondary met

and only

cytoskeleton, and the acquisition and evolution of anaerobic organisms were able to live. Cyanobacteria

organelles. Nuclear eukaryotic genes appear to have began the oxygenation of the atmosphere. The

had an origin in the Archaea, whereas the energy increase in oxygen concentration allowed the evolution

machinery of eukaryotic cells appears to be bacterial in of other life forms.

origin. The mitochondria and plastids originated from Prokaryotes (domains Archaea and Bacteria) are endosymbiotic events when ancestral cells engulfed an single-celled organisms lacking a nucleus. They have a aerobic bacterium (in the case of mitochondria) and a single piece of circular DNA in the nucleoid area of the photosynthetic bacterium (in the case of chloroplasts). cell. Most prokaryotes have cell wall outside the The evolution of mitochondria likely preceded the plasma membrane. Bacteria and Archaea differ in the evolution of chloroplasts. There is evidence of compositions of their cell membranes and the secondary endosymbiotic events in which plastids characteristics of their cell walls. appear to be the result of endosymbiosis after a

previous endosymbiotic event.

Bacterial cell walls contain peptidoglycan. Archaeal cell walls do not have peptidoglycan. Bacteria can be 13.3 Protists divided into two major groups: Gram-positive and

Gram-negative. Gram-positive organisms have a thick Protists are extremely diverse in terms of biological cell wall. Gram-negative organisms have a thin cell wall and ecological characteristics due in large part to the and an outer membrane. Prokaryotes use diverse fact that they are an artificial assemblage of sources of energy to assemble macromolecules from phylogenetically unrelated groups. Protists display smaller molecules. Phototrophs obtain their energy highly varied cell structures, several types of from sunlight, whereas chemotrophs obtain it from reproductive strategies, virtually every possible type of chemical compounds. nutrition, and varied habitats. Most single-celled

protists are motile, but these organisms use diverse

Infectious diseases caused by bacteria remain among structures for transportation.

the leading causes of death worldwide. The excessive

use of antibiotics to control bacterial infections has The process of classifying protists into meaningful resulted in resistant forms of bacteria being selected. groups is ongoing, but genetic data in the past 20 years Foodborne diseases result from the consumption of have clarified many relationships that were previously contaminated food, pathogenic bacteria, viruses, or unclear or mistaken. The majority view at present is to parasites that contaminate food. Prokaryotes are used order all eukaryotes into six supergroups. The goal of in human food products. Microbial bioremediation is this classification scheme is to create clusters of the use of microbial metabolism to remove pollutants. species that all are derived from a common ancestor. The human

it b: modification of work by Stephen Ausmus, USDA ARS; credit c:
modification of work by David Nance, USDA ARS; credit d: modification of work by Jolly Janner)

CHAPTER OUTLINE

14.1 The Plant Kingdom

14.2 Seedless Plants

14.3 Seed Plants: Gymnosperms

14.4 Seed Plants: Angiosperms

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INTRODUCTION Plants play an integral role in all aspects of life on the planet, shaping the physical terrain, influencing the climate, and maintaining life as we know it. For millennia, human societies have depended on plants for nutrition and medicinal compounds, and for many industrial by-products, such as timber, paper, dyes, and textiles. Palms provide materials including rattans, oils, and dates. Wheat is grown to feed both human and animal populations. The cotton boll flower is harvested and its fibers transformed into clothing or pulp for paper. The showy opium poppy is valued both as an ornamental flower and as a source of potent opiate compounds.

Current evolutionary thought holds that all plants are monophyletic: that is, descendants of a single common ancestor. The evolutionary transition from water to land imposed severe constraints on the ancestors of contemporary plants. Plants had to evolve strategies to avoid drying out, to disperse reproductive cells in air, for structural support, and to filter sunlight. While seed plants developed adaptations that allowed them to populate even the most arid habitats on Earth, full independence from water did not happen in all plants, and most seedless plants still require a moist environment.

14.1 The Plant Kingdom

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe the major characteristics of the plant kingdom
- ? Discuss the challenges to plant life on land
- ? Describe the adaptations that allowed plants to colonize land

Plants are a large and varied group of organisms. There are close to 300,000 species of catalogued plants.¹ Of these, about 260,000 are plants that produce seeds. Mosses, ferns, conifers, and flowering plants are all members of the plant kingdom. The plant kingdom contains mostly photosynthetic organisms; a few parasitic forms have lost the ability to photosynthesize. The process of photosynthesis uses chlorophyll, which is located in organelles called chloroplasts. Plants possess cell walls containing cellulose. Most plants reproduce sexually, but they also have diverse methods of asexual reproduction. Plants exhibit indeterminate growth, meaning they do not have a final body form, but continue to grow body mass until they die.

Plant sex cells and sex organs are classified using the same system as those of animals. Plants reproduce through a union of two sex cells or gametes of different sizes. The larger gametes, called eggs, are classified as female gametes; the organs that produce them, called pistils, are classified as female organs. The smaller gametes, called sperm, are classified as male gametes; the organs that produces them, c

alled the stamens, are classified as male organs. In many plants, pistils and stamens are found on the same plant body, making the plant capable of both self-fertilization or cross-fertilization with another individual.

Plant Adaptations to Life on Land

As organisms adapt to life on land, they have to contend with several challenges in the terrestrial environment. Water has been described as "the stuff of life." The cell's interior—the medium in which most small molecules dissolve and diffuse, and in which the majority of the chemical reactions of metabolism take place—is a watery soup. Desiccation, or drying out, is a constant danger for an organism exposed to air. Even when parts of a plant are close to a source of water, their aerial structures are likely to dry out. Water provides buoyancy to organisms that live in aquatic habitats. On land, plants need to develop structural support in air—a medium that does not give the same lift. Additionally, the male and female gametes must reach one another using new strategies because swimming is no longer possible. Finally, both gametes and zygotes must be protected from drying out. The successful land plants evolved strategies to deal with all of

1 A.D. Chapman (2009) Numbers of Living Species in Australia and the World. 2nd edition. A Report for the Australian Biological Resources Study. Australian Biodiversity Information Services, Toowoomba, Australia. Available online at <http://www.environment.gov.au/biodiversity/abrs/publications/other/species-numbers/2009/04-03-groups-plants.html>.

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these challenges, although not all adaptations appeared at once. Some species did not move far from an aquatic environment, whereas others left the water and went on to conquer the driest environments on Earth.

To balance these survival challenges, life on land offers several advantages. First, sunlight is abundant. On land, the spectral quality of light absorbed by the photosynthetic pigment, chlorophyll, is not filtered out by water or competing photosynthetic species in the water column above. Second, carbon dioxide is more readily available because its concentration is higher in air than in water. Additionally, land plants evolved before land animals; therefore, until dry land was colonized by animals, no predators threatened the well-being of plants. This situation changed as animals emerged from the water and found abundant sources of nutrients in the established flora. In turn, plants evolved strategies to deter predation: from spines and thorns to toxic chemicals.

The early land plants, like the early land animals, did not live far from an abundant source of water and developed survival strategies to combat dryness. One of these strategies is drought tolerance. Mosses, for example, can dry out to a brown and brittle mat, but as soon as rain makes water available, mosses will soak it up and regain their healthy, green appearance. Another str

le to prevent desiccation. (c) To grow taller, like these maple trees, plants had to evolve new structural chemicals to strengthen their stems and vascular systems to transport water and minerals from the soil and nutrients from the leaves. (d) Plants developed physical and chemical defenses to avoid being eaten by animals. (credit a, b: Zanker; credit c: modification of work by Christine Cimala; credit d: modification of work by Jo Naylor)

EVOLUTION CONNECTION

Paleobotany

How organisms acquired traits that allow them to colonize new environments, and how the contemporary ecosystem is shaped, are fundamental questions of evolution. Paleobotany addresses these questions by specializing in the study of extinct plants. Paleobotanists analyze specimens retrieved from field studies, reconstituting the morphology of organisms that have long disappeared. They trace the evolution of plants by following the modifications in plant morphology, and shed light on the connection between existing plants by identifying common ancestors that display the same traits. This field seeks to find transitional species that bridge gaps in the path to the development of modern organisms. Fossils are formed when organisms are trapped in sediments or environments where their shapes are preserved (Figure 14.7). Paleobotanists determine the geological age of specimens and the nature of their environment using the geological sediments and fossil organisms surrounding them. The activity requires great care to preserve the integrity of the delicate fossils and the layers in which they are found.

One of the most exciting recent developments in paleobotany is the use of analytical chemistry and molecular biology to study fossils. Preservation of molecular structures requires an environment free of oxygen, since oxidation and degradation of material through the activity of microorganisms depend on the presence of oxygen.

One example of the use of analytical chemistry and molecular biology is in the identification of oleanane, a compound that deters pests and which, up to this point, appears to be unique to flowering plants. Oleanane was recovered from sediments dating from the Permian, much earlier than the current dates given for the appearance of the first flowering plants. Fossilized nucleic acids?DNA and RNA?yield the most information. Their sequences are analyzed and compared to those of living and related organisms. Through this analysis, evolutionary relationships can be built for plant lineages.

Some paleobotanists are skeptical of the conclusions drawn from the analysis of molecular fossils. For one, the chemical materials of interest degrade rapidly during initial isolation when exposed to air, as well as in further

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manipulations. There is always a high risk of contaminating the specimens with extraneous material, mostly from microorganisms. Nevertheless, as technology is refined, the analysis of DNA from f

an array of required skills, a landscape designer's education includes a solid background in botany, soil science, plant pathology, entomology, and horticulture. Coursework in architecture and design software is also required for the completion of the degree. The successful design of a landscape rests on an extensive knowledge of

plant growth requirements, such as light and shade, moisture levels, compatibility of different species, and susceptibility to pathogens and pests. For example, mosses and ferns will thrive in a shaded area where fountains

provide moisture; cacti, on the other hand, would not fare well in that environment. The future growth of the individual plants must be taken into account to avoid crowding and competition for light and nutrients. The appearance of the space over time is also of concern. Shapes, colors, and biology must be balanced for a well-maintained and sustainable green space. Art, architecture, and biology blend in a beautifully designed and

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14.3 ? Seed Plants: Gymnosperms 333

implemented landscape.

FIGURE 14.18 This campus garden was designed by students in the horticulture and landscaping department of the college. (credit: Myriam Feldman)

14.3 Seed Plants: Gymnosperms

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Discuss the type of seeds produced by gymnosperms, as well as other characteristics of gymnosperms
- ? List the four groups of modern-day gymnosperms and provide examples of each

The first plants to colonize land were most likely closely related to modern-day mosses (bryophytes) and are thought to have appeared about 500 million years ago. They were followed by liverworts (also bryophytes) and primitive vascular plants, the pterophytes, from which modern ferns are derived. The life cycle of bryophytes and

pterophytes is characterized by the alternation of generations. The completion of the life cycle requires water, as the

male gametes must swim to the female gametes. The male gametophyte releases sperm, which must swim?propelled by their flagella?to reach and fertilize the female gamete or egg. After fertilization, the zygote matures and grows into a sporophyte, which in turn will form sporangia, or "spore vessels," in which mother cells

undergo meiosis and produce haploid spores. The release of spores in a suitable environment will lead to germination and a new generation of gametophytes.

The Evolution of Seed Plants

In seed plants, the evolutionary trend led to a dominant sporophyte generation, in which the larger and more ecologically significant generation for a species is the diploid plant. At the same time, the trend led to a reduction in

the size of the gametophyte, from a conspicuous structure to a microscopic cluster of cells enclosed in the tissues of

the sporophyte. Lower vascular plants, such as club mosses and ferns, are mostly homosporous (produce only one

type of spore). In contrast, all seed plants, or spermatophytes, a

Animal evolution began in the ocean over 600 million years ago, with tiny creatures that probably do not resemble any living organism today. Since then, animals have evolved into a highly diverse kingdom. Although over one million currently living species of animals have been identified, scientists are continually discovering more species. The number of described living animal species is estimated to be about 1.4 million,¹ and there may be as many as 6.8 million.

Understanding and classifying the variety of living species helps us to better understand how to conserve and benefit from this diversity. The animal classification system characterizes animals

1 ?Number of Living Species in Australia and the World,? A.D. Chapman, Australia Biodiversity Information Services, last modified August 26, 2010, <http://www.environment.gov.au/biodiversity/abrs/publications/other/species-numbers/2009/03-exec-summary.html>.

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based on their anatomy, features of embryological development, and genetic makeup. Scientists are faced with the task of classifying animals within a system of taxonomy that reflects their evolutionary history. Additionally, they must identify traits that are common to all animals as well as traits that can be used to distinguish among related groups of animals. However, animals vary in the complexity of their organization and exhibit a huge diversity of body forms, so the classification scheme is constantly changing as new information about species is learned.

15.1 Features of the Animal Kingdom

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? List the features that distinguish the animal kingdom from other kingdoms
- ? Explain the processes of animal reproduction and embryonic development
- ? Describe the hierarchy of basic animal classification
- ? Compare and contrast the embryonic development of protostomes and deuterostomes

Even though members of the animal kingdom are incredibly diverse, animals share common features that distinguish them from organisms in other kingdoms. All animals are eukaryotic, multicellular organisms, and almost all animals have specialized tissues. Most animals are motile, at least during certain life stages. Animals require a source of food to grow and develop. All animals are heterotrophic, ingesting living or dead organic matter. This form of obtaining energy distinguishes them from autotrophic organisms, such as most plants, which make their own nutrients through photosynthesis and from fungi that digest their food externally. Animals may be carnivores, herbivores, omnivores, or parasites (Figure 15.2). Most animals reproduce sexually: The offspring pass through a series of developmental stages that establish a determined body plan, unlike plants, for example, in which the exact shape of the body is indeterminate. The body plan refers to the shape of an animal.

FIGURE 15.2 All animals that derive energy from food are heterotrophs. The (a) black bear is a

re shared in groups of related animal species, and these events are one of the main ways scientists classify high-level groups of animals. During development, animal cells specialize and form tissues, determining their future morphology and physiology. In many animals, such as mammals, the young resemble the adult.

Other

animals, such as some insects and amphibians, undergo complete metamorphosis in which individuals enter one or

more larval stages. For these animals, the young and the adult have different diets and sometimes habitats. In other

species, a process of incomplete metamorphosis occurs in which the young somewhat resemble the adults and go

through a series of stages separated by molts (shedding of the skin) until they reach the final adult form.

Asexual Reproduction

Asexual reproduction, unlike sexual reproduction, produces offspring genetically identical to each other and to the

parent. A number of animal species?especially those without backbones, but even some fish, amphibians, and reptiles?are capable of asexual reproduction. Asexual reproduction, except for occasional identical twinning, is absent in birds and mammals. The most common forms of asexual reproduction for stationary aquatic animals include budding and fragmentation, in which part of a parent individual can separate and grow into a new individual.

In contrast, a form of asexual reproduction found in certain invertebrates and rare vertebrates is called parthenogenesis (or ?virgin beginning?), in which unfertilized eggs develop into new offspring.

Classification Features of Animals

Animals are classified according to morphological and developmental characteristics, such as a body plan. With the

exception of sponges, the animal body plan is symmetrical. This means that their distribution of body parts is balanced along an axis. Additional characteristics that contribute to animal classification include the number of tissue layers formed during development, the presence or absence of an internal body cavity, and other features of

embryological development.

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VISUAL CONNECTION

FIGURE 15.3 The phylogenetic tree of animals is based on morphological, fossil, and genetic evidence.

Which of the following statements is false?

- a. Eumetazoa have specialized tissues and Parazoa do not.
- b. Both acoelomates and pseudocoelomates have a body cavity.
- c. Chordates are more closely related to echinoderms than to rotifers according to the figure.
- d. Some animals have radial symmetry, and some animals have bilateral symmetry.

Body Symmetry

Animals may be asymmetrical, radial, or bilateral in form (Figure 15.4). Asymmetrical animals are animals with no

pattern or symmetry; an example of an asymmetrical animal is a sponge (Figure 15.4a). An organism with radial symmetry (Figure 15.4b) has a longitudinal (up-and-down) orientation: Any plane cut along this up?down axis produces roughly mirror-image halves. An example of an organism with radial symmetry is a

ve a layer of longitudinal muscle. The mesodermal tissues include support cells and secretory cells that secrete mucus and other materials to the surface. The flatworms are acoelomate, so their bodies contain no cavities or spaces between the outer surface and the inner digestive tract.

Physiological Processes of Flatworms

Free-living species of flatworms are predators or scavengers, whereas parasitic forms feed from the tissues of their hosts. Most flatworms have an incomplete digestive system with an opening, the "mouth," that is also used to expel digestive system wastes. Some species also have an anal opening. The gut may be a simple sac or highly branched.

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Digestion is extracellular, with enzymes secreted into the space by cells lining the tract, and digested materials taken into the same cells by phagocytosis. One group, the cestodes, does not have a digestive system, because their parasitic lifestyle and the environment in which they live (suspended within the digestive cavity of their host) allows them to absorb nutrients directly across their body wall. Flatworms have an excretory system with a network of tubules throughout the body that open to the environment and nearby flame cells, whose cilia beat to direct waste fluids concentrated in the tubules out of the body. The system is responsible for regulation of dissolved salts and excretion of nitrogenous wastes. The nervous system consists of a pair of nerve cords running the length of the body with connections between them and a large ganglion or concentration of nerve cells at the anterior end of the worm; here, there may also be a concentration of photosensory and chemosensory cells (Figure 15.15).

FIGURE 15.15 This planarian is a free-living flatworm that has an incomplete digestive system, an excretory system with a network of tubules throughout the body, and a nervous system made up of nerve cords running the length of the body with a concentration of nerves and photosensory and chemosensory cells at the anterior end.

Since there is no circulatory or respiratory system, gas and nutrient exchange is dependent on diffusion and intercellular junctions. This necessarily limits the thickness of the body in these organisms, constraining them to be "flat" worms. Most flatworm species are monoecious (hermaphroditic, possessing both sets of sex organs), and fertilization is typically internal. Asexual reproduction is common in some groups in which an entire organism can be regenerated from just a part of itself.

Diversity of Flatworms

Flatworms are traditionally divided into four classes: Turbellaria, Monogenea, Trematoda, and Cestoda (Figure 15.16). The turbellarians include mainly free-living marine species, although some species live in freshwater or moist terrestrial environments. The simple planarians found in freshwater ponds and aquaria are examples. The epidermal layer of the underside of turbellarians is ciliated, and this helps them move. Som

e turbellarians are capable of remarkable feats of regeneration in which they may regrow the body, even from a small fragment.

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15.3 ? Flatworms, Nematodes, and Arthropods 365

FIGURE 15.16 Phylum Platyhelminthes is divided into four classes: (a) Bedford's Flatworm (*Pseudobiceros bedfordi*) and the (b) planarian belong to class Turbellaria; (c) the Trematoda class includes about 20,000 species, most of which are parasitic; (d) class Cestoda includes tapeworms such as this *Taenia saginata*; and the parasitic class Monogenea (not shown). (credit a: modification of work by Jan Derk; credit c: modification of work by ?Sahaquiel9102?/Wikimedia Commons; credit d: modification of work by CDC)

The monogeneans are external parasites mostly of fish with life cycles consisting of a free-swimming larva that attaches to a fish to begin transformation to the parasitic adult form. They have only one host during their life, typically of just one species. The worms may produce enzymes that digest the host tissues or graze on surface mucus and skin particles. Most monogeneans are hermaphroditic, but the sperm develop first, and it is typical for them to mate between individuals and not to self-fertilize.

The trematodes, or flukes, are internal parasites of mollusks and many other groups, including humans. Trematodes have complex life cycles that involve a primary host in which sexual reproduction occurs and one or more secondary hosts in which asexual reproduction occurs. The primary host is almost always a mollusk. Trematodes are responsible for serious human diseases including schistosomiasis, caused by a blood fluke (*Schistosoma*). The disease infects an estimated 200 million people in the tropics and leads to organ damage and chronic symptoms including fatigue. Infection occurs when a human enters the water, and a larva, released from the primary snail host, locates and penetrates the skin. The parasite infects various organs in the body and feeds on red blood cells before reproducing. Many of the eggs are released in feces and find their way into a waterway where they are able to reinfect the primary snail host.

The cestodes, or tapeworms, are also internal parasites, mainly of vertebrates. Tapeworms live in the intestinal tract of the primary host and remain fixed using a sucker on the anterior end, or scolex, of the tapeworm body. The remaining body of the tapeworm is made up of a long series of units called proglottids, each of which may contain an excretory system with flame cells, but will contain reproductive structures, both male and female. Tapeworms do not have a digestive system, they absorb nutrients from the food matter passing through in the host's intestine. Proglottids are produced at the scolex and are pushed to the end of the tapeworm as new proglottids form, at which point, they are "mature" and all structures except fertilized eggs have degenerated. Most reproduction occurs by cross-fertilization. The proglot

sexes), or may reproduce asexually by parthenogenesis. *Caenorhabditis elegans* is nearly unique among animals in having both self-fertilizing hermaphrodites and a male sex that can mate with the hermaphrodite.

Arthropoda

The name "arthropoda" means "jointed legs," which aptly describes each of the enormous number of species belonging to this phylum. Arthropoda dominate the animal kingdom with an estimated 85 percent of known species, with many still undiscovered or undescribed. The principal characteristics of all the animals in this phylum

are functional segmentation of the body and the presence of jointed appendages (Figure 15.18). As members of Ecdysozoa, arthropods also have an exoskeleton made principally of chitin. Arthropoda is the largest phylum in the

animal world in terms of numbers of species, and insects form the single largest group within this phylum. Arthropods are true coelomate animals and exhibit protostomic development.

FIGURE 15.18 Trilobites, like the one in this fossil, are an extinct group of arthropods. (credit: Kevin Walsh)

Physiological Processes of Arthropods

A unique feature of arthropods is the presence of a segmented body with fusion of certain sets of segments to give

rise to functional segments. Fused segments may form a head, thorax, and abdomen, or a cephalothorax and abdomen, or a head and trunk. The coelom takes the form of a hemocoel (or blood cavity). The open circulatory system, in which blood bathes the internal organs rather than circulating in vessels, is regulated by a two-chambered heart. Respiratory systems vary, depending on the group of arthropod: Insects and myriapods use a series of tubes (tracheae) that branch throughout the body, open to the outside through openings called spiracles,

and perform gas exchange directly between the cells and air in the tracheae. Aquatic crustaceans use gills, arachnids employ "book lungs," and aquatic chelicerates use "book gills." The book lungs of arachnids are internal

stacks of alternating air pockets and hemocoel tissue shaped like the pages of a book. The book gills of crustaceans

are external structures similar to book lungs with stacks of leaf-like structures that exchange gases with the surrounding water (Figure 15.19).

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FIGURE 15.19 The book lungs of (a) arachnids are made up of alternating air pockets and hemocoel tissue shaped like a stack of books.

The book gills of (b) crustaceans are similar to book lungs but are external so that gas exchange can occur with the surrounding water.

(credit a: modification of work by Ryan Wilson based on original work by John Henry Comstock; credit b: modification of work by Angel Schatz)

Arthropod Diversity

Phylum Arthropoda includes animals that have been successful in colonizing terrestrial, aquatic, and aerial habitats.

The phylum is further classified into five subphyla: Trilobitomorpha (trilobites), Hexapoda (insects and relatives), Myriapoda (millipedes, centipedes, and relati

he Crocodilia (?small lizard?) arose approximately 84 million years ago, and living species include alligators, crocodiles, and caimans. Crocodilians (Figure 15.43a) live throughout the tropics of Africa, South America, the southeastern United States, Asia, and Australia. They are found in freshwater habitats, such as rivers and lakes, and spend most of their time in water. Some species are able to move on land due to their semi-erect posture.

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FIGURE 15.43 (a) Crocodilians, such as this Siamese crocodile, provide parental care for their offspring. (b) This Jackson?s chameleon blends in with its surroundings. (c) The garter snake belongs to the genus *Thamnophis*, the most widely distributed reptile genus in North America. (d) The African spurred tortoise lives at the southern edge of the Sahara Desert. It is the third largest tortoise in the world. (credit a: modification of work by Keshav Mukund Kandhadai; credit c: modification of work by Steve Jurvetson; credit d: modification of work by Jim Bowen)

The Sphenodontia (?wedge tooth?) arose in the Mesozoic Era and includes only one living genus, *Tuatara*, with two species that are found in New Zealand. There are many fossil species extending back to the Triassic period (250?200 million years ago). Although the tuataras resemble lizards, they are anatomically distinct and share characteristics that are found in birds and turtles.

Squamata (?scaly?) arose in the late Permian; living species include lizards and snakes, which are the largest extant clade of reptiles (Figure 15.43b). Lizards differ from snakes by having four limbs, eyelids, and external ears, which are lacking in snakes. Lizard species range in size from chameleons and geckos that are a few centimeters in length to the Komodo dragon, which is about 3 meters in length.

Snakes are thought to have descended from either burrowing lizards or aquatic lizards over 100 million years ago (Figure 15.43c). Snakes comprise about 3,000 species and are found on every continent except Antarctica. They range in size from 10 centimeter-long thread snakes to 7.5 meter-long pythons and anacondas. All snakes are carnivorous and eat small animals, birds, eggs, fish, and insects.

Turtles are members of the clade Testudines (?having a shell?) (Figure 15.43d). Turtles are characterized by a bony or cartilaginous shell, made up of the carapace on the back and the plastron on the ventral surface, which develops from the ribs. Turtles arose approximately 200 million years ago, predating crocodiles, lizards, and snakes. Turtles lay eggs on land, although many species live in or near water. Turtles range in size from the speckled padloper tortoise at 8 centimeters (3.1 inches) to the leatherback sea turtle at 200 centimeters (over 6 feet). The term ?turtle? is sometimes used to describe only those species of Testudines that live in the sea, with the terms ?tortoise? and ?terrapin? used to refer to species that live on land and in

e a lipid mixture called

sebum that is secreted onto the hair and skin for water resistance and lubrication. Sebaceous glands are located

over most of the body. Sudoriferous glands produce sweat and scent, which function in thermoregulation and communication, respectively. Mammary glands produce milk that is used to feed newborns. While male and female

monotremes and eutherians possess mammary glands, some male marsupials do not.

The skeletal system of mammals possesses unique features that differentiate them from other vertebrates.

Most

mammals have heterodont teeth, meaning they have different types and shapes of teeth that allow them to feed on

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different kinds of foods. These different types of teeth include the incisors, the canines, premolars, and molars.

The

first two types are for cutting and tearing, whereas the latter two types are for crushing and grinding. Different groups have different proportions of each type, depending on their diet. Most mammals are also diphyodonts, meaning they have two sets of teeth in their lifetime: deciduous or "baby" teeth, and permanent teeth. In other vertebrates, the teeth can be replaced throughout life.

Modern mammals are divided into three broad groups: monotremes, marsupials, and eutherians (or placental mammals). The eutherians, or placental mammals, and the marsupials collectively are called therian mammals, whereas monotremes are called prototherians.

There are three living species of monotremes: the platypus and two species of echidnas, or spiny anteaters (Figure

15.45). The platypus and one species of echidna are found in Australia, whereas the other species of echidna is found in New Guinea. Monotremes are unique among mammals, as they lay leathery eggs, similar to those of reptiles, rather than giving birth to live young. However, the eggs are retained within the mother's reproductive tract

until they are almost ready to hatch. Once the young hatch, the female begins to secrete milk from pores in a ridge

of mammary tissue along the ventral side of her body. Like other mammals, monotremes are endothermic but regulate body temperatures somewhat lower (90 °F, 32 °C) than placental mammals do (98 °F, 37 °C). Like reptiles,

monotremes have one posterior opening for urinary, fecal, and reproductive products, rather than three separate

openings like placental mammals do. Adult monotremes lack teeth.

FIGURE 15.45 The platypus (left), a monotreme, possesses a leathery beak and lays eggs rather than giving birth to live young. An echidna, another monotreme, is shown in the right photo. (credit "echidna": modification of work by Barry Thomas)

Marsupials are found primarily in Australia and nearby islands, although about 100 species of opossums and a few

species of two other families are found in the Americas. Australian marsupials number over 230 species and include

the kangaroo, koala, bandicoot, and Tasmanian devil (Figure 15.46). Most species of marsupials possess

ke, sucking mouth pharyngeal slit an opening in the pharynx

lancelet a member of Cephalochordata; named for its pneumatic bone an air-filled bone

blade-like shape polyp the stalk-like, sessile life form of a cnidarians

lateral line the sense organ that runs the length of a with mouth and tentacles facing upward, usually

fish's body, used to detect vibration in the water sessile but may be able to glide along a surface

Lophotrochozoa a clade of invertebrate organisms Porifera a phylum of animals with no true tissues, but

that is a sister group to the Ecdysozoa a porous body with a rudimentary endoskeleton

madreporite a pore for regulating entry and exit of post-anal tail a muscular, posterior elongation of the

water into the water vascular system body extending beyond the anus in chordates

mammal one of the groups of endothermic Primates includes lemurs, tarsiers, monkeys, apes,

vertebrates that possess hair and mammary glands and humans

mammary gland in female mammals, a gland that prosimians a group of primates that includes bush

produces milk for newborns babies of Africa, lemurs of Madagascar, and lorises,

mantle a specialized epidermis that encloses all pottos, and tarsiers of southeast Asia

visceral organs and secretes shells in mollusks protostome describing an animal in which the mouth

marsupial one of the groups of mammals that develops first during embryogenesis and a second

includes the kangaroo, koala, bandicoot, Tasmanian opening developing into the anus

devil, and several other species; young develop pseudocoelomate an animal with a coelom that is

within a pouch not completely lined with tissues derived from the

medusa a free-floating cnidarian body plan with a mesoderm as in eucoelomate animals

mouth on the underside and tentacles hanging radial symmetry a type of symmetry with multiple

down from a bell planes of symmetry all cross at an axis through the

mesoglea the non-living, gel-like matrix present in center of the organism

between ectoderm and endoderm in cnidarians radula a tongue-like scraping organ with chitinous

mesohyl the collagen-like gel containing suspended ornamentation found in most mollusks

cells that perform various functions in sponges salamander a tailed amphibian that belongs to the

metamerism having a series of body structures that clade Urodela

are similar internally and externally, such as Sarcopterygii lobe-finned fishes

segments sebaceous gland in mammals, a skin gland that

Mollusca a phylum of protostomes with soft bodies produce a lipid mixture called sebum

and no segmentation Sphenodontia the reptilian clade that includes the

monoecious having both sexes in one body, tuataras

hermaphroditic spicule a short sliver or spike-like structure, in

monotreme an egg-laying mammal sponges, they are formed of silicon dioxide, calcium

Myxini hagfishes carbonate, or protein, and are found in the mesohyl

nacre a calcareous secretion produced by bivalve spiracle a respiratory openings in insects that all

OW

mollusks to line the inner side of shells as well as to air into the tracheae

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spongocoel the central cavity within the body of trachea in some arthropods, such as insects, a some sponges respiratory tube that conducts air from the spiracles

Squamata the reptilian clade of lizards and snakes to the tissues

stereoscopic vision two overlapping fields of vision triploblast an animal that develops from three germ

from the eyes that produces depth perception layers

sudoriferous gland a gland in mammals that tunicate a sessile chordate that is a member of

produces sweat and scent molecules Urochordata

swim bladder in fishes, a gas filled organ that helps Urochordata the clade composed of the tunicates

to control the buoyancy of the fish Urodela salamanders

tadpole the larval stage of a frog vertebral column a series of separate bones that

Testudines turtles surround the spinal cord in vertebrates

tetrapod a four-footed animal; includes amphibians, water vascular system a system in echinoderms in

reptiles, birds, and mammals which water is the circulatory fluid

Chapter Summary

15.1 Features of the Animal Kingdom cell for delivering toxins to prey and predators.

Cnidarians have separate sexes. They have a life cycle

Animals constitute a diverse kingdom of organisms. that involves morphologically distinct forms?medusoid

Although animals range in complexity from simple sea and polypoid?at various stages in their life cycle.

sponges to human beings, most members share

certain features. Animals are eukaryotic, multicellular, 15.3 Flatworms, Nematodes, and

heterotrophic organisms that ingest their food and Arthropods

usually develop into motile creatures with a fixed body

plan. Most members of the animal kingdom have Flatworms are acoelomate, triploblastic animals. They

differentiated tissues of four main classes?nervous, lack circulatory and respiratory systems, and have a

muscular, connective, and epithelial?that are rudimentary excretory system. The digestive system is

specialized to perform different functions. Most incomplete in most species. There are four traditional

animals reproduce sexually, leading to a classes of flatworms, the largely free-living

developmental sequence that is relatively similar turbellarians, the ectoparasitic monogeneans, and the

across the animal kingdom. endoparasitic trematodes and cestodes. Trematodes

have complex life cycles involving a secondary mollusk

Organisms in the animal kingdom are classified based host and a primary host in which sexual reproduction

on their body morphology and development. True takes place. Cestodes, or tapeworms, infect the

animals are divided into those with radial versus digestive systems of primary vertebrate hosts.

bilateral symmetry. Animals with three germ layers,

called triploblasts, are further characterized by the Nematodes are pseudocoelomate members of the

presence or absence of an internal body cavity called a clade Ecdysozoa. They have a compl

ete digestive

coelom. Animals with a body cavity may be either system and a pseudocoelomic body cavity. This phylum coelomates or pseudocoelomates, depending on which includes free-living as well as parasitic organisms. tissue gives rise to the coelom. Coelomates are further They include dioecious and hermaphroditic species. divided into two groups called protostomes and Nematodes have a poorly developed excretory system. deuterostomes, based on a number of developmental Embryonic development is external and proceeds characteristics. through larval stages separated by molts.

15.2 Sponges and Cnidarians Arthropods represent the most successful phylum of animals on Earth, in terms of number of species as well

Animals included in phylum Porifera are parazoans and as the number of individuals. They are characterized by do not possess true tissues. These organisms show a a segmented body and jointed appendages. In the simple organization. Sponges have multiple cell types basic body plan, a pair of appendages is present per that are geared toward executing various metabolic body segment. Within the phylum, classification is functions. based on mouthparts, number of appendages, and

modifications of appendages. Arthropods bear a
Cnidarians have outer and inner tissue layers

chitinous exoskeleton. Gills, tracheae, and book lungs
sandwiching a noncellular mesoglea. Cnidarians

facilitate respiration. Embryonic development may
possess a well-formed digestive system and carry out

include multiple larval stages.
extracellular digestion. The cnidocyte is a specialized

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15.4 Mollusks and Annelids and some species are parasitic on other fishes.
Gnathostomes include the jawed fishes (cartilaginous

The phylum Mollusca is a large, mainly marine group of and bony fishes) as well as all other tetrapods. invertebrates. Mollusks show a variety of Cartilaginous fishes include sharks, rays, skates, and morphologies. Many mollusks secrete a calcareous ghost sharks. Bony fishes can be further divided into shell for protection, but in other species, the shell is ray-finned and lobe-finned fishes. reduced or absent. Mollusks are protostomes. The dorsal epidermis in mollusks is modified to form the As tetrapods, most amphibians are characterized by mantle, which encloses the mantle cavity and visceral four well-developed limbs, although some species of organs. This cavity is distinct from the coelomic cavity, salamanders and all caecilians are limbless. which the adult animal retains, surrounding the heart. Amphibians have a moist, permeable skin used for Respiration is facilitated by gills known as ctenidia. A cutaneous respiration. Amphibia can be divided into chitinous scraper called the radula is present in most three clades: salamanders (Urodela), frogs (Anura), mollusks. Mollusks are mostly dioecious and are and caecilians (Apoda). The life cycle of amphibians divided into seven classes. consists of two

distinct stages: the larval stage and

metamorphosis to an adult stage.

The phylum Annelida includes worm-like, segmented

animals. Segmentation is both external and internal. The amniotes are distinguished from amphibians by

which is called metamerism. Annelids are the presence of a terrestrially adapted egg protected

protostomes. The presence of chitinous hairs called by amniotic membranes. The amniotes include

chaetae is characteristic of most members. These reptiles, birds, and mammals. A key adaptation that

animals have well-developed nervous and digestive permitted reptiles to live on land was the development

systems. Polychaete annelids have parapodia that of scaly skin. Reptilia includes four living clades:

participate in locomotion and respiration. Suckers are Crocodilia (crocodiles and alligators), Sphenodontia

seen in the order Hirudinea. Breeding systems include (tuataras), Squamata (lizards and snakes), and

separate sexes and hermaphroditism. Testudines (turtles).

15.5 Echinoderms and Chordates Birds are endothermic amniotes. Feathers act as insulation and allow for flight. Birds have pneumatic

Echinoderms are deuterostome marine organisms. This bones that are hollow rather than tissue-filled. Airflow

phylum of animals bear a calcareous endoskeleton through bird lungs travels in one direction. Birds

composed of ossicles covered by a spiny skin. evolved from dinosaurs.

Echinoderms possess a water-based circulatory

system. The madreporite is the point of entry and exit Mammals have hair and mammary glands. Mammalian

for water for the water vascular system. skin includes various secretory glands. Mammals are

endothermic, like birds. There are three groups of

The characteristic features of Chordata are a mammals living today: monotremes, marsupials, and

notochord, a dorsal hollow nerve cord, pharyngeal slits, eutherians. Monotremes are unique among mammals

and a post-anal tail. Chordata contains two clades of as they lay eggs, rather than giving birth to live young.

invertebrates: Urochordata (tunicates) and Eutherian mammals have a complex placenta.

Cephalochordata (lancelets), together with the

vertebrates. Most tunicates live on the ocean floor and There are 16 extant (living) orders of eutherian

are suspension feeders. Lancelets are suspension mammals. Humans are most closely related to

feeders that feed on phytoplankton and other Primates, all of which have adaptations for climbing

microorganisms. trees, although not all species are arboreal. Other

characteristics of primates are brains that are larger

15.6 Vertebrates than those of other mammals, claws that have been

The earliest vertebrates that diverged from the modified into flattened nails, and typically one young

invertebrate chordates were the jawless fishes. per pregnancy, stereoscopic vision, and a trend toward

Hagfishes are eel-like scavengers that feed on dead holding the body upright. Primates are divided into two

invertebrates and other fishes. Lampreys are groups: p

ly are the level of blood glucose, body temperature, blood calcium level. These conditions remain stable because of physiologic processes that result in negative feedback relationships. If the blood glucose or calcium rises, this sends a signal to organs responsible for lowering blood glucose or calcium. The signals that restore the normal levels are examples of negative feedback. When homeostatic mechanisms fail, the results can be unfavorable for the animal. Homeostatic mechanisms keep the body in dynamic equilibrium by constantly adjusting to the changes that the body's systems encounter. Even an animal that is apparently inactive is maintaining this homeostatic equilibrium. Two examples of factors that are regulated homeostatically are temperature and water content. The processes that maintain homeostasis of these two factors are called thermoregulation and osmoregulation.

Homeostasis

The goal of homeostasis is the maintenance of equilibrium around a specific value of some aspect of the body or its cells called a set point. While there are normal fluctuations from the set point, the body's systems will usually attempt to go back to this point. A change in the internal or external environment is called a stimulus and is detected by a receptor; the response of the system is to adjust the activities of the system so the value moves back toward the set point. For instance, if the body becomes too warm, adjustments are made to cool the animal. If glucose levels in the blood rise after a meal, adjustments are made to lower them and to get the nutrient into tissues that need it or to store it for later use.

When a change occurs in an animal's environment, an adjustment must be made so that the internal environment of the body and cells remains stable. The receptor that senses the change in the environment is part of a feedback mechanism. The stimulus—temperature, glucose, or calcium levels—is detected by the receptor. The receptor sends information to a control center, often the brain, which relays appropriate signals to an effector organ that is able to cause an appropriate change, either up or down, depending on the information the sensor was sending.

Thermoregulation

Animals can be divided into two groups: those that maintain a constant body temperature in the face of differing environmental temperatures, and those that have a body temperature that is the same as their environment and thus varies with the environmental temperature. Animals that do not have internal control of their body temperature are called ectotherms. The body temperature of these organisms is generally similar to the temperature of the environment, although the individual organisms may do things that keep their bodies slightly below or above the environmental temperature. This can include burrowing underground on a hot day or resting in the sunlight on a cold day. The ectotherms have been called cold-blooded, a term that may not apply to an animal in the desert with a very

are

centered in the hypothalamus of the advanced animal brain. The hypothalamus maintains the set point for body temperature through reflexes that cause vasodilation or vasoconstriction and shivering or sweating. The sympathetic nervous system under control of the hypothalamus directs the responses that effect the changes in temperature loss or gain that return the body to the set point. The set point may be adjusted in some instances. During an infection, compounds called pyrogens are produced and circulate to the hypothalamus resetting the thermostat to a higher value. This allows the body's temperature to increase to a new homeostatic equilibrium point

in what is commonly called a fever. The increase in body heat makes the body less optimal for bacterial growth and

increases the activities of cells so they are better able to fight the infection.

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VISUAL CONNECTION

FIGURE 16.2 The body is able to regulate temperature in response to signals from the nervous system.

When bacteria are destroyed by leukocytes, pyrogens are released into the blood. Pyrogens reset the body's thermostat to a higher temperature, resulting in fever. How might pyrogens cause the body temperature to rise?

Osmoregulation

Osmoregulation is the process of maintaining salt and water balance (osmotic balance) across membranes within

the body. The fluids inside and surrounding cells are composed of water, electrolytes, and nonelectrolytes. An electrolyte is a compound that dissociates into ions when dissolved in water. A nonelectrolyte, in contrast, does not

dissociate into ions in water. The body's fluids include blood plasma, fluid that exists within cells, and the interstitial fluid that exists in the spaces between cells and tissues of the body. The membranes of the body (both

the membranes around cells and the "membranes" made of cells lining body cavities) are semipermeable membranes. Semipermeable membranes are permeable to certain types of solutes and to water, but typically cell

membranes are impermeable to solutes.

The body does not exist in isolation. There is a constant input of water and electrolytes into the system. Excess water, electrolytes, and wastes are transported to the kidneys and excreted, helping to maintain osmotic balance.

Insufficient fluid intake results in fluid conservation by the kidneys. Biological systems constantly interact and exchange water and nutrients with the environment by way of consumption of food and water and through excretion

in the form of sweat, urine, and feces. Without a mechanism to regulate osmotic pressure, or when a disease damages this mechanism, there is a tendency to accumulate toxic waste and water, which can have dire consequences.

Mammalian systems have evolved to regulate not only the overall osmotic pressure across membranes, but also

specific concentrations of important electrolytes in the three major fluid compartments: blood plasma, interstitial fluid, and intracellular fluid.

Since osmotic pressure is regulated by the movement of water across membranes, the volume of the fluid compartments can also change temporarily. Since blood plasma is one of the fluid components, osmotic pressures have a direct bearing on blood pressure.

Excretory System

The human excretory system functions to remove waste from the body through the skin as sweat, the lungs in the form of exhaled carbon dioxide, and through the urinary system in the form of urine. All three of these systems participate in osmoregulation and waste removal. Here we focus on the urinary system, which is comprised of the

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paired kidneys, the ureter, urinary bladder and urethra (Figure 16.3). The kidneys are a pair of bean-shaped structures that are located just below the liver in the body cavity. Each of the kidneys contains more than a million tiny units called nephrons that filter blood containing the metabolic wastes from cells. All the blood in the human body is filtered about 60 times a day by the kidneys. The nephrons remove wastes, concentrate them, and form urine that is collected in the bladder.

Internally, the kidney has three regions—an outer cortex, a medulla in the middle, and the renal pelvis, which is the expanded end of the ureter. The renal cortex contains the nephrons—the functional unit of the kidney. The renal pelvis collects the urine and leads to the ureter on the outside of the kidney. The ureters are urine-bearing tubes that exit the kidney and empty into the urinary bladder.

FIGURE 16.3 The human excretory system is made up of the kidneys, ureter, urinary bladder, and urethra. The kidneys filter blood and form urine, which is stored in the bladder until it is eliminated through the urethra. On the right, the internal structure of the kidney is shown.
(credit: modification of work by NCI, NIH)

Blood enters each kidney from the aorta, the main artery supplying the body below the heart, through a renal artery. It is distributed in smaller vessels until it reaches each nephron in capillaries. Within the nephron the blood comes in intimate contact with the waste-collecting tubules in a structure called the glomerulus. Water and many solutes present in the blood, including ions of sodium, calcium, magnesium, and others; as well as wastes and valuable substances such as amino acids, glucose and vitamins, leave the blood and enter the tubule system of the nephron. As materials pass through the tubule much of the water, required ions, and useful compounds are reabsorbed back into the capillaries that surround the tubules leaving the wastes behind. Some of this reabsorption requires active transport and consumes ATP. Some wastes, including ions and some drugs remaining in the blood, diffuse out of the capillaries into the interstitial fluid and are taken up by the tubule cells. These wastes are then actively secreted into the tubules. The blood then collects in larger and

and an involuntary event. How often a breath is taken and how much air is inhaled or exhaled is regulated by the respiratory center in the brain in response to signals it receives about the carbon dioxide content of the blood. However, it is possible to override this automatic regulation for activities such as speaking, singing and swimming under water.

During inhalation the diaphragm descends creating a negative pressure around the lungs and they begin to inflate, drawing in air from outside the body. The air enters the body through the nasal cavity located just inside the nose (Figure 16.9). As the air passes through the nasal cavity, the air is warmed to body temperature and humidified by moisture from mucous membranes. These processes help equilibrate the air to the body conditions, reducing any damage that cold, dry air can cause. Particulate matter that is floating in the air is removed in the nasal passages by hairs, mucus, and cilia. Air is also chemically sampled by the sense of smell.

From the nasal cavity, air passes through the pharynx (throat) and the larynx (voice box) as it makes its way to the trachea (Figure 16.9). The main function of the trachea is to funnel the inhaled air to the lungs and the exhaled air back out of the body. The human trachea is a cylinder, about 25 to 30 cm (9.8?11.8 in) long, which sits in front of the esophagus and extends from the pharynx into the chest cavity to the lungs. It is made of incomplete rings of cartilage and smooth muscle. The cartilage provides strength and support to the trachea to keep the passage open. The trachea is lined with cells that have cilia and secrete mucus. The mucus catches particles that have been inhaled, and the cilia move the particles toward the pharynx.

The end of the trachea divides into two bronchi that enter the right and left lung. Air enters the lungs through the primary bronchi. The primary bronchus divides, creating smaller and smaller diameter bronchi until the passages are under 1 mm (.03 in) in diameter when they are called bronchioles as they split and spread through the lung. Like the trachea, the bronchus and bronchioles are made of cartilage and smooth muscle. Bronchi are innervated by nerves of both the parasympathetic and sympathetic nervous systems that control muscle contraction (parasympathetic) or relaxation (sympathetic) in the bronchi and bronchioles, depending on the nervous system's cues. The final bronchioles are the respiratory bronchioles. Alveolar ducts are attached to the end of each respiratory bronchiole. At the end of each duct are alveolar sacs, each containing 20 to 30 alveoli. Gas exchange occurs only in the alveoli. The alveoli are thin-walled and look like tiny bubbles within the sacs. The alveoli are in direct contact with capillaries of the circulatory system. Such intimate contact ensures that oxygen will diffuse from the alveoli into the blood. In addition, carbon dioxide will diffuse from the blood into the alveoli to be exhal

sease), and disorders of the adrenal glands (Cushing's disease and Addison's disease).

Endocrinologists are required to assess patients and diagnose endocrine disorders through extensive use of laboratory tests. Many endocrine diseases are diagnosed using tests that stimulate or suppress endocrine organ functioning. Blood samples are then drawn to determine the effect of stimulating or suppressing an endocrine organ on the production of hormones. For example, to diagnose diabetes mellitus, patients are required to fast for 12 to 24 hours. They are then given a sugary drink, which stimulates the pancreas to produce insulin to decrease blood-glucose levels. A blood sample is taken one to two hours after the sugar drink is consumed. If the pancreas is functioning properly, the blood-glucose level will be within a normal range. Another example is the A1C test, which can be performed during blood screening. The A1C test measures average blood-glucose levels over the past two to three months. The A1C test is an indicator of how well blood glucose is being managed over a long time.

Once a disease such as diabetes has been diagnosed, endocrinologists can prescribe lifestyle changes and medications to treat the disease. Some cases of diabetes mellitus can be managed by exercise, weight loss, and a healthy diet; in other cases, medications may be required to enhance insulin's production or effect. If the disease cannot be controlled by these means, the endocrinologist may prescribe insulin injections.

In addition to clinical practice, endocrinologists may also be involved in primary research and development activities. For example, ongoing islet transplant research is investigating how healthy pancreas islet cells may be transplanted into diabetic patients. Successful islet transplants may allow patients to stop taking insulin injections.

How Hormones Work

Hormones cause changes in target cells by binding to specific cell-surface or intracellular hormone receptors, molecules embedded in the cell membrane or floating in the cytoplasm with a binding site that matches a binding site on the hormone molecule. In this way, even though hormones circulate throughout the body and come into contact with many different cell types, they only affect cells that possess the necessary receptors. Receptors for a specific hormone may be found on or in many different cells or may be limited to a small number of specialized cells. For example, thyroid hormones act on many different tissue types, stimulating metabolic activity throughout the body. Cells can have many receptors for the same hormone but often also possess receptors for different types of hormones. The number of receptors that respond to a hormone determines the cell's sensitivity to that hormone,

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and the resulting cellular response. Additionally, the number of receptors available to respond to a hormone can change over time, resulting in increased or decreased cell sensitivity

y. In up-regulation, the number of receptors increases in response to rising hormone levels, making the cell more sensitive to the hormone and allowing for more cellular activity. When the number of receptors decreases in response to rising hormone levels, called down-regulation, cellular activity is reduced.

Endocrine Glands

The endocrine glands secrete hormones into the surrounding interstitial fluid; those hormones then diffuse into blood and are carried to various organs and tissues within the body. The endocrine glands include the pituitary, thyroid, parathyroid, adrenal glands, gonads, pineal, and pancreas.

The pituitary gland, sometimes called the hypophysis, is located at the base of the brain (Figure 16.13a). It is attached to the hypothalamus. The posterior lobe stores and releases oxytocin and antidiuretic hormone produced by the hypothalamus. The anterior lobe responds to hormones produced by the hypothalamus by producing its own hormones, most of which regulate other hormone-producing glands.

FIGURE 16.13 (a) The pituitary gland sits at the base of the brain, just above the brain stem. (b) The parathyroid glands are located on the posterior of the thyroid gland. (c) The adrenal glands are on top of the kidneys. d) The pancreas is found between the stomach and the small intestine. (credit: modification of work by NCI, NIH)

The anterior pituitary produces six hormones: growth hormone, prolactin, thyroid-stimulating hormone, adrenocorticotropic hormone, follicle-stimulating hormone, and luteinizing hormone. Growth hormone stimulates cellular activities like protein synthesis that promote growth. Prolactin stimulates the production of milk by the

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mammary glands. The other hormones produced by the anterior pituitary regulate the production of hormones by other endocrine tissues (Table 16.1). The posterior pituitary is significantly different in structure from the anterior pituitary. It is a part of the brain, extending down from the hypothalamus, and contains mostly nerve fibers that extend from the hypothalamus to the posterior pituitary.

The thyroid gland is located in the neck, just below the larynx and in front of the trachea (Figure 16.13b). It is a butterfly-shaped gland with two lobes that are connected. The thyroid follicle cells synthesize the hormone thyroxine, which is also known as T₄ because it contains four atoms of iodine, and triiodothyronine, also known as T₃

because it contains three atoms of iodine. T₃ and T₄ are released by the thyroid in response to thyroid-stimulating hormone produced by the anterior pituitary, and both T₃ and T₄ have the effect of stimulating metabolic activity in the body and increasing energy use. A third hormone, calcitonin, is also produced by the thyroid. Calcitonin is released in response to rising calcium ion concentrations in the blood and has the effect of reducing those levels.

Most people have four parathyroid glands;

however, the number can vary from two to six. These glands are located on the posterior surface of the thyroid gland (Figure 16.13b).

The parathyroid glands produce parathyroid hormone. Parathyroid hormone increases blood calcium concentrations when calcium ion levels fall below normal.

The adrenal glands are located on top of each kidney (Figure 16.13c). The adrenal glands consist of an outer adrenal cortex and an inner adrenal medulla. These regions secrete different hormones.

The adrenal cortex produces mineralocorticoids, glucocorticoids, and androgens. The main mineralocorticoid is aldosterone, which regulates the concentration of ions in urine, sweat, and saliva. Aldosterone release from the adrenal cortex is stimulated by a decrease in blood concentrations of sodium ions, blood volume, or blood pressure, or by an increase in blood potassium levels. The glucocorticoids maintain proper blood-glucose levels between meals. They also control a response to stress by increasing glucose synthesis from fats and proteins and interact with epinephrine to cause vasoconstriction.

The testes and the adrenal cortex both secrete androgens, and the testes produce a larger amount that mostly drives development in a testicular reproductive system. When androgens are produced, some of them are later converted to estrogens. Androgens and estrogens resemble one another in chemical structure and originate from the same molecule. Minute amounts of estrogen occur in all humans through actions in adipose, brain, skin, and bone, which convert testosterone to estrogen. The small amount of androgens produced in the adrenal cortex may supplement hormones released from the gonads. The adrenal medulla contains two types of secretory cells: one that produces epinephrine (adrenaline) and another that produces norepinephrine (noradrenaline). Epinephrine and norepinephrine cause immediate, short-term changes in response to stressors, inducing the so-called fight-or-flight response. The responses include increased heart rate, breathing rate, cardiac muscle contractions, and blood-glucose levels. They also accelerate the breakdown of glucose in skeletal muscles and stored fats in adipose tissue, and redirect blood flow toward skeletal muscles and away from skin and viscera. The release of epinephrine and norepinephrine is stimulated by neural impulses from the sympathetic nervous system that originate from the hypothalamus.

The pancreas is an elongated organ located between the stomach and the proximal portion of the small intestine (Figure 16.13d). It contains both exocrine cells that excrete digestive enzymes and endocrine cells that release hormones.

The endocrine cells of the pancreas form clusters called pancreatic islets or the islets of Langerhans. Among the cell types in each pancreatic islet are the alpha cells, which produce the hormone glucagon, and the beta cells, which produce the hormone insulin. These hormones regulate blood-glucose levels. Alpha cells r

packaged into orderly units. The orderly arrangement of the proteins in each unit, shown as red and blue lines, gives the cell its striated appearance.

The striated appearance of skeletal muscle tissue is a result of repeating bands of the proteins actin and myosin that occur along the length of myofibrils.

Myofibrils are composed of smaller structures called myofilaments. There are two main types of myofilaments: thick filaments and thin filaments. Thick filaments are composed of the protein myosin. The primary component of thin filaments is the protein actin.

The thick and thin filaments alternate with each other in a structure called a sarcomere. The sarcomere is the unit of contraction in a muscle cell. Contraction is stimulated by an electrochemical signal from a nerve cell associated with the muscle fiber. For a muscle cell to contract, the sarcomere must shorten. However, thick and thin filaments do not shorten. Instead, they slide by one another, causing the sarcomere to shorten while the filaments remain the same length. The sliding is accomplished when a molecular extension of myosin, called the myosin head, temporarily binds to an actin filament next to it and through a change in conformation, bends, dragging the two filaments in opposite directions. The myosin head then releases its actin filament, relaxes, and then repeats the process, dragging the two filaments further along each other. The combined activity of many binding sites and repeated movements within the sarcomere causes it to contract. The coordinated contractions of many sarcomeres in a myofibril leads to contraction of the entire muscle cell and ultimately the muscle itself. The movement of the myosin head requires ATP, which provides the energy for the contraction.

LINK TO LEARNING

View this animation (http://openstax.org//skeletal_muscl2) to see how muscle fibers are organized.

16.6 Nervous System

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe the form and function of a neuron
- ? Describe the basic parts and functions of the central nervous system
- ? Describe the basic parts and functions of the peripheral nervous system

As you read this, your nervous system is performing several functions simultaneously. The visual system is processing what is seen on the page; the motor system controls your eye movements and the turn of the pages (or click of the mouse); the prefrontal cortex maintains attention. Even fundamental functions, like breathing and regulation of body temperature, are controlled by the nervous system. The nervous system is one of two systems that exert control over all the organ systems of the body; the other is the endocrine system. The nervous system's control is much more specific and rapid than the hormonal system. It communicates signals through cells and the tiny gaps between them rather than through the circulatory system as in the endocrine system. It uses a combination of chemical and electrochemical signals,

on. If it is strong enough (that is, if the threshold of excitation, a depolarization to around -60mV is reached), then depolarization creates a positive feedback loop: as more Na^+ ions enter the cell, the axon becomes further depolarized, opening even more sodium channels at further distances from the cell body. This will cause voltage dependent Na^+ channels further down the axon to open and more positive ions to enter the cell. In the axon, this 'signal' will become a self-propagating brief reversal of the resting membrane potential called an action potential.

An action potential is an all-or-nothing event; it either happens or it does not. The threshold of excitation must be reached for the neuron to 'fire' an action potential. As sodium ions rush into the cell, depolarization actually reverses the charge across the membrane from -70mV to $+30\text{mV}$. This change in the membrane potential causes voltage-gated K^+ channels to open, and K^+ begins to leave the cell, repolarizing it. At the same time, Na^+ channels inactivate so no more Na^+ enters the cell. K^+ ions continue to leave the cell and the membrane potential returns to the resting potential. At the resting potential, the K^+ channels close and Na^+ channels reset. The depolarization of the membrane proceeds in a wave down the length of the axon. It travels in only one direction because the sodium channels have been inactivated and unavailable until the membrane potential is near the resting potential again; at this point they are reset to closed and can be opened again.

An axon is a tube-like structure that propagates the signal from the cell body to specialized endings called axon terminals. These terminals in turn then synapse with other neurons, muscle, or target organs. When the action potential reaches the axon terminal, this causes the release of neurotransmitter onto the dendrite of another neuron. Neurotransmitters released at axon terminals allow signals to be communicated to these other cells, and the process begins again. Neurons usually have one or two axons, but some neurons do not contain any axons.

Some axons are covered with a special structure called a myelin sheath, which acts as an insulator to keep the electrical signal from dissipating as it travels down the axon. This insulation is important, as the axon from a human motor neuron can be as long as a meter (3.2 ft) from the base of the spine to the toes. The myelin sheath is produced by glial cells. Along the axon there are periodic gaps in the myelin sheath. These gaps are called nodes of Ranvier and are sites where the signal is 'recharged' as it travels along the axon.

It is important to note that a single neuron does not act alone—neuronal communication depends on the connections that neurons make with one another (as well as with other cells, like muscle cells). Dendrites from a single neuron may receive synaptic contact from many other neurons. For example, dendrites from a Purkinje cell in the cerebellum are though

treat severe epilepsy, the corpus callosum is cut instead of removing an entire hemisphere. This causes a condition called split-brain, which gives insights into unique functions of the two hemispheres. For example, when an object is presented to patients' left visual field, they may be unable to verbally name the object (and may claim to not have seen an object at all). This is because the visual input from the left visual field crosses and enters the right hemisphere and cannot then signal to the speech center, which generally is found in the left side of the brain. Remarkably, if a split-brain patient is asked to pick up a specific object out of a group of objects with the left hand, the patient will be able to do so but will still be unable to verbally identify it.

LINK TO LEARNING

Visit the following website (<http://openstax.org//split-brain2>) to learn more about split-brain patients and to play a game where you can model split-brain experiments yourself.

Each hemisphere contains regions called lobes that are involved in different functions. Each hemisphere of the mammalian cerebral cortex can be broken down into four functionally and spatially defined lobes: frontal, parietal, temporal, and occipital (Figure 16.22).

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FIGURE 16.22 The human cerebral cortex includes the frontal, parietal, temporal, and occipital lobes.

The frontal lobe is located at the front of the brain, over the eyes. This lobe contains the olfactory bulb, which processes smells. The frontal lobe also contains the motor cortex, which is important for planning and implementing movement. Areas within the motor cortex map to different muscle groups. Neurons in the frontal lobe also control cognitive functions like maintaining attention, speech, and decision-making. Studies of humans who have damaged their frontal lobes show that parts of this area are involved in personality, socialization, and assessing risk. The parietal lobe is located at the top of the brain. Neurons in the parietal lobe are involved in speech and also reading. Two of the parietal lobe's main functions are processing somatosensation—touch sensations like pressure, pain, heat, cold—and processing proprioception—the sense of how parts of the body are oriented in space. The parietal lobe contains a somatosensory map of the body similar to the motor cortex. The occipital lobe is located at the back of the brain. It is primarily involved in vision—seeing, recognizing, and identifying the visual world. The temporal lobe is located at the base of the brain and is primarily involved in processing and interpreting sounds. It also contains the hippocampus (named from the Greek for “seahorse,” which it resembles in shape) a structure that processes memory formation. The role of the hippocampus in memory was partially determined by studying one famous epileptic patient, HM, who had both sides of his hippocampus removed in an attempt to cure his epilepsy. His seizures went away, but

meninges (singular: meninx) the membranes that diffuse into blood and are carried to various organs cover and protect the central nervous system and tissues within the body mineral an inorganic, elemental molecule that carries

endotherm an organism that relies primarily on out important roles in the body internal heat sources to maintain its body myelin sheath a cellular extension containing a fatty temperature substance produced by glia that surrounds and

esophagus a tubular organ that connects the mouth insulates axons to the stomach myofibril the long cylindrical structures that lie

essential nutrient a nutrient that cannot be parallel to the muscle fiber synthesized by the body; it must be obtained from myofilament the small structures that make up food myofibrils

exocrine gland the gland that secretes chemicals nasal cavity an opening of the respiratory system to through ducts that lead to skin surfaces, body the outside environment cavities, and organ cavities. nephron the functional unit of the kidney

fibrous joint a joint held together by fibrous neuron a specialized cell that can receive and connective tissue transmit electrical and chemical signals

frontal lobe the part of the cerebral cortex that occipital lobe the part of the cerebral cortex that contains the motor cortex and areas involved in contains visual cortex and processes visual stimuli planning, attention, and language open circulatory system a circulatory system that

gallbladder the organ that stores and concentrates has the blood mixed with interstitial fluid in the bile body cavity and directly bathes the organs

glia (also, glial cells) the cells that provide support oral cavity the point of entry of food into the digestive functions for neurons system

hippocampus the brain structure in the temporal osmoregulation the mechanism by which water and lobe involved in processing memories solute concentrations are maintained at desired

hormone a chemical released by cells in one area of levels the body that affects cells in other parts of the body osmotic balance the appropriate values of water and

hyoid bone the bone that lies below the mandible in solute concentrations for a healthy organism the front of the neck pancreas a gland that secretes digestive juices

hypothalamus the brain structure that controls pancreas the organ located between the stomach hormone release and body homeostasis and the small intestine that contains exocrine and

inferior vena cava the major vein of the body endocrine cells returning blood from the lower parts of the body to parasympathetic nervous system the division of the right atrium autonomic nervous system that regulates visceral

interstitial fluid the fluid found between cells in the functions during relaxation body, similar in constitution to the fluid component parathyroid gland the gland located on the surface of of blood, but without the high concentrations of the thyroid that produces parathyroid hormone proteins parietal

hormones thyroxine and the liver and muscle cells, and in adipose tissue. triiodothyronine. The thyroid also produces calcitonin. Excess adipose storage can lead to obesity and serious health problems. The parathyroid glands lie on the posterior surface of the thyroid gland and produce parathyroid hormone.

16.3 Circulatory and Respiratory Systems The adrenal glands are located on top of the kidneys and consist of the adrenal cortex and adrenal medulla.

Animal respiratory systems are designed to facilitate gas exchange. In mammals, air is warmed and humidified in the nasal cavity. Air then travels down the pharynx and larynx, through the trachea, and into the lungs. In the lungs, air passes through the branching bronchi, reaching the respiratory bronchioles. The respiratory bronchioles open up into the alveolar sacs, and alveoli. Because there are so many cells in the alveoli and alveolar sacs in the lung, the surface area which contains alpha cells that release glucagon and beta cells that release insulin. Some organs possess

endocrine activity as a secondary function but have

The mammalian circulatory system is a closed system with double circulation passing through the lungs and the body. The heart produces the hormone atrial natriuretic peptide, which functions to

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reduce blood volume, pressure, and Na^+ concentration. 16.6 Nervous System

The gastrointestinal tract produces various hormones

that aid in digestion. The kidneys produce erythropoietin. The thymus produces hormones that aid in the development of the immune system. The gonads produce large amounts of steroid hormones, including testosterone in testes and estrogen and progesterone in ovaries. Adipose tissue produces leptin, which promotes satiety signals in the brain.

The nervous system is made up of neurons and glia. Neurons are specialized cells that are capable of sending electrical as well as chemical signals. Most neurons contain dendrites, which receive these signals, and axons that send signals to other neurons or tissues. Glia are non-neuronal cells in the nervous system that support neuronal development and

signaling. There are several types of glia that serve

16.5 Musculoskeletal System different functions.

The human skeleton is an endoskeleton that is composed of the axial and appendicular skeleton. The axial skeleton is composed of the bones of the skull, the hyoid bone, vertebral column, and the ossicles of the ear. Neurons have a resting potential across their membranes and when they are stimulated by a strong enough signal from another neuron an action potential

rkings of the body in the early 18th century in Europe, the practice of inoculation as a method to prevent the often-deadly effects of smallpox was introduced from the courts of the Ottoman Empire. The method involved causing limited infection with the smallpox virus by introducing the pus of an affected individual to a scratch in an uninfected person. The resulting infection was milder than if it had been caught naturally and mortality rates were shown to be about two percent rather than 30 percent from natural infections. Moreover, the inoculation gave the individual immunity to the disease. It was from these early experiences with inoculation that the methods of vaccination were developed, in which a weakened or relatively harmless (killed) derivative of a pathogen is introduced into the

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individual. The vaccination induces immunity to the disease with few of the risks of being infected. A modern understanding of the causes of the infectious disease and the mechanisms of the immune system began in the late 19th century and continues to grow today.

17.1 Viruses

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe how viruses were first discovered and how they are detected
- ? Explain the detailed steps of viral replication
- ? Describe how vaccines are used in prevention and treatment of viral diseases

FIGURE 17.2 (a) The tobacco mosaic virus, seen by transmission electron microscopy, was the first virus to be discovered. (b) The leaves of an infected plant are shown. (credit a: scale-bar data from Matt Russell; credit b: modification of work by USDA, Department of Plant Pathology Archive, North Carolina State University)

No one knows exactly when viruses emerged or from where they came, since viruses do not leave historical footprints such as fossils. Modern viruses are thought to be a mosaic of bits and pieces of nucleic acids picked up from various sources along their respective evolutionary paths. Viruses are acellular, parasitic entities that are not classified within any domain because they are not considered alive. They have no plasma membrane, internal organelles, or metabolic processes, and they do not divide. Instead, they infect a host cell and use the host's replication processes to produce progeny virus particles. Viruses infect all forms of organisms including bacteria, archaea, fungi, plants, and animals. Living things grow, metabolize, and reproduce. Viruses replicate, but to do so, they are entirely dependent on their host cells. They do not metabolize or grow, but are assembled in their mature form.

Viruses are diverse. They vary in their structure, their replication methods, and in their target hosts or even host cells. While most biological diversity can be understood through evolutionary history, such as how species have adapted to conditions and environments, much about virus origins and evolution remains unknown.

How Viruses Replicate

Viruses were first

It was discovered after the development of a porcelain filter, called the Chamberland-Pasteur filter, which could remove all bacteria visible under the microscope from any liquid sample. In 1886, Adolph Meyer demonstrated that a disease of tobacco plants, tobacco mosaic disease, could be transferred from a diseased plant to a healthy one through liquid plant extracts. In 1892, Dmitri Ivanowski showed that this disease could be transmitted in this way even after the Chamberland-Pasteur filter had removed all viable bacteria from the extract. Still, it was many years before it was proven that these "filterable" infectious agents were not simply very small bacteria but were a new type of tiny, disease-causing particle.

Virions, single virus particles, are very small, about 20–250 nanometers (1 nanometer = 1/1,000,000 mm). These individual virus particles are the infectious form of a virus outside the host cell. Unlike bacteria (which are about 100 times larger), we cannot see viruses with a light microscope, with the exception of some large virions of the poxvirus family (Figure 17.3).

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FIGURE 17.3 The size of a virus is very small relative to the size of cells and organelles.

It was not until the development of the electron microscope in the 1940s that scientists got their first good view of the structure of the tobacco mosaic virus (Figure 17.2) and others. The surface structure of virions can be observed by both scanning and transmission electron microscopy, whereas the internal structures of the virus can only be observed in images from a transmission electron microscope (Figure 17.4).

FIGURE 17.4 The ebola virus is shown here as visualized through (a) a scanning electron micrograph and (b) a transmission electron micrograph. (credit a: modification of work by Cynthia Goldsmith, CDC; credit b: modification of work by Thomas W. Geisbert, Boston University School of Medicine; scale-bar data from Matt Russell)

The use of this technology has allowed for the discovery of many viruses of all types of living organisms. They were initially grouped by shared morphology, meaning their size, shape, and distinguishing structures. Later, groups of viruses were classified by the type of nucleic acid they contained, DNA or RNA, and whether their nucleic acid was single- or double-stranded. More recently, molecular analysis of viral replication cycles has further refined their classification.

A virion consists of a nucleic-acid core, an outer protein coating, and sometimes an outer envelope made of protein and phospholipid membranes derived from the host cell. The most visible difference between members of viral families is their morphology, which is quite diverse. An interesting feature of viral complexity is that the complexity of the host does not correlate to the complexity of the virion. Some of the most complex virion structures are observed in bacteriophages, viruses that infect the simplest living

organisms, bacteria.

Viruses come in many shapes and sizes, but these are consistent and distinct for each viral family (Figure 17.5). All

virions have a nucleic-acid genome covered by a protective layer of protein, called a capsid. The capsid is made of protein subunits called capsomeres. Some viral capsids are simple polyhedral "spheres," whereas others are quite

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complex in structure. The outer structure surrounding the capsid of some viruses is called the viral envelope. All viruses use some sort of glycoprotein to attach to their host cells at molecules on the cell called viral receptors. The

virus exploits these cell-surface molecules, which the cell uses for some other purpose, as a way to recognize and

infect specific cell types. For example, the measles virus uses a cell-surface glycoprotein in humans that normally

functions in immune reactions and possibly in the sperm-egg interaction at fertilization. Attachment is a requirement for viruses to later penetrate the cell membrane, inject the viral genome, and complete their replication

inside the cell.

The T4 bacteriophage, which infects the *E. coli* bacterium, is among the most complex virion known; T4 has a protein tail structure that the virus uses to attach to the host cell and a head structure that houses its DNA.

Adenovirus, a nonenveloped animal virus that causes respiratory illnesses in humans, uses protein spikes protruding from its capsomeres to attach to the host cell. Nonenveloped viruses also include those that cause polio

(poliovirus), plantar warts (papillomavirus), and hepatitis A (hepatitis A virus). Nonenveloped viruses tend to be more robust and more likely to survive under harsh conditions, such as the gut.

Enveloped virions like HIV (human immunodeficiency virus), the causative agent in AIDS (acquired immune deficiency syndrome), consist of nucleic acid (RNA in the case of HIV) and capsid proteins surrounded by a phospholipid bilayer envelope and its associated proteins (Figure 17.5). Chicken pox, influenza, and mumps are examples of diseases caused by viruses with envelopes. Because of the fragility of the envelope, nonenveloped viruses are more resistant to changes in temperature, pH, and some disinfectants than enveloped viruses.

Overall, the shape of the virion and the presence or absence of an envelope tells us little about what diseases the

viruses may cause or what species they might infect, but is still a useful means to begin viral classification.

VISUAL CONNECTION

FIGURE 17.5 Viruses can be complex in shape or relatively simple. This figure shows three relatively complex virions: the bacteriophage

T4, with its DNA-containing head group and tail fibers that attach to host cells; adenovirus, which uses spikes from its capsid to bind to the

host cells; and HIV, which uses glycoproteins embedded in its envelope to do so. Notice that HIV has proteins called matrix proteins,

internal to the envelope, which help stab

hat cell.

Steps of Virus Infections

A virus must "take over" a cell to replicate. The viral replication cycle can produce dramatic biochemical and structural changes in the host cell, which may cause cell damage. These changes, called cytopathic effects, can change cell functions or even destroy the cell. Some infected cells, such as those infected by the common cold virus

(rhinovirus), die through lysis (bursting) or apoptosis (programmed cell death or "cell suicide?"), releasing all the progeny virions at once. The symptoms of viral diseases result from the immune response to the virus, which attempts to control and eliminate the virus from the body, and from cell damage caused by the virus. Many animal

viruses, such as HIV (human immunodeficiency virus), leave the infected cells of the immune system by a process

known as budding, where virions leave the cell individually. During the budding process, the cell does not undergo

lysis and is not immediately killed. However, the damage to the cells that HIV infects may make it impossible for the

cells to function as mediators of immunity, even though the cells remain alive for a period of time. Most productive

viral infections follow similar steps in the virus replication cycle: attachment, penetration, uncoating, replication, assembly, and release.

A virus attaches to a specific receptor site on the host-cell membrane through attachment proteins in the capsid or

proteins embedded in its envelope. The attachment is specific, and typically a virus will only attach to cells of one or

a few species and only certain cell types within those species with the appropriate receptors.

LINK TO LEARNING

View this video (<http://openstax.org//influenza2>) for a visual explanation of how influenza attacks the body.

Unlike animal viruses, the nucleic acid of bacteriophages is injected into the host cell naked, leaving the capsid outside the cell. Plant and animal viruses can enter their cells through endocytosis, in which the cell membrane surrounds and engulfs the entire virus. Some enveloped viruses enter the cell when the viral envelope fuses directly

with the cell membrane. Once inside the cell, the viral capsid is degraded and the viral nucleic acid is released, which then becomes available for replication and transcription.

The replication mechanism depends on the viral genome. DNA viruses usually use host cell proteins and enzymes to

make additional DNA that is used to copy the genome or be transcribed to messenger RNA (mRNA), which is then

used in protein synthesis. RNA viruses, such as the influenza virus, usually use the RNA core as a template for

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synthesis of viral genomic RNA and mRNA. The viral mRNA is translated into viral enzymes and capsid proteins to

assemble new virions (Figure 17.6). Of course, there are exceptions to this pattern. If a host cell does not provide

the enzymes necessary for viral replication, viral genes supply the information to dire

that represented a milestone in the use of vaccines. Polio epidemics occurred with increasing frequency and impact as the twentieth century progressed, becoming a terrifying and tragic event each summer. Tens of thousands of people died and many more were paralyzed; children made up a large portion of the victims. Using killed virus tested on the HeLa cell line (originally obtained from Henrietta Lacks and then mass produced to meet the need), Jonas Salk developed a successful vaccine. Mass immunization campaigns in the U.S. in the 1950s (killed vaccine) and 1960s (live vaccine) essentially eradicated the disease. The success of the polio vaccine paved the way for the routine dispensation of childhood vaccines against measles, mumps, rubella, chickenpox, and other diseases.

Live vaccines are usually made by attenuation (weakening) of the "wild-type" (disease-causing) virus by growing it in the laboratory in tissues or at temperatures different from what the virus is accustomed to in the host. For example, the virus may be grown in cells in a test tube, in bird embryos, or in live animals. The adaptation to these new cells or temperature induces mutations in the virus' genomes, allowing them to grow better in the laboratory while inhibiting their ability to cause disease when reintroduced into the conditions found in the host. These attenuated viruses thus still cause an infection, but they do not grow very well, allowing the immune response to develop in time to prevent major disease. The danger of using live vaccines, which are usually more effective than killed vaccines, is the low but significant risk that these viruses will revert back to their disease-causing form by back mutations. Back mutations occur when the vaccine undergoes mutations in the host such that it readapts to the host and can again cause disease, which can then be spread to other humans in an epidemic. This happened as recently as 2007 in Nigeria where mutations in a polio vaccine led to an epidemic of polio in that country.

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Some vaccines are in continuous development because certain viruses, such as influenza and HIV, have a high mutation rate compared to other viruses or host cells. With influenza, mutation in genes for the surface molecules helps the virus evade the protective immunity that may have been obtained in a previous influenza season, making it necessary for individuals to get vaccinated every year. Other viruses, such as those that cause the childhood diseases measles, mumps, and rubella, mutate so little that the same vaccine is used year after year.

Other vaccines are developed because of a new mutation of a well known type of virus. Coronaviruses are very common in nature and various diseases, but a specific mutation of a particular coronavirus led to the COVID-19 pandemic. The rapid development of successful vaccines was due in part to scientists who had been working with other coronaviruses prior to the pandemi

c. Kizzmekia S. Corbett, a research fellow and scientific lead, had deep experience and knowledge of coronaviruses, which was instrumental in developing one of the first vaccines (Moderna). She is now applying that experience to other respiratory diseases and vaccine development processes.

Vaccines and Antiviral Drugs for Treatment

In some cases, vaccines can be used to treat an active viral infection. In the case of rabies, a fatal neurological disease transmitted in the saliva of rabies virus-infected animals, the progression of the disease from the time of the animal bite to the time it enters the central nervous system may be two weeks or longer. This is enough time to vaccinate an individual who suspects being bitten by a rabid animal, and the boosted immune response from the vaccination is enough to prevent the virus from entering nervous tissue. Thus, the fatal neurological consequences of the disease are averted and the individual only has to recover from the infected bite. This approach is also being used for the treatment of Ebola, one of the fastest and most deadly viruses affecting humans, though usually infecting limited populations. Ebola is also a leading cause of death in gorillas. Transmitted by bats and great apes, this virus can cause death in 70-90 percent of the infected within two weeks. Using newly developed vaccines that boost the immune response, there is hope that immune systems of affected individuals will be better able to control the virus, potentially reducing mortality rates.

Another way of treating viral infections is the use of antiviral drugs. These drugs often have limited ability to cure viral disease but have been used to control and reduce symptoms for a wide variety of viral diseases. For most viruses, these drugs inhibit the virus by blocking the actions of one or more of its proteins. It is important that the targeted proteins be encoded for by viral genes and that these molecules are not present in a healthy host cell. In fact, for many years, scientists thought that drugs capable of impacting viruses would be too toxic for the body to endure. To meet this challenge, researcher Gertrude Elion sought to develop drugs that would target only the virus through processes such as inhibiting only viral DNA replication. For example, some of her medicines focused on purines, while others affected DNA polymerase. There are large numbers of antiviral drugs available to treat infections, some specific for a particular virus and others that can affect multiple viruses.

Antivirals have been developed to treat genital herpes (herpes simplex II) and influenza. For genital herpes, drugs such as acyclovir, developed by Elion, can reduce the number and duration of the episodes of active viral disease during which patients develop viral lesions in their skin cells. As the virus remains latent in nervous tissue of the body for life, this drug is not a cure but can make the symptoms of the disease more manageable. For infl

blood and lymph and develops into a macrophage after it moves into infected tissue. A macrophage is a large cell that engulfs foreign particles and pathogens. Mast cells are produced in the same way as white blood cells, but unlike circulating white blood cells, mast cells take up residence in connective tissues and especially mucosal tissues. They are responsible for releasing chemicals in response to physical injury. They also play a role in the allergic response, which will be discussed later in the chapter.

When a pathogen is recognized as foreign, chemicals called cytokines are released. A cytokine is a chemical messenger that regulates cell differentiation (form and function), proliferation (production), and gene expression to produce a variety of immune responses. Approximately 40 types of cytokines exist in humans. In addition to being released from white blood cells after pathogen recognition, cytokines are also released by the infected cells and bind to nearby uninfected cells, inducing those cells to release cytokines. This positive feedback loop results in a burst of cytokine production.

One class of early-acting cytokines is the interferons, which are released by infected cells as a warning to nearby uninfected cells. An interferon is a small protein that signals a viral infection to other cells. The interferons stimulate uninfected cells to produce compounds that interfere with viral replication. Interferons also activate macrophages and other cells.

The Inflammatory Response and Phagocytosis

The first cytokines to be produced encourage inflammation, a localized redness, swelling, heat, and pain. Inflammation is a response to physical trauma, such as a cut or a blow, chemical irritation, and infection by pathogens (viruses, bacteria, or fungi). The chemical signals that trigger an inflammatory response enter the extracellular fluid and cause capillaries to dilate (expand) and capillary walls to become more permeable, or leaky.

The serum and other compounds leaking from capillaries cause swelling of the area, which in turn causes pain. Various kinds of white blood cells are attracted to the area of inflammation. The types of white blood cells that arrive at an inflamed site depend on the nature of the injury or infecting pathogen. For example, a neutrophil is an early arriving white blood cell that engulfs and digests pathogens. Neutrophils are the most abundant white blood cells of the immune system (Figure 17.9). Macrophages follow neutrophils and take over the phagocytosis function and are involved in the resolution of an inflamed site, cleaning up cell debris and pathogens.

FIGURE 17.9 White blood cells (leukocytes) release chemicals to stimulate the inflammatory response following a cut in the skin.

Cytokines also send feedback to cells of the nervous system to bring about the overall symptoms of feeling sick, which include lethargy, muscle pain, and nausea. Cytokines also increase the core body temperature, causing a

to the pathogen. When they attach, the antibodies change shape providing a binding site for one of the complement proteins. After the first few complement proteins bind, a cascade of binding in a specific sequence of proteins follows in which the pathogen rapidly becomes coated in complement proteins.

Complement proteins perform several functions, one of which is to serve as a marker to indicate the presence of a pathogen to phagocytic cells and enhance engulfment. Certain complement proteins can combine to open pores in microbial cell membranes and cause lysis of the cells.

17.3 Adaptive Immunity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Explain adaptive immunity
- ? Describe cell-mediated immune response and humoral immune response
- ? Describe immune tolerance

The adaptive, or acquired, immune response takes days or even weeks to become established?much longer than the innate response; however, adaptive immunity is more specific to an invading pathogen. Adaptive immunity is an immunity that occurs after exposure to an antigen either from a pathogen or a vaccination. An antigen is a molecule that stimulates a response in the immune system. This part of the immune system is activated when the innate immune response is insufficient to control an infection. In fact, without information from the innate immune system, the adaptive response could not be mobilized. There are two types of adaptive responses: the cell-mediated immune response, which is controlled by activated T cells, and the humoral immune response, which is controlled by activated B cells and antibodies. Activated T and B cells whose surface binding sites are specific to the molecules on the pathogen greatly increase in numbers and attack the invading pathogen. Their attack can kill pathogens directly or they can secrete antibodies that enhance the phagocytosis of pathogens and disrupt the infection. Adaptive immunity also involves a memory to give the host long-term protection from reinfection with the same type of pathogen; on reexposure, this host memory will facilitate a rapid and powerful response.

B and T Cells

Lymphocytes, which are white blood cells, are formed with other blood cells in the red bone marrow found in many flat bones, such as the shoulder or pelvic bones. The two types of lymphocytes of the adaptive immune response are B and T cells (Figure 17.12). Whether an immature lymphocyte becomes a B cell or T cell depends on where in the body it matures. The B cells remain in the bone marrow to mature (hence the name ?B? for ?bone marrow?), while T cells migrate to the thymus, where they mature (hence the name ?T? for ?thymus?).

Maturation of a B or T cell involves becoming immunocompetent, meaning that it can recognize, by binding, a specific molecule or antigen (discussed below). During the maturation process, B and T cells that bind too strongly to the body's own cells are eliminated in order to minimize an immune response against the bod

tigen, these complexes signal an invader.

FIGURE 17.15 An antigen-presenting cell (APC), such as a macrophage, engulfs a foreign antigen, partially digests it in a lysosome, and then embeds it in an MHC class II molecule for presentation at the cell surface. Lymphocytes of the adaptive immune response must interact with antigen-embedded MHC class II molecules to mature into functional immune cells.

LINK TO LEARNING

View this animation from Rockefeller University (http://openstax.org//immune_system2) to see how dendritic cells act as sentinels in the body's immune system.

T cells have many functions. Some respond to APCs of the innate immune system and indirectly induce immune responses by releasing cytokines. Others stimulate B cells to start the humoral response as described previously.

Another type of T cell detects APC signals and directly kills the infected cells, while some are involved in suppressing inappropriate immune reactions to harmless or "self" antigens.

There are two main types of T cells: helper T lymphocytes (TH) and the cytotoxic T lymphocytes (TC). The TH lymphocytes function indirectly to tell other immune cells about potential pathogens. TH lymphocytes recognize specific antigens presented by the MHC class II complexes of APCs. There are two populations of TH cells: TH1 and TH2. TH1 cells secrete cytokines to enhance the activities of macrophages and other T cells. TH2 cells stimulate naïve B cells to secrete antibodies. Whether a TH1 or a TH2 immune response develops depends on the specific types of cytokines secreted by cells of the innate immune system, which in turn depends on the nature of the invading pathogen.

Cytotoxic T cells (TC) are the key component of the cell-mediated part of the adaptive immune system and attack and destroy infected cells. TC cells are particularly important in protecting against viral infections; this is because viruses replicate within cells where they are shielded from extracellular contact with circulating antibodies. Once activated, the TC creates a large clone of cells with one specific set of cell-surface receptors, as in the case with proliferation of activated B cells. As with B cells, the clone includes active TC cells and inactive memory TC cells. The resulting active TC cells then identify infected host cells. Because of the time required to generate a population of clonal T and B cells, there is a delay in the adaptive immune response compared to the innate immune response.

TC cells attempt to identify and destroy infected cells before the pathogen can replicate and escape, thereby halting the progression of intracellular infections. TC cells also support NK lymphocytes to destroy early cancers. Cytokines secreted by the TH1 response that stimulates macrophages also stimulate TC cells and enhance their ability to identify and destroy infected cells and tumors. A summary of how the humoral and cell-mediated immune responses are activated appears in Figure 17.16.

B plasma cell

e system, leading to host-cell damage that can become fatal.

Immunodeficiency

Immunodeficiency is a failure, insufficiency, or delay in the response of the immune system, which may be acquired or inherited. Immunodeficiency can allow pathogens or tumor cells to gain a foothold and replicate or proliferate to high enough levels so that the immune system becomes overwhelmed. Immunodeficiency can be acquired as a result of infection with certain pathogens that attack the cells of the immune system itself (such as HIV), chemical exposure (including certain medical treatments such as chemotherapy), malnutrition, or extreme stress. For instance, radiation exposure can destroy populations of lymphocytes and elevate an individual's susceptibility to infections and cancer. Rarely, primary immunodeficiencies that are present from birth may also occur. For example, severe combined immunodeficiency disease (SCID) is a condition in which children are born without functioning B or T cells.

Hypersensitivities

A maladaptive immune response toward harmless foreign substances or self-antigens that occur after tissue sensitization is termed a hypersensitivity. Types of hypersensitivities include immediate, delayed, and autoimmune. A large proportion of the human population is affected by one or more types of hypersensitivity.

Allergies

The immune reaction that results from immediate hypersensitivities in which an antibody-mediated immune response occurs within minutes of exposure to a usually harmless antigen is called an allergy. In the United States, 20 percent of the population exhibits symptoms of allergy or asthma, whereas 55 percent test positive against one or more allergens. On initial exposure to a potential allergen, an allergic individual synthesizes antibodies through the typical process of APCs presenting processed antigen to TH cells that stimulate B cells to produce the antibodies. The antibody molecules interact with mast cells embedded in connective tissues. This process primes, or sensitizes, the tissue. On subsequent exposure to the same allergen, antibody molecules on mast cells bind the antigen and stimulate the mast cell to release histamine and other inflammatory chemicals; these chemical mediators then recruit eosinophils (a type of white blood cell), which also appear to be adapted to responding to parasitic worms (Figure 17.22). Eosinophils release factors that enhance the inflammatory response and the secretions of mast cells. The effects of an allergic reaction range from mild symptoms like sneezing and itchy, watery eyes to more severe or even life-threatening reactions involving intensely itchy welts or hives, airway constriction with severe respiratory distress, and plummeting blood pressure caused by dilating blood vessels and fluid loss from the circulatory system. This extreme reaction, typically in response to an allergen introduced to the circulatory system, is known as anaphylactic shock. Antihistamines are an insufficient co

incorrectly marks because of genetic factors or physiology, and is not
?self? components as foreign and stimulates the caused by infection or vaccination
immune response interferon a cytokine that inhibits viral replication

autoimmunity a type of hypersensitivity to self- lymph the watery fluid present in the lymphatic
antigens circulatory system that bathes tissues and organs

B cell a lymphocyte that matures in the bone marrow with protective white blood cells and does not
capsid the protein coating of the viral core contain erythrocytes
cell-mediated immune response an adaptive lymphocyte a type of white blood cell that includes

immune response that is controlled by T cells natural killer cells of the innate immune system and
complement system an array of approximately 20 B and T cells of the adaptive immune system

soluble proteins of the innate immune system that macrophage a large phagocytic cell that engulfs
enhance phagocytosis, bore holes in pathogens, and foreign particles and pathogens
recruit lymphocytes major histocompatibility complex (MHC) I a group

cytokine a chemical messenger that regulates cell of proteins found on the surface of all nucleated
differentiation, proliferation, and gene expression to cells that signals to immune cells whether the cell is
effect immune responses normal or is infected or cancerous; it also provides

cytopathic causing cell damage the appropriate sites into which antigens can be
cytotoxic T lymphocyte (TC) an adaptive immune cell loaded for recognition by lymphocytes

that directly kills infected cells via enzymes, and major histocompatibility complex (MHC) II
that releases cytokines to enhance the immune molecule a protein found on the surface of
response antigen-presenting cells that signals to immune

dendritic cell an immune cell that processes antigen cells whether the cell is normal or is infected or
material and presents it on the surface of its cell in cancerous; it provides the appropriate template into
MHC class II molecules and induces an immune which antigens can be loaded for recognition by

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lymphocytes by transfer of antibodies from one individual to
mast cell a leukocyte that produces inflammatory another

molecules, such as histamine, in response to large primary immune response the response of the
pathogens adaptive immune system to the first exposure to an

memory cell an antigen-specific B or T lymphocyte antigen
that does not differentiate into an effector cell secondary immune response the response of the
during the primary immune response but that can adaptive immune system to a second or later
immediately become an effector cell on reexposure exposure to an antigen mediated by memory cells
to the same pathogen T cell a lymphocyte that matures in the thymus gland

monocyte a type of white blood cell that circulates in vaccine a weakened solution of virus components,
the blood and lymph and differentiates into a viruses, or other agents that produce

ely in sex-specific

morphology and behavior, there is almost always one type of body that produces smaller gametes, and one type of body that produces larger gametes. The individuals who produce smaller gametes are classified with a male sex, while those who produce larger gametes are classified with a female sex. During sexual reproduction the male gamete (sperm) may be placed inside the female's body for internal fertilization, the sperm may be left in the environment for the female to pick up and place in their own body, or both sperm and eggs may be released into the environment for external fertilization. Seahorses provide an example of the latter, but with a twist

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(Figure 18.1). Following a mating dance, the female releases eggs into the male seahorse's abdominal brood pouch and the male releases sperm into the water, which then find their way into the brood pouch to fertilize the eggs. The fertilized eggs develop in the pouch for several weeks.

18.1 How Animals Reproduce

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Describe advantages and disadvantages of asexual and sexual reproduction
- ? Discuss asexual reproduction methods
- ? Discuss sexual reproduction methods
- ? Discuss internal and external methods of fertilization

Some animals produce offspring through asexual reproduction while other animals produce offspring through sexual reproduction. Both methods have advantages and disadvantages. Asexual reproduction produces offspring that are genetically identical to the parent because the offspring are all clones of the original parent. A single individual can produce offspring asexually and large numbers of offspring can be produced quickly; these are two advantages that asexually reproducing organisms have over sexually reproducing organisms. In a stable or predictable environment, asexual reproduction is an effective means of reproduction because all the offspring will be adapted to that environment. In an unstable or unpredictable environment, species that reproduce asexually may be at a disadvantage because all the offspring are genetically identical and may not be adapted to different conditions.

During sexual reproduction, the genetic material of two individuals is combined to produce genetically diverse offspring that differ from their parents. The genetic diversity of sexually produced offspring is thought to give sexually reproducing individuals greater fitness because more of their offspring may survive and reproduce in an unpredictable or changing environment. Species that reproduce sexually (and have separate sexes) must maintain two different types of individuals, males and females. Only half the population (females) can produce the offspring, so fewer offspring will be produced when compared to asexual reproduction. This is a disadvantage of sexual reproduction compared to asexual reproduction.

Asexual Reproduction

Asexual reproduction occurs in prok

aryotic microorganisms (bacteria and archaea) and in many eukaryotic, single-celled and multi-celled organisms. There are several ways that animals reproduce asexually, the details of which vary among individual species.

Fission

Fission, also called binary fission, occurs in some invertebrate, multi-celled organisms. It is in some ways analogous to the process of binary fission of single-celled prokaryotic organisms. The term fission is applied to instances in which an organism appears to split itself into two parts and, if necessary, regenerate the missing parts of each new organism. For example, species of turbellarian flatworms commonly called the planarians, such as *Dugesia dorotocephala*, are able to separate their bodies into head and tail regions and then regenerate the missing half in each of the two new organisms. Sea anemones (Cnidaria), such as species of the genus *Anthopleura* (Figure 18.2), will divide along the oral-aboral axis, and sea cucumbers (Echinodermata) of the genus *Holothuria*, will divide into two halves across the oral-aboral axis and regenerate the other half in each of the resulting individuals.

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FIGURE 18.2 The *Anthopleura artemisia* sea anemone can reproduce through fission.

Budding

Budding is a form of asexual reproduction that results from the outgrowth of a part of the body leading to a separation of the "bud" from the original organism and the formation of two individuals, one smaller than the other.

Budding occurs commonly in some invertebrate animals such as hydras and corals. In hydras, a bud forms that develops into an adult and breaks away from the main body (Figure 18.3).

FIGURE 18.3 (a) Hydra reproduce asexually through budding: a bud forms on the tubular body of an adult hydra, develops a mouth and tentacles, and then detaches from its parent. The new hydra is fully developed and will find its own location for attachment. (b) Some coral, such as the *Lophelia pertusa* shown here, can reproduce through budding. (credit b: modification of work by Ed Bowlby, NOAA/Olympic Coast NMS; NOAA/OAR/Office of Ocean Exploration)

LINK TO LEARNING

View this video (http://openstax.org//budding_hydra2) to see a hydra budding.

Fragmentation

Fragmentation is the breaking of an individual into parts followed by regeneration. If the animal is capable of fragmentation, and the parts are big enough, a separate individual will regrow from each part. Fragmentation may

occur through accidental damage, damage from predators, or as a natural form of reproduction. Reproduction through fragmentation is observed in sponges, some cnidarians, turbellarians, echinoderms, and annelids. In some

sea stars, a new individual can be regenerated from a broken arm and a piece of the central disc. This sea star (Figure 18.4) is in the process of growing a complete sea star from an arm that has been cut off. Fisheries workers

have been known to try to kill the sea stars eating the

by Assaf Shtilman; credit b: modification of work by "Schristia"/Flickr)

Sex Determination

Pioneering scientist Nettie Stevens was the first to observe chromosomal differences between the different sexes of organisms. Using a microscope to observe mealworm cells, she noted that one chromosome was notably different between females and males, and she concluded that those chromosomes were the most likely determinants of the worms' sex. She later studied other insects, and her discoveries were confirmed by other scientists. (Stevens was initially denied credit due to her gender, but later publications acknowledged her critical role.) Mammalian sex is determined genetically by the combination of X and Y chromosomes. Individuals homozygous for X (XX) are female and heterozygous individuals (XY) are male. In mammals, the presence of a Y chromosome causes the development of male characteristics and its absence results in female characteristics. The XY system is also found in some insects and plants.

Bird sex determination is dependent on the combination of Z and W chromosomes. Homozygous for Z (ZZ) results in a male and heterozygous (ZW) results in a female. Notice that this system is the opposite of the mammalian system because in birds the female is the sex with the different sex chromosomes. The W appears to be essential in determining the sex of the individual, similar to the Y chromosome in mammals. Some fish, crustaceans, insects (such as butterflies and moths), and reptiles use the ZW system.

More complicated chromosomal sex determining systems also exist. For example, some swordtail fish have three sex chromosomes in a population.

The sex of some other species is not determined by chromosomes, but by some aspect of the environment.

Sex

determination in alligators, some turtles, and tuataras, for example, is dependent on the temperature during the middle third of egg development. This is referred to as environmental sex determination, or more specifically, as temperature-dependent sex determination. In many turtles, cooler temperatures during egg incubation produce males and warm temperatures produce females, while in many other species of turtles, the reverse is true. In some crocodiles and some turtles, moderate temperatures produce males and both warm and cool temperatures produce females.

Individuals of some species change their entire set of reproductive organs during their lives, switching from one to the other. If the individual is born with an ovarian system of organs first, it is termed protogyny or "first female," if it is born with a testicular system of organs first, it is termed protandry or "first male." Oysters are born with male morphology, grow in size, and change body parts and lay eggs. The wrasses, a family of reef fishes, are all sequential hermaphrodites. Some of these species live in closely coordinated schools with a dominant male and a large number of smaller females. If the male dies, a female increases in size, chan

the birth of the individual. The length of time of gestation, or the gestation period, in humans is 266 days and is similar in other great apes.

Within 24 hours of fertilization, the egg nucleus has finished meiosis and the egg and sperm nuclei fuse. With fusion, the cell is known as a zygote. The zygote initiates cleavage and the developing embryo travels through the oviduct to the uterus. The developing embryo must implant into the wall of the uterus within seven days, or it will deteriorate and die. The outer layers of the developing embryo or blastocyst grow into the endometrium by digesting the endometrial cells, and healing of the endometrium closes up the blastocyst into the tissue. Another layer of the blastocyst, the chorion, begins releasing a hormone called human beta chorionic gonadotropin (β -HCG), which makes its way to the corpus luteum and keeps that structure active. This ensures adequate levels of progesterone that will maintain the endometrium of the uterus for the support of the developing embryo. Pregnancy tests determine the level of β -HCG in urine or serum. If the hormone is present, the test is positive.

The gestation period is divided into three equal periods or trimesters. During the first two-to-four weeks of the first trimester, nutrition and waste are handled by the endometrial lining through diffusion. As the trimester progresses, the outer layer of the embryo begins to merge with the endometrium, and the placenta forms. The placenta takes over the nutrient and waste requirements of the embryo and fetus, with the gestational parent's blood passing nutrients to the placenta and removing waste from it. Chemicals from the fetus, such as bilirubin, are processed by the gestational parent's liver for elimination. Some of the pregnant person's immunoglobulins will pass through the placenta, providing passive immunity against some potential infections.

Internal organs and body structures begin to develop during the first trimester. By five weeks, limb buds, eyes, the heart, and liver have been basically formed. By eight weeks, the term fetus applies, and the body is essentially formed (Figure 18.18a). The individual is about five centimeters (two inches) in length and many of the organs, such as the lungs and liver, are not yet functioning. Exposure to any toxins is especially dangerous during the first trimester, as all of the body's organs and structures are going through initial development. Anything that interferes with chemical signaling during that development can have a severe effect on the fetus's survival.

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FIGURE 18.18 (a) Fetal development is shown at nine weeks gestation. (b) This fetus is just entering the second trimester, when the placenta takes over more of the functions performed as the baby develops. (c) There is rapid fetal growth during the third trimester. (credit a: modification of work by Ed Uthman; credit b: modification of work by National Museum of He

Internal fertilization the fertilization of eggs by
from the outgrowth of a part of an organism leading sperm inside the body of the female
to a separation from the original animal into two interstitial cells of Leydig a cell type found next to the
individuals seminiferous tubules that makes testosterone

bulbourethral gland the paired glands in the human labia majora the large folds of tissue covering
male that produce a secretion that cleanses the inguinal area
urethra prior to ejaculation labia minora the smaller folds of tissue within labia

clitoris a sensory and erectile structure in female majora
mammals, homologous to the male penis, luteinizing hormone (LH) a reproductive hormone
stimulated during sexual arousal that causes testosterone production in males and

corpus luteum the endocrine tissue that develops ovulation and lactation in females
from an ovarian follicle after ovulation; secretes menstrual cycle the cycle of the degradation and re-
progesterone and estrogen during pregnancy growth of the endometrium

estrogen a reproductive hormone, usually present in oogenesis the process of producing haploid eggs
animals of any sex, which in females assists in organogenesis the process of organ formation during
endometrial regrowth, ovulation, and calcium development
absorption ovarian cycle the cycle of preparation of egg for

external fertilization the fertilization of eggs by ovulation and the conversion of the follicle to the
sperm outside an animal's body, often during corpus luteum
spawning oviduct (also, fallopian tube) the muscular tube

fission (also, binary fission) a form of asexual connecting uterus with ovary area
reproduction in which an organism splits into two oviparity a process by which fertilized eggs are laid
separate organisms or two parts that regenerate the outside the female's body and develop there,
missing portions of the body receiving nourishment from the yolk that is a part of

follicle stimulating hormone (FSH) a reproductive the egg
hormone that causes sperm production in males ovoviparity a process by which fertilized eggs are
and follicle development in females retained within the female; the embryo obtains its

fragmentation the breaking of an organism into parts nourishment from the egg's yolk, and the young are
and the growth of a separate individual from each fully developed when they are hatched
part ovulation the release of an oocyte from a mature

gastrulation the process in which the blastula folds follicle in the ovary of a vertebrate
over itself to form the three germ layers parthenogenesis a form of asexual reproduction in

gestation the development before birth of a which an egg develops into a complete individual
viviparous animal without being fertilized

gestation period the length of time of development, penis the male reproductive structure for urine
from conception to birth, of the young of a elimination and copulation
viviparous animal placenta the organ that supports the transport of

gonadotro

Species Distribution

In addition to measuring density, further information about a population can be obtained by looking at the distribution of the individuals throughout their range. A species distribution pattern is the distribution of individuals within a habitat at a particular point in time? broad categories of patterns are used to describe them.

Individuals within a population can be distributed at random, in groups, or equally spaced apart (more or less). These are known as random, clumped, and uniform distribution patterns, respectively (Figure 19.3). Different distributions reflect important aspects of the biology of the species; they also affect the mathematical methods required to estimate population sizes. An example of random distribution occurs with dandelion and other plants that have wind-dispersed seeds that germinate wherever they happen to fall in favorable environments. A clumped distribution, may be seen in plants that drop their seeds straight to the ground, such as oak trees; it can also be seen in animals that live in social groups (schools of fish or herds of elephants). Uniform distribution is observed in plants that secrete substances inhibiting the growth of nearby individuals (such as the release of toxic chemicals by sage plants). It is also seen in territorial animal species, such as penguins that maintain a defined territory for nesting. The territorial defensive behaviors of each individual create a regular pattern of distribution of similar-sized territories and individuals within those territories. Thus, the distribution of the individuals within a population provides more information about how they interact with each other than does a simple density measurement. Just as lower density species might have more difficulty finding a mate, solitary species with a random distribution might have a similar difficulty when compared to social species clumped together in groups.

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FIGURE 19.3 Species may have a random, clumped, or uniform distribution. Plants such as (a) dandelions with wind-dispersed seeds tend to be randomly distributed. Animals such as (b) elephants that travel in groups exhibit a clumped distribution. Territorial birds such as (c) penguins tend to have a uniform distribution. (credit a: modification of work by Rosendahl; credit b: modification of work by Rebecca Wood; credit c: modification of work by Ben Tubby)

Demography

While population size and density describe a population at one particular point in time, scientists must use demography to study the dynamics of a population. Demography is the statistical study of population changes over time: birth rates, death rates, and life expectancies. These population characteristics are often displayed in a life table.

Life Tables

Life tables provide important information about the life history of an organism and the life expectancy of individuals at each age. They are modeled after actuarial tables used by the ins

is seen in bacteria. Bacteria are prokaryotes that reproduce largely by binary fission. This division takes about an hour for many bacterial species. If 1000 bacteria are placed in a large flask with an abundant supply of nutrients (so the nutrients will not become quickly depleted), the number of bacteria will have doubled from 1000 to 2000 after just an hour. In another hour, each of the 2000 bacteria will

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divide, producing 4000 bacteria. After the third hour, there should be 8000 bacteria in the flask. The important concept of exponential growth is that the growth rate—the number of organisms added in each reproductive generation—is itself increasing; that is, the population size is increasing at a greater and greater rate. After 24 of these cycles, the population would have increased from 1000 to more than 16 billion bacteria. When the population size, N , is plotted over time, a J-shaped growth curve is produced (Figure 19.5a).

The bacteria-in-a-flask example is not truly representative of the real world where resources are usually limited. However, when a species is introduced into a new habitat that it finds suitable, it may show exponential growth for a while. In the case of the bacteria in the flask, some bacteria will die during the experiment and thus not reproduce; therefore, the growth rate is lowered from a maximal rate in which there is no mortality. The growth rate of a population is largely determined by subtracting the death rate, D , (number organisms that die during an interval) from the birth rate, B , (number organisms that are born during an interval). The growth rate can be expressed in a simple equation that combines the birth and death rates into a single factor: r . This is shown in the following formula:

The value of r can be positive, meaning the population is increasing in size (the rate of change is positive); or negative, meaning the population is decreasing in size; or zero, in which case the population size is unchanging, a condition known as zero population growth.

Logistic Growth

Extended exponential growth is possible only when infinite natural resources are available; this is not the case in the real world. Charles Darwin recognized this fact in his description of the “struggle for existence,” which states that individuals will compete (with members of their own or other species) for limited resources. The successful ones are more likely to survive and pass on the traits that made them successful to the next generation at a greater rate (natural selection). To model the reality of limited resources, population ecologists developed the logistic growth model.

Carrying Capacity and the Logistic Model

In the real world, with its limited resources, exponential growth cannot continue indefinitely. Exponential growth may occur in environments where there are few individuals and plentiful resources, but when the number of individuals ge

ts large enough, resources will be depleted and the growth rate will slow down. Eventually, the growth rate will plateau or level off (Figure 19.5b). This population size, which is determined by the maximum population size that a particular environment can sustain, is called the carrying capacity, or K . In real populations, a growing population often overshoots its carrying capacity, and the death rate increases beyond the birth rate causing the population size to decline back to the carrying capacity or below it. Most populations usually fluctuate around the carrying capacity in an undulating fashion rather than existing right at it.

The formula used to calculate logistic growth adds the carrying capacity as a moderating force in the growth rate.

The expression $(K - N)$ is equal to the number of individuals that may be added to a population at a given time, and

$(K - N)$ divided by K is the fraction of the carrying capacity available for further growth. Thus, the exponential growth model is restricted by this factor to generate the logistic growth equation:

Notice that when N is almost zero the quantity in brackets is almost equal to 1 (or K/K) and growth is close to exponential. When the population size is equal to the carrying capacity, or $N = K$, the quantity in brackets is equal to

zero and growth is equal to zero. A graph of this equation (logistic growth) yields the S-shaped curve (Figure 19.5b).

It is a more realistic model of population growth than exponential growth. There are three different sections to an S-

shaped curve. Initially, growth is exponential because there are few individuals and ample resources available. Then, as resources begin to become limited, the growth rate decreases. Finally, the growth rate levels off at the carrying capacity of the environment, with little change in population number over time.

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FIGURE 19.5 When resources are unlimited, populations exhibit (a) exponential growth, shown in a J-shaped curve. When resources are limited, populations exhibit (b) logistic growth. In logistic growth, population expansion decreases as resources become scarce, and it levels off when the carrying capacity of the environment is reached. The logistic growth curve is S-shaped.

Role of Intraspecific Competition

The logistic model assumes that every individual within a population will have equal access to resources and, thus, an equal chance for survival. For plants, the amount of water, sunlight, nutrients, and space to grow are the important resources, whereas in animals, important resources include food, water, shelter, nesting space, and mates.

In the real world, phenotypic variation among individuals within a population means that some individuals will be better adapted to their environment than others. The resulting competition for resources among population members of the same species is termed intraspecific competition. Intraspecific competition may not affect populations

ese are grouped into density-dependent factors, in which the density of the population affects growth rate and mortality, and density-independent factors, which cause mortality in a population regardless of population density. Wildlife biologists, in particular, want to understand both types because this helps them manage populations and prevent extinction or overpopulation.

Density-dependent Regulation

Most density-dependent factors are biological in nature and include predation, inter- and intraspecific competition, and parasites. Usually, the denser a population is, the greater its mortality rate. For example, during intra- and interspecific competition, the reproductive rates of the species will usually be lower, reducing their populations? rate of growth. In addition, low prey density increases the mortality of its predator because it has more difficulty locating its food source. Also, when the population is denser, diseases spread more rapidly among the members of the population, which affect the mortality rate.

Density dependent regulation was studied in a natural experiment with wild donkey populations on two sites in Australia.² On one site the population was reduced by a population control program; the population on the other site received no interference. The high-density plot was twice as dense as the low-density plot. From 1986 to 1987 the high-density plot saw no change in donkey density, while the low-density plot saw an increase in donkey density. The difference in the growth rates of the two populations was caused by mortality, not by a difference in birth rates. The researchers found that numbers of offspring birthed by each mother was unaffected by density. Growth rates in the two populations were different mostly because of juvenile mortality caused by the mother?s malnutrition due to scarce high-quality food in the dense population. Figure 19.7 shows the difference in age-specific mortalities in the two populations.

FIGURE 19.7 This graph shows the age-specific mortality rates for wild donkeys from high- and low-density populations. The juvenile mortality is much higher in the high-density population because of maternal malnutrition caused by a shortage of high-quality food.

Density-independent Regulation and Interaction with Density-dependent Factors

Many factors that are typically physical in nature cause mortality of a population regardless of its density. These factors include weather, natural disasters, and pollution. An individual deer will be killed in a forest fire regardless of how many deer happen to be in that area. Its chances of survival are the same whether the population density is high or low. The same holds true for cold winter weather.

In real-life situations, population regulation is very complicated and density-dependent and independent factors can interact. A dense population that suffers mortality from a density-independent cause will be able to recover differently than a sparse population. For exam

tes, age at first reproduction, the numbers of offspring, and even death rates evolve just like anatomy or behavior, leading to adaptations that affect population growth. Population ecologists have described a continuum of life-history ?strategies? with K-selected species on one end and r-selected species on the other. K-selected species are adapted to stable, predictable environments. Populations of K-selected species tend to exist close to their carrying capacity. These species tend to have larger, but fewer, offspring and contribute large amounts of resources to each offspring. Elephants would be an example of a K-selected species. r-selected species are adapted to unstable and unpredictable environments. They have large numbers of small offspring. Animals that are r-selected do not provide a lot of resources or parental care to offspring, and the offspring are relatively self-sufficient at birth. Examples of r-selected species are marine invertebrates such as jellyfish and plants such as the dandelion. The two extreme strategies are at two ends of a continuum on which real species life histories will exist. In addition, life history strategies do not need to evolve as suites, but can evolve independently of each other, so each species may have some characteristics that trend toward one extreme or the other.

19.3 The Human Population

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Discuss how human population growth can be exponential
- ? Explain how humans have expanded the carrying capacity of their habitat
- ? Relate population growth and age structure to the level of economic development in different countries
- ? Discuss the long-term implications of unchecked human population growth

Concepts of animal population dynamics can be applied to human population growth. Humans are not unique in their ability to alter their environment. For example, beaver dams alter the stream environment where they are built.

Humans, however, have the ability to alter their environment to increase its carrying capacity, sometimes to the detriment of other species. Earth?s human population and their use of resources are growing rapidly, to the extent

that some worry about the ability of Earth?s environment to sustain its human population. Long-term exponential growth carries with it the potential risks of famine, disease, and large-scale death, as well as social consequences of crowding such as increased crime.

Human technology and particularly our harnessing of the energy contained in fossil fuels have caused unprecedented changes to Earth?s environment, altering ecosystems to the point where some may be in danger of collapse. Changes on a global scale including depletion of the ozone layer, desertification and topsoil loss, and global climate change are caused by human activities.

The world?s human population is presently growing exponentially (Figure 19.9).

FIGURE 19.9 Human population growth since 1000 AD is exponential.

A consequenc

people will starve to death in spite of any crash programs embarked upon now. At this late date nothing can prevent a substantial increase in the world death rate.²⁶ While many critics view this statement as an exaggeration, the laws of exponential population growth are still in effect, and unchecked human population growth cannot continue indefinitely.

Efforts to moderate population control led to the one-child policy in China, which imposes fines on urban couples who have more than one child. Due to the fact that some couples wish to have a male heir, many Chinese couples continue to have more than one child. The effectiveness of the policy in limiting overall population growth is controversial, as is the policy itself. Moreover, there are stories of female infanticide having occurred in some of the more rural areas of the country. Family planning education programs in other countries have had highly positive effects on limiting population growth rates and increasing standards of living. In spite of population control policies, the human population continues to grow. Because of the subsequent need to produce more and more food to feed our population, inequalities in access to food and other resources will continue to widen. The United Nations estimates the future world population size could vary from 6 billion (a decrease) to 16 billion people by the year 2100. There is no way to know whether human population growth will moderate to the point where the crisis described by Dr. Ehrlich will be averted.

Another consequence of population growth is the change and degradation of the natural environment. Many countries have attempted to reduce the human impact on climate change by limiting their emission of greenhouse gases. However, a global climate change treaty remains elusive, and many underdeveloped countries trying to improve their economic condition may be less likely to agree with such provisions without compensation if it means slowing their economic development. Furthermore, the role of human activity in causing climate change has become a hotly debated socio-political issue in some developed countries, including the United States. Thus, we enter the future with considerable uncertainty about our ability to curb human population growth and protect our environment to maintain the carrying capacity for the human species.

LINK TO LEARNING

Visit this website (<http://openstax.org/l/populations2>) and select ?Launch the movie? for an animation discussing the global impacts of human population growth.

19.4 Community Ecology

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- ? Discuss the predator-prey cycle
- ? Give examples of defenses against predation and herbivory
- ? Describe the competitive exclusion principle
- ? Give examples of symbiotic relationships between species
- ? Describe community structure and succession

In general, populations of one species never live in isolation from populations of other species. The interactin

the aquatic ecosystem. The relationships between populations in the community, and possibly the biodiversity, would change dramatically if these fish were to become extinct.

FIGURE 19.25 The *Pisaster ochraceus* sea star is a keystone species. (credit: Jerry Kirkhart)

EVERYDAY CONNECTION

Invasive Species

Invasive species are non-native organisms that, when introduced to an area out of its native range, alter the community they invade. In the United States, invasive species like the purple loosestrife (*Lythrum salicaria*) and the zebra mussel (*Dreissena polymorpha*) have altered aquatic ecosystems, and some forests are threatened by the spread of common buckthorn (*Rhamnus cathartica*) and garlic mustard (*Alliaria petiolata*). Some well-known invasive animals include the emerald ash borer (*Agrilus planipennis*) and the European starling (*Sturnus vulgaris*). Whether enjoying a forest hike, taking a summer boat trip, or simply walking down an urban street, you

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have likely encountered an invasive species.

One of the many recent proliferations of an invasive species concerns the Asian carp in the United States. Asian

carp were introduced to the United States in the 1970s by fisheries (commercial catfish ponds) and by sewage treatment facilities that used the fish's excellent filter feeding abilities to clean their ponds of excess plankton. Some of the fish escaped, and by the 1980s they had colonized many waterways of the Mississippi River basin, including the Illinois and Missouri Rivers.

Voracious feeders and rapid reproducers, Asian carp may outcompete native species for food and could lead to their extinction. One species, the grass carp, feeds on phytoplankton and aquatic plants. It competes with native species for these resources and alters nursery habitats for other fish by removing aquatic plants. Another species, the silver carp, competes with native fish that feed on zooplankton. In some parts of the Illinois River, Asian carp constitute 95 percent of the community's biomass. Although edible, the fish is bony and not desired in the United States. Moreover, their presence now threatens the native fish and fisheries of the Great Lakes, which are important to local economies and recreational anglers. Asian carp have even injured humans. The fish, frightened by the sound of approaching motorboats, thrust themselves into the air, often landing in the boat or directly hitting boaters.

The Great Lakes and their prized salmon and lake trout fisheries are being threatened by Asian carp. The carp are not yet present in the Great Lakes, and attempts are being made to prevent its access to the lakes through the Chicago Ship and Sanitary Canal, which is the only connection between the Mississippi River and Great Lakes basins. To prevent the Asian carp from leaving the canal, a series of electric barriers have been used to discourage their migration; however, the threat is significant enough that several states an

Is human population growing exponentially?

of populations: size, density, and distribution pattern. Humans have increased their carrying capacity through technology, urbanization, and harnessing the energy of fossil fuels. Life tables are useful to calculate life expectancies of individual population members. Survivorship curves show the number of individuals surviving at each age to predict population growth. The age structure of a population allows us to show the number of individuals surviving at each age to predict population growth. Unchecked human population growth could have dire long-term effects on human welfare and Earth's ecosystems.

Populations with unlimited resources grow exponentially

with an accelerating growth rate. When resources become limiting, populations follow a logistic growth curve in which population size will level off at the carrying capacity. Many organisms have developed defenses against predation and herbivory, including mechanical defenses, warning coloration, and mimicry. Populations are regulated by a variety of density-dependent and density-independent factors. Two species cannot exist indefinitely in the same habitat competing directly for the same resources. Species may form symbiotic relationships such as commensalism, mutualism, or parasitism. Community structure is described by its foundation and keystone species. Communities respond to environmental disturbances by succession: the predictable appearance of different types of plant species, until a stable community structure is established.

Visual Connection Questions

1. Figure 19.2 As this graph shows, population density typically decreases with increasing body size. Why are growing, slow growing, and stable populations shown in stages 1 through 3. What type of

population change do you think stage 4 represents?

2. Figure 19.6 If the major food source of seals declines due to pollution or overfishing, which of the following would likely occur?

a. The carrying capacity of seals would decrease,

as would the seal population.

b. The carrying capacity of seals would decrease,

but the seal population would remain the same.

c. The number of seal deaths would increase, but

the number of births would also increase, so the population size would remain the same.

d. The carrying capacity of seals would remain the same, but the seal population would increase.

and organisms present.

Ecology of Ecosystems

Life in an ecosystem often involves competition for limited resources, which occurs both within a single species and between different species. Organisms compete for food, water, sunlight, space, and mineral nutrients. These resources provide the energy for metabolic processes and the matter to make up organisms' physical structures. Other critical factors influencing community dynamics are the components of its physical environment: a habitat's climate (seasons, sunlight, and rainfall), elevation, and geology. These can all be important environmental variables that determine which organisms can exist within a particular area.

Freshwater ecosystems are the least common, occurring on only 1.8 percent of Earth's surface. These systems comprise lakes, rivers, streams, and springs; they are quite diverse, and support a variety of animals, plants, fungi, protists and prokaryotes.

Marine ecosystems are the most common, comprising 75 percent of Earth's surface and consisting of three basic types: shallow ocean, deep ocean water, and deep ocean bottom. Shallow ocean ecosystems include extremely biodiverse coral reef ecosystems, yet the deep ocean water is known for large numbers of plankton and krill (small crustaceans) that support it. These two environments are especially important to aerobic respirators worldwide, as the phytoplankton perform 40 percent of all photosynthesis on Earth. Although not as diverse as the other two, deep

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ocean bottom ecosystems contain a wide variety of marine organisms. Such ecosystems exist even at depths where light is unable to penetrate through the water.

Terrestrial ecosystems, also known for their diversity, are grouped into large categories called biomes. A biome is a large-scale community of organisms, primarily defined on land by the dominant plant types that exist in geographic regions of the planet with similar climatic conditions. Examples of biomes include tropical rainforests, savannas, deserts, grasslands, temperate forests, and tundras. Grouping these ecosystems into just a few biome categories obscures the great diversity of the individual ecosystems within them. For example, the saguaro cacti (*Carnegiea gigantea*) and other plant life in the Sonoran Desert, in the United States, are relatively diverse compared with the desolate rocky desert of Boa Vista, an island off the coast of Western Africa (Figure 20.3).

FIGURE 20.3 Desert ecosystems, like all ecosystems, can vary greatly. The desert in (a) Saguaro National Park, Arizona, has abundant plant life, while the rocky desert of (b) Boa Vista island, Cape Verde, Africa, is devoid of plant life. (credit a: modification of work by Jay Galvin; credit b: modification of work by Ingo Wölbern)

Ecosystems and Disturbance

Ecosystems are complex with many interacting parts. They are routinely exposed to various disturbances: c

hanges

in the environment that affect their compositions, such as yearly variations in rainfall and temperature. Many disturbances are a result of natural processes. For example, when lightning causes a forest fire and destroys part of a forest ecosystem, the ground is eventually populated with grasses, followed by bushes and shrubs, and later mature trees: thus, the forest is restored to its former state. This process is so universal that ecologists have given it a name?succession. The impact of environmental disturbances caused by human activities is now as significant as the changes wrought by natural processes. Human agricultural practices, air pollution, acid rain, global deforestation, overfishing, oil spills, and illegal dumping on land and into the ocean all have impacts on ecosystems.

Equilibrium is a dynamic state of an ecosystem in which, despite changes in species numbers and occurrence, biodiversity remains somewhat constant. In ecology, two parameters are used to measure changes in ecosystems:

resistance and resilience. The ability of an ecosystem to remain at equilibrium in spite of disturbances is called resistance. The speed at which an ecosystem recovers equilibrium after being disturbed is called resilience. Ecosystem resistance and resilience are especially important when considering human impact. The nature of an ecosystem may change to such a degree that it can lose its resilience entirely. This process can lead to the complete destruction or irreversible altering of the ecosystem.

Food Chains and Food Webs

A food chain is a linear sequence of organisms through which nutrients and energy pass as one organism eats another; the levels in the food chain are producers, primary consumers, higher-level consumers, and finally decomposers. These levels are used to describe ecosystem structure and dynamics. There is a single path through a food chain. Each organism in a food chain occupies a specific trophic level (energy level), its position in the food chain or food web.

In many ecosystems, the base, or foundation, of the food chain consists of photosynthetic organisms (plants or phytoplankton), which are called producers. The organisms that consume the producers are herbivores: the primary consumers. Secondary consumers are usually carnivores that eat the primary consumers. Tertiary consumers are carnivores that eat other carnivores. Higher-level consumers feed on the next lower trophic levels,

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and so on, up to the organisms at the top of the food chain: the apex consumers. In the Lake Ontario food chain, shown in Figure 20.4, the Chinook salmon is the apex consumer at the top of this food chain.

FIGURE 20.4 These are the trophic levels of a food chain in Lake Ontario at the United States?Canada border. Energy and nutrients flow from photosynthetic green algae at the base to the top of the food chain: the Chinook salmon. (credit: modification of work by National Oceanic and Atmospheric

"mold": modification of work by Dr. Lucille Georg, CDC; credit "earthworm": modification of work by Rob Hille; credit "bacteria": modification of work by Don Stalons, CDC)

LINK TO LEARNING

Head to this online interactive simulator (http://openstax.org/l/food_web) to investigate food web function. In the Interactive Labs box, under Foodweb, click on the link. Read the instructions first, and then click Step 2 for additional instructions. When you are ready to create a simulation, in the upper-right corner of the Interactive Labs box, click OPEN SIMULATOR.

Two general types of food webs are often shown interacting within a single ecosystem. A grazing food web has plants or other photosynthetic organisms at its base, followed by herbivores and various carnivores. A detrital food web consists of a base of organisms that feed on decaying organic matter (dead organisms), including decomposers (which break down dead and decaying organisms) and detritivores (which consume organic detritus). These organisms are usually bacteria, fungi, and invertebrate animals that recycle organic material back into the biotic part of the ecosystem as they themselves are consumed by other organisms. As ecosystems require a method to recycle material from dead organisms, grazing food webs have an associated detrital food web. For example, in a meadow ecosystem, plants may support a grazing food web of different organisms, primary and other levels of consumers, while at the same time supporting a detrital food web of bacteria and fungi feeding off dead plants and animals. Simultaneously, a detrital food web can contribute energy to a grazing food web, as when a robin eats an earthworm.

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How Organisms Acquire Energy in a Food Web

All living things require energy in one form or another. Energy is used by most complex metabolic pathways (usually in the form of ATP), especially those responsible for building large molecules from smaller compounds. Living organisms would not be able to assemble macromolecules (proteins, lipids, nucleic acids, and complex carbohydrates) from their monomers without a constant energy input.

Food-web diagrams illustrate how energy flows directionally through ecosystems. They can also indicate how efficiently organisms acquire energy, use it, and how much remains for use by other organisms of the food web. Energy is acquired by living things in two ways: autotrophs harness light or chemical energy and heterotrophs acquire energy through the consumption and digestion of other living or previously living organisms.

Photosynthetic and chemosynthetic organisms are autotrophs, which are organisms capable of synthesizing their own food (more specifically, capable of using inorganic carbon as a carbon source). Photosynthetic autotrophs (photoautotrophs) use sunlight as an energy source, and chemosynthetic autotrophs (chemoautotrophs) use inorganic m

olecules as an energy source. Autotrophs are critical for most ecosystems: they are the producer trophic level. Without these organisms, energy would not be available to other living organisms, and life itself would not be possible.

Photoautotrophs, such as plants, algae, and photosynthetic bacteria, are the energy source for a majority of the world's ecosystems. These ecosystems are often described by grazing and detrital food webs.

Photoautotrophs

harness the Sun's solar energy by converting it to chemical energy in the form of ATP (and NADP). The energy stored

in ATP is used to synthesize complex organic molecules, such as glucose. The rate at which photosynthetic producers incorporate energy from the Sun is called gross primary productivity. However, not all of the energy incorporated by producers is available to the other organisms in the food web because producers must also grow

and reproduce, which consumes energy. Net primary productivity is the energy that remains in the producers after

accounting for these organisms' respiration and heat loss. The net productivity is then available to the primary consumers at the next trophic level.

Chemoautotrophs are primarily bacteria and archaea that are found in rare ecosystems where sunlight is not available, such as those associated with dark caves or hydrothermal vents at the bottom of the ocean (Figure 20.7).

Many chemoautotrophs in hydrothermal vents use hydrogen sulfide (H_2S), which is released from the vents as a

source of chemical energy; this allows them to synthesize complex organic molecules, such as glucose, for their own

energy and, in turn, supplies energy to the rest of the ecosystem.

FIGURE 20.7 Swimming shrimp, a few squat lobsters, and hundreds of vent mussels are seen at a hydrothermal vent at the bottom of the ocean. As no sunlight penetrates to this depth, the ecosystem is supported by chemoautotrophic bacteria and organic material that sinks from the ocean's surface. This picture was taken in 2006 at the submerged NW Eifuku volcano off the coast of Japan by the National Oceanic and Atmospheric Administration (NOAA). The summit of this highly active volcano lies 1535 m below the surface.

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Consequences of Food Webs: Biological Magnification

One of the most important consequences of ecosystem dynamics in terms of human impact is biomagnification. Biomagnification is the increasing concentration of persistent, toxic substances in organisms at each successive trophic level. These are substances that are fat soluble, not water soluble, and are stored in the fat reserves of each

organism. Many substances have been shown to biomagnify, including classical studies with the pesticide dichlorodiphenyltrichloroethane (DDT), which were described in the 1960s bestseller, *Silent Spring* by Rachel Carson. DDT was a commonly used pesticide before its dangers to apex consumers, such as the bald eagle, became

known. In aquatic ecosystems, organisms from

mber of

the Army Corps of Engineers holds a clump of oysters being used as a part of the oyster restoration effort in the bay. (credit a: modification of work by NASA/MODIS; credit b: modification of work by U.S. Army)

The Chesapeake Bay (Figure 20.15a) is one of the most scenic areas on Earth; it is now in distress and is recognized

as a case study of a declining ecosystem. In the 1970s, the Chesapeake Bay was one of the first aquatic ecosystems

to have identified dead zones, which continue to kill many fish and bottom-dwelling species such as clams, oysters,

and worms. Several species have declined in the Chesapeake Bay because surface water runoff contains excess

nutrients from artificial fertilizer use on land. The source of the fertilizers (with high nitrogen and phosphate content)

is not limited to agricultural practices. There are many nearby urban areas and more than 150 rivers and streams

empty into the bay that are carrying fertilizer runoff from lawns and gardens. Thus, the decline of the Chesapeake

Bay is a complex issue and requires the cooperation of industry, agriculture, and individual homeowners.

Of particular interest to conservationists is the oyster population (Figure 20.15b); it is estimated that more than 200,000 acres of oyster reefs existed in the bay in the 1700s, but that number has now declined to only 36,000 acres. Oyster harvesting was once a major industry for Chesapeake Bay, but it declined 88 percent between 1982

and 2007. This decline was caused not only by fertilizer runoff and dead zones, but also because of overharvesting.

Oysters require a certain minimum population density because they must be in close proximity to reproduce. Human activity has altered the oyster population and locations, thus greatly disrupting the ecosystem.

The restoration of the oyster population in the Chesapeake Bay has been ongoing for several years with mixed success. Not only do many people find oysters good to eat, but the oysters also clean up the bay. They are filter feeders, and as they eat, they clean the water around them. Filter feeders eat by pumping a continuous stream of

water over finely divided appendages (gills in the case of oysters) and capturing prokaryotes, plankton, and fine organic particles in their mucus. In the 1700s, it was estimated that it took only a few days for the oyster population

to filter the entire volume of the bay. Today, with the changed water conditions, it is estimated that the present population would take nearly a year to do the same job.

Restoration efforts have been ongoing for several years by non-profit organizations such as the Chesapeake Bay

Foundation. The restoration goal is to find a way to increase population density so the oysters can reproduce more

efficiently. Many disease-resistant varieties (developed at the Virginia Institute of Marine Science for the College of

ases larger amounts of hydrogen sulfide gas into the atmosphere.

As rain falls through this gas, it creates the phenomenon known as acid rain, which damages the natural environment by lowering the pH of lakes, thus killing many of the resident plants and animals. Acid rain is corrosive

rain caused by rainwater falling to the ground through sulfur dioxide gas, turning it into weak sulfuric acid, which causes damage to aquatic ecosystems. Acid rain also affects the man-made environment through the chemical degradation of buildings. For example, many marble monuments, such as the Lincoln Memorial in Washington, DC,

have suffered significant damage from acid rain over the years. These examples show the wide-ranging effects of

human activities on our environment and the challenges that remain for our future.

20.3 Terrestrial Biomes

LEARNING OBJECTIVES

By the end of this section, you will be able to:

? Identify the two major abiotic factors that determine the type of terrestrial biome in an area

? Recognize distinguishing characteristics of each of the eight major terrestrial biomes

Earth's biomes can be either terrestrial or aquatic. Terrestrial biomes are based on land, while aquatic biomes include both ocean and freshwater biomes. The eight major terrestrial biomes on Earth are each distinguished by

characteristic temperatures and amount of precipitation. Annual totals and fluctuations of precipitation affect the kinds of vegetation and animal life that can exist in broad geographical regions. Temperature variation on a daily and

seasonal basis is also important for predicting the geographic distribution of a biome. Since a biome is defined by

climate, the same biome can occur in geographically distinct areas with similar climates (Figure 20.18). There are

also large areas on Antarctica, Greenland, and in mountain ranges that are covered by permanent glaciers and support very little life. Strictly speaking, these are not considered biomes and in addition to extremes of cold, they

are also often deserts with very low precipitation.

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FIGURE 20.18 Each of the world's eight major biomes is distinguished by characteristic temperatures and amount of precipitation. Polar ice caps and mountains are also shown.

Tropical Forest

Tropical rainforests are also referred to as tropical wet forests. This biome is found in equatorial regions (Figure 20.18). Tropical rainforests are the most diverse terrestrial biome. This biodiversity is still largely unknown to science and is under extraordinary threat primarily through logging and deforestation for agriculture. Tropical rainforests have also been described as nature's pharmacy because of the potential for new drugs that is largely

hidden in the chemicals produced by the huge diversity of plants, animals, and other organisms. The vegetation is

characterized by plants with spreading roots and broad leaves that fall off throughout the year, unlike the trees of

deciduous fo

rests that lose their leaves in one season. These forests are "evergreen," year-round.

The temperature and sunlight profiles of tropical rainforests are stable in comparison to that of other terrestrial biomes, with average temperatures ranging from 20°C to 34°C (68°F to 93°F). Month-to-month temperatures are relatively constant in tropical rainforests, in contrast to forests further from the equator. This lack of temperature seasonality leads to year-round plant growth, rather than the seasonal growth seen in other biomes. In contrast to other ecosystems, a more constant daily amount of sunlight (11–12 hours per day) provides more solar radiation, thereby a longer period of time for plant growth.

The annual rainfall in tropical rainforests ranges from 250 cm to more than 450 cm (8.2–14.8 ft) with considerable seasonal variation. Tropical rainforests have wet months in which there can be more than 30 cm (11–12 in) of precipitation, as well as dry months in which there are fewer than 10 cm (3.5 in) of rainfall. However, the driest month of a tropical rainforest can still exceed the annual rainfall of some other biomes, such as deserts.

Tropical rainforests have high net primary productivity because the annual temperatures and precipitation values support rapid plant growth (Figure 20.19). However, the high rainfall quickly leaches nutrients from the soils of these forests, which are typically low in nutrients. Tropical rainforests are characterized by vertical layering of vegetation and the formation of distinct habitats for animals within each layer. On the forest floor is a sparse layer of plants and decaying plant matter. Above that is an understory of short, shrubby foliage. A layer of trees rises above this understory and is topped by a closed upper canopy—the uppermost overhead layer of branches and leaves. Some additional trees emerge through this closed upper canopy. These layers provide diverse and complex habitats for the variety of plants, animals, and other organisms within the tropical wet forests. Many species of animals use the variety of plants and the complex structure of the tropical wet forests for food and shelter. Some organisms live several meters above ground rarely ever descending to the forest floor.

Rainforests are not the only forest biome in the tropics; there are also tropical dry forests, which are characterized by a dry season of varying lengths. These forests commonly experience leaf loss during the dry season to one degree or another. The loss of leaves from taller trees during the dry season opens up the canopy and allows sunlight to the forest floor that allows the growth of thick ground-level brush, which is absent in tropical rainforests. Extensive tropical dry forests occur in Africa (including Madagascar), India, southern Mexico, and South America.

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FIGURE 20.19 Species diversity is very high in tropical wet forests, such as these forests o

ter itself.

Finally, all natural water contains dissolved solids, or salts. Fresh water contains low levels of such dissolved substances because the water is rapidly recycled through evaporation and precipitation. The oceans have a relatively constant high salt content. Aquatic habitats at the interface of marine and freshwater ecosystems have complex and variable salt environments that range between freshwater and marine levels. These are known as brackish water environments. Lakes located in closed drainage basins concentrate salt in their waters and can have

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extremely high salt content that only a few and highly specialized species are able to inhabit.

Marine Biomes

The ocean is a continuous body of salt water that is relatively uniform in chemical composition. It is a weak solution

of mineral salts and decayed biological matter. Within the ocean, coral reefs are a second type of marine biome. Estuaries, coastal areas where salt water and fresh water mix, form a third unique marine biome.

The ocean is categorized by several zones (Figure 20.28). All of the ocean's open water is referred to as the pelagic realm (or zone). The benthic realm (or zone) extends along the ocean bottom from the shoreline to the deepest parts of the ocean floor. From the surface to the bottom or the limit to which photosynthesis occurs is the photic zone (approximately 200 m or 650 ft). At depths greater than 200 m, light cannot penetrate; thus, this is referred to as the aphotic zone. The majority of the ocean is aphotic and lacks sufficient light for photosynthesis. The deepest part of the ocean, the Challenger Deep (in the Mariana Trench, located in the western Pacific Ocean), is about 11,000 m (about 6.8 mi) deep. To give some perspective on the depth of this trench, the ocean is, on average, 4267 m or 14,000 ft deep.

Ocean

The physical diversity of the ocean has a significant influence on the diversity of organisms that live within it. The

ocean is categorized into different zones based on how far light reaches into the water. Each zone has a distinct group of species adapted to the biotic and abiotic conditions particular to that zone.

The intertidal zone (Figure 20.28) is the oceanic region that is closest to land. With each tidal cycle, the intertidal

zone alternates between being inundated with water and left high and dry. Generally, most people think of this portion of the ocean as a sandy beach. In some cases, the intertidal zone is indeed a sandy beach, but it can also be

rocky, muddy, or dense with tangled roots in mangrove forests. The intertidal zone is an extremely variable environment because of tides. Organisms may be exposed to air at low tide and are underwater during high tide.

Therefore, living things that thrive in the intertidal zone are often adapted to being dry for long periods of time. The

shore of the intertidal zone is also repeatedly struck by waves and the or

daily temperatures, very low and

permafrost a perennially frozen portion of the Arctic unpredictable precipitation, and characterized by a tundra soil limited dry-adapted vegetation

photic zone the upper layer of ocean water in which temperate forest a biome found in temperate regions photosynthesis is able to take place with moderate rainfall and dominated structurally

photoautotroph an organism that uses sunlight as an by deciduous trees energy source to synthesize its own food molecules temperate grassland a biome dominated by grasses

planktivore an animal that eats plankton and herbaceous plants due to low precipitation, primary consumer the trophic level that obtains its periodic fires, and grazing

energy from the producers of an ecosystem tertiary consumer a trophic level in an ecosystem, producer the trophic level that obtains its energy usually carnivores that eat other carnivores

from sunlight, inorganic chemicals, or dead or trophic level the position of a species or group of decaying organic material species in a food chain or a food web

resilience (ecological) the speed at which an tropical rainforest a biome found near the equator ecosystem recovers equilibrium after being characterized by stable temperatures with abundant disturbed and seasonal rainfall in which trees form the

resistance (ecological) the ability of an ecosystem to structurally important vegetation remain at equilibrium in spite of disturbances wetland environment in which the soil is either

savanna a biome located in the tropics with an permanently or periodically saturated with water extended dry season and characterized by a

Chapter Summary

20.1 Waterford's Energy Flow through damaged by a variety of human activities that alter the Ecosystems natural biogeochemical cycles due to pollution, oil

spills, and events causing global climate change. The Ecosystems exist underground, on land, at sea, and in health of the biosphere depends on understanding the air. Organisms in an ecosystem acquire energy in a these cycles and how to protect the environment from variety of ways, which is transferred between trophic irreversible damage.

levels as the energy flows from the base to the top of

the food web, with energy being lost at each transfer. 20.3 Terrestrial Biomes

There is energy lost at each trophic level, so the Earth has terrestrial and aquatic biomes. Aquatic lengths of food chains are limited because there is a biomes include both freshwater and marine point where not enough energy remains to support a environments. There are eight major terrestrial biomes: population of consumers. Fat soluble compounds tropical rainforests, savannas, subtropical deserts, biomagnify up a food chain causing damage to top chaparral, temperate grasslands, temperate forests, consumers. even when environmental concentrations boreal forests, and Arctic tundra. The same biome can of a toxin are low. occur in different geographic locations with similar

20.2 Biogeochemical Cycle

anet in

the next century or two. The losses will include many species we know today. Although it is sometimes difficult to predict which species will become extinct, many are listed as endangered (at great risk of extinction). However, the majority of extinctions will be of species that science has not yet even described.

Most of these "invisible" species that will become extinct currently live in tropical rainforests like those of the Amazon basin. These rainforests are the most diverse ecosystems on the planet and are being destroyed rapidly by deforestation, which biologists believe is driving many rare species with limited distributions extinct. Between 1970 and 2011, almost 20 percent of the Amazon rainforest was lost. Rates are higher in other tropical rainforests. What we are likely to notice on a day-to-day basis as a result of biodiversity loss is that food will be more difficult to produce, clean water will be more difficult to find, and the rate of development of new medicines will become slower, as we depend upon other species for much of these services. This increased loss of biodiversity is almost entirely a result of human activities as we destroy species' habitats,

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introduce disruptive species into ecosystems, hunt some species to extinction, continue to warm the planet with greenhouse gases, and influence nature in other ways. Slowing the loss of biodiversity is within our abilities if we make dramatic changes in our consumptive behavior and identify and protect the elements of our ecosystems that we depend on for our lives and welfare.

21.1 Importance of Biodiversity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

? Describe biodiversity as the equilibrium of naturally fluctuating rates of extinction and speciation

? Identify benefits of biodiversity to humans

FIGURE 21.2 This tropical lowland rainforest in Madagascar is an example of a high biodiversity habitat. This particular location is protected within a national forest, yet only 10 percent of the original coastal lowland forest remains, and research suggests half the original biodiversity has been lost. (credit: Frank Vassen)

Biodiversity is a broad term for biological variety, and it can be measured at a number of organizational levels. Traditionally, ecologists have measured biodiversity by taking into account both the number of species and the number of individuals in each of those species. However, biologists are using measures of biodiversity at several levels of biological organization (including genes, populations, and ecosystems) to help focus efforts to preserve the biologically and technologically important elements of biodiversity.

When biodiversity loss through extinction is thought of as the loss of the passenger pigeon, the dodo, or, even, the woolly mammoth there seems to be no reason to care about it because these events happened long ago. How is the loss practically important for the we

lfare of the human

species? Would these species have made our lives any better? From the perspective of evolution and ecology, the loss of a particular individual species, with some exceptions, may seem unimportant, but the current accelerated extinction rate means the loss of tens of thousands of species within our lifetimes. Much of this loss is occurring in tropical rainforests like the one pictured in Figure 21.2, which are especially high-diversity ecosystems that are being cleared for timber and agriculture. This is likely to have dramatic effects on human welfare through the collapse of ecosystems and in added costs to maintain food production, clean air and water, and improve human health.

Biologists recognize that human populations are embedded in ecosystems and are dependent on them, just as is every other species on the planet. Agriculture began after early hunter-gatherer societies first settled in one place and heavily modified their immediate environment: the ecosystem in which they existed. This cultural transition has made it difficult for humans to recognize their dependence on living things other than crops and domesticated animals on the planet. Today our technology smoothes out the extremes of existence and allows many of us to

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live longer, more comfortable lives, but ultimately the human species cannot exist without its surrounding ecosystems. Our ecosystems provide our food. This includes living plants that grow in soil ecosystems and the animals that eat these plants (or other animals) as well as photosynthetic organisms in the oceans and the other organisms that eat them. Our ecosystems have provided and will provide many of the medications that maintain our health, which are commonly made from compounds found in living organisms. Ecosystems provide our clean water, which is held in lake and river ecosystems or passes through terrestrial ecosystems on its way into groundwater.

Types of Biodiversity

A common meaning of biodiversity is simply the number of species in a location or on Earth; for example, the American Ornithologists' Union lists 2078 species of birds in North and Central America. This is one measure of the bird biodiversity on the continent. More sophisticated measures of diversity take into account the relative abundances of species. For example, a forest with 10 equally common species of trees is more diverse than a forest that has 10 species of trees wherein just one of those species makes up 95 percent of the trees rather than them being equally distributed. Biologists have also identified alternate measures of biodiversity, some of which are important in planning how to preserve biodiversity.

Genetic and Chemical Biodiversity

Genetic diversity is one alternate concept of biodiversity. Genetic diversity (or variation) is the raw material for adaptation in a species. A species' future potential for adaptation depends on the genet

c diversity held in the

genomes of the individuals in populations that make up the species. The same is true for higher taxonomic categories. A genus with very different types of species will have more genetic diversity than a genus with species

that look alike and have similar ecologies. The genus with the greatest potential for subsequent evolution is the most genetically diverse one.

Most genes code for proteins, which in turn carry out the metabolic processes that keep organisms alive and reproducing. Genetic diversity can also be conceived of as chemical diversity in that species with different genetic

makeups produce different assortments of chemicals in their cells (proteins as well as the products and byproducts

of metabolism). This chemical diversity is important for humans because of the potential uses for these chemicals,

such as medications. For example, the drug eptifibatide is derived from rattlesnake venom and is used to prevent

heart attacks in individuals with certain heart conditions.

At present, it is far cheaper to discover compounds made by an organism than to imagine them and then synthesize

them in a laboratory. Chemical diversity is one way to measure diversity that is important to human health and welfare. Through selective breeding, humans have domesticated animals, plants, and fungi, but even this diversity is

suffering losses because of market forces and increasing globalism in human agriculture and migration. For example, international seed companies produce only a very few varieties of a given crop and provide incentives around the world for farmers to buy these few varieties while abandoning their traditional varieties, which are far more diverse. The human population depends on crop diversity directly as a stable food source and its decline is

troubling to biologists and agricultural scientists.

Ecosystems Diversity

It is also useful to define ecosystem diversity: the number of different ecosystems on Earth or in a geographical area. Whole ecosystems can disappear even if some of the species might survive by adapting to other ecosystems.

The loss of an ecosystem means the loss of the interactions between species, the loss of unique features of coadaptation, and the loss of biological productivity that an ecosystem is able to create. An example of a largely extinct ecosystem in North America is the prairie ecosystem (Figure 21.3). Prairies once spanned central North America from the boreal forest in northern Canada down into Mexico. They are now all but gone, replaced by crop

fields, pasture lands, and suburban sprawl. Many of the species survive, but the hugely productive ecosystem that

was responsible for creating our most productive agricultural soils is now gone. As a consequence, their soils are

now being depleted unless they are maintained artificially at greater expense. The decline in soil productivity occurs

because the interactions in the original ecosystem have been lost; this was a far more important

eristics and whether or not that organism belongs to any other described species. It allows biologists to find and recognize the species after the initial discovery to follow up on questions about its biology. That subsequent research will produce the discoveries that make the species valuable to humans and to our ecosystems. Without a name and description, a species cannot be studied in depth and in a coordinated way by multiple scientists.

Patterns of Biodiversity

Biodiversity is not evenly distributed on the planet. Lake Victoria contained almost 500 species of cichlids (only one family of fishes present in the lake) before the introduction of an exotic species in the 1980s and 1990s caused a mass extinction. All of these species were found only in Lake Victoria, which is to say they were endemic.

Endemic

species are found in only one location. For example, the blue jay is endemic to North America, while the Barton Springs salamander is endemic to the mouth of one spring in Austin, Texas. Endemics with highly restricted distributions, like the Barton Springs salamander, are particularly vulnerable to extinction. Higher taxonomic levels, such as genera and families, can also be endemic.

Lake Huron contains about 79 species of fish, all of which are found in many other lakes in North America.

What

accounts for the difference in diversity between Lake Victoria and Lake Huron? Lake Victoria is a tropical lake, while

Lake Huron is a temperate lake. Lake Huron in its present form is only about 7,000 years old, while Lake Victoria in

its present form is about 15,000 years old. These two factors, latitude and age, are two of several hypotheses biogeographers have suggested to explain biodiversity patterns on Earth.

CAREER CONNECTION

Biogeography

Biogeography is the study of the distribution of the world's species both in the past and in the present. The work of

biogeographers is critical to understanding our physical environment, how the environment affects species, and how

changes in environment impact the distribution of a species.

There are three main fields of study under the heading of biogeography: ecological biogeography, historical biogeography (called paleobiogeography), and conservation biogeography. Ecological biogeography studies the current factors affecting the distribution of plants and animals. Historical biogeography, as the name implies, studies

the past distribution of species. Conservation biogeography, on the other hand, is focused on the protection and restoration of species based upon the known historical and current ecological information. Each of these fields considers both zoogeography and phylogeography—the past and present distribution of animals and plants.

One of the oldest observed patterns in ecology is that biodiversity in almost every taxonomic group of organism increases as latitude declines. In other words, biodiversity increases closer to the equator (Figure 21.4).

ration (IISE), 2011 State of Observed Species (SOS). Tempe, AZ: IISE, 2011. Accessed May, 20, 2012. <http://species.asu.edu/SOS>.

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FIGURE 21.4 This map illustrates the number of amphibian species across the globe and shows the trend toward higher biodiversity at lower latitudes. A similar pattern is observed for most taxonomic groups.

It is not yet clear why biodiversity increases closer to the equator, but hypotheses include the greater age of the ecosystems in the tropics versus temperate regions, which were largely devoid of life or drastically impoverished during the last ice age. The greater age provides more time for speciation. Another possible explanation is the greater energy the tropics receive from the sun versus the lesser energy input in temperate and polar regions. But scientists have not been able to explain how greater energy input could translate into more species. The complexity of tropical ecosystems may promote speciation by increasing the habitat heterogeneity, or number of ecological niches, in the tropics relative to higher latitudes. The greater heterogeneity provides more opportunities for coevolution, specialization, and perhaps greater selection pressures leading to population differentiation. However, this hypothesis suffers from some circularity?ecosystems with more species encourage speciation, but how did they get more species to begin with? The tropics have been perceived as being more stable than temperate regions, which have a pronounced climate and day-length seasonality. The tropics have their own forms of seasonality, such as rainfall, but they are generally assumed to be more stable environments and this stability might promote speciation.

Regardless of the mechanisms, it is certainly true that biodiversity is greatest in the tropics. The number of endemic species is higher in the tropics. The tropics also contain more biodiversity hotspots. At the same time, our knowledge of the species living in the tropics is lowest and because of recent, heavy human activity the potential for biodiversity loss is greatest.

Importance of Biodiversity

Loss of biodiversity eventually threatens other species we do not impact directly because of their interconnectedness; as species disappear from an ecosystem other species are threatened by the changes in available resources. Biodiversity is important to the survival and welfare of human populations because it has impacts on our health and our ability to feed ourselves through agriculture and harvesting populations of wild animals.

Human Health

Many medications are derived from natural chemicals made by a diverse group of organisms. For example, many plants produce secondary plant compounds, which are toxins used to protect the plant from insects and other animals that eat them. Some of these secondary plant compounds also work as human medicines.

Contemporary

societies that live close to the land often have a broad knowledge of the

d grow, and those resources are being removed unsustainably from the environment. The three greatest proximate threats to biodiversity are habitat loss, overharvesting, and introduction of exotic species. The first two of these are a direct result of human population growth and resource use. The third results from increased mobility and trade. A fourth major cause of extinction, anthropogenic (human-caused) climate change, has not yet had a large impact, but it is predicted to become significant during this century. Global climate change is also a consequence of human population needs for energy and the use of fossil fuels to meet those needs (Figure 21.7).

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Environmental issues, such as toxic pollution, have specific targeted effects on species, but are not generally seen as threats at the magnitude of the others.

FIGURE 21.7 Atmospheric carbon dioxide levels fluctuate in a cyclical manner. However, the burning of fossil fuels in recent history has caused a dramatic increase in the levels of carbon dioxide in the Earth's atmosphere, which have now reached levels never before seen on Earth. Scientists predict that the addition of this "greenhouse gas" to the atmosphere is resulting in climate change that will significantly impact biodiversity in the coming century.

Habitat Loss

Humans rely on technology to modify their environment and replace certain functions that were once performed by the natural ecosystem. Other species cannot do this. Elimination of their habitat—whether it is a forest, coral reef, grassland, or flowing river—will kill the individuals in the species. Remove the entire habitat within the range of a species and, unless they are one of the few species that do well in human-built environments, the species will become extinct. Human destruction of habitats (habitats generally refer to the part of the ecosystem required by a particular species) accelerated in the latter half of the twentieth century. Consider the exceptional biodiversity of Sumatra: it is home to one species of orangutan, a species of critically endangered elephant, and the Sumatran tiger, but half of Sumatra's forest is now gone. The neighboring island of Borneo, home to the other species of orangutan, has lost a similar area of forest. Forest loss continues in protected areas of Borneo. The orangutan in Borneo is listed as endangered by the International Union for Conservation of Nature (IUCN), but it is simply the most visible of thousands of species that will not survive the disappearance of the forests of Borneo. The forests are removed for timber and to plant palm oil plantations (Figure 21.8). Palm oil is used in many products including food products, cosmetics, and biodiesel in Europe. A 5-year estimate of global forest cover loss for the years from 2000 to 2005 was 3.1 percent. Much loss (2.4 percent) occurred in the humid tropics where forest loss is primarily from timber extraction. These losses certainly also

represent the extinction of species unique to those areas.

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FIGURE 21.8 An oil palm plantation in Sabah province Borneo, Malaysia, replaces native forest habitat that a variety of species depended on to live. (credit: Lian Pin Koh)

EVERYDAY CONNECTION

Preventing Habitat Destruction with Wise Wood Choices

Most consumers do not imagine that the home improvement products they buy might be contributing to habitat loss and species extinctions. Yet the market for illegally harvested tropical timber is huge, and the wood products often find themselves in building supply stores in the United States. One estimate is that 10 percent of the imported timber stream in the United States, which is the world's largest consumer of wood products, is potentially illegally logged. In 2006, this amounted to \$3.6 billion in wood products. Most of the illegal products are imported from countries that act as intermediaries and are not the originators of the wood.

How is it possible to determine if a wood product, such as flooring, was harvested sustainably or even legally? The Forest Stewardship Council (FSC) certifies sustainably harvested forest products; therefore, looking for their

certification on flooring and other hardwood products is one way to ensure that the wood has not been taken illegally from a tropical forest. Certification applies to specific products, not to a producer; some producers' products may not have certification while other products are certified. There are certifications other than the FSC, but these are run by timber companies creating a conflict of interest. Another approach is to buy domestic wood species. While it would be great if there was a list of legal versus illegal woods, it is not that simple. Logging and forest management laws vary from country to country; what is illegal in one country may be legal in another. Where and how a product is harvested and whether the forest from which it comes is being sustainably maintained all factor into whether a wood product will be certified by the FSC. It is always a good idea to ask questions about where a wood product came from and how the supplier knows that it was harvested legally.

Habitat destruction can affect ecosystems other than forests. Rivers and streams are important ecosystems and are

frequently the target of habitat modification through building and from damming or water removal. Damming of rivers affects flows and access to all parts of a river. Altering a flow regime can reduce or eliminate populations that

are adapted to seasonal changes in flow. For example, an estimated 91 percent of river lengths in the United States

have been modified with damming or bank modifications. Many fish species in the United States, especially rare species or species with restricted distributions, have seen declines caused by river damming and habitat loss.

Research has confirmed that species of amphibians that must carry

out parts of their life cycles in both aquatic and terrestrial habitats are at greater risk of population declines and extinction because of the increased likelihood that one of their habitats or access between them will be lost. This is of particular concern because amphibians have been declining in numbers and going extinct more rapidly than many other groups for a variety of possible reasons. Habitat destruction, even when undertaken on behalf of humans, can lead to negative effects for us as well. Excessive soil erosion after forest removal, for example, can remove fertile soil and make river water toxic. Scientists and conservationists such as Wangari Maathai, who founded the Green Belt movement in Kenya, focus on replanting trees to repair habitats and prevent damage from deforestation. Maathai was awarded a Nobel Prize for

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her work, but unfortunately passed away in 2011.

Overharvesting

Overharvesting is a serious threat to many species, but particularly to aquatic species. There are many examples of regulated fisheries (including hunting of marine mammals and harvesting of crustaceans and other species) monitored by fisheries scientists that have nevertheless collapsed. The western Atlantic cod fishery is the most spectacular recent collapse. While it was a hugely productive fishery for 400 years, the introduction of modern factory trawlers in the 1980s and the pressure on the fishery led to it becoming unsustainable. The causes of fishery collapse are both economic and political in nature. Most fisheries are managed as a common resource, available to anyone willing to fish, even when the fishing territory lies within a country's territorial waters. Common resources are subject to an economic pressure known as the tragedy of the commons, in which fishers have little motivation to exercise restraint in harvesting a fishery when they do not own the fishery. The general outcome of harvests of resources held in common is their overexploitation. While large fisheries are regulated to attempt to avoid this pressure, it still exists in the background. This overexploitation is exacerbated when access to the fishery is open and unregulated and when technology gives fishers the ability to overfish. In a few fisheries, the biological growth of the resource is less than the potential growth of the profits made from fishing if that time and money were invested elsewhere. In these cases?whales are an example?economic forces will drive toward fishing the population to extinction.

LINK TO LEARNING

Explore a U.S. Fish & Wildlife Service interactive map (http://openstax.org/l/habitat_map2) of critical habitat for endangered and threatened species in the United States. To begin, select ?Visit the online mapper.?

For the most part, fishery extinction is not equivalent to biological extinction?the last fish of a species is rarely fished out of the ocean. But there are some instances in which true extinction is a po

dangerous climate changes in the coming decades. However, there is still debate and a lack of understanding about specific outcomes. Scientists disagree about the likely magnitude of the effects on extinction rates, with estimates ranging from 15 to 40 percent of species committed to extinction by 2050. Scientists do agree that climate change will alter regional climates, including rainfall and snowfall patterns, making habitats less hospitable to the species living in them. The warming trend will shift colder climates toward the north and south poles, forcing species to move with their adapted climate norms, but also to face habitat gaps along the way. The shifting ranges will impose new competitive regimes on species as they find themselves in contact with other species not present in their historic range. One such unexpected species contact is between polar bears and grizzly bears. Previously, these two species had separate ranges. Now, their ranges are overlapping and there are documented cases of these two species mating and producing viable offspring. Changing climates also throw off the delicate timing adaptations that species have to seasonal food resources and breeding times. Scientists have already documented many contemporary mismatches to shifts in resource availability and timing.

Range shifts are already being observed: for example, on average, European bird species ranges have moved 91 km (56.5 mi) northward. The same study suggested that the optimal shift based on warming trends was double that distance, suggesting that the populations are not moving quickly enough. Range shifts have also been observed in plants, butterflies, other insects, freshwater fishes, reptiles, amphibians, and mammals.

Climate gradients will also move up mountains, eventually crowding species higher in altitude and eliminating the habitat for those species adapted to the highest elevations. Some climates will completely disappear. The rate of warming appears to be accelerated in the arctic, which is recognized as a serious threat to polar bear populations that require sea ice to hunt seals during the winter months: seals are the only source of protein available to polar bears. A trend to decreasing sea ice coverage has occurred since observations began in the mid-twentieth century. The rate of decline observed in recent years is far greater than previously predicted by climate models (Figure 21.12).

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FIGURE 21.12 The effect of global warming can be seen in the continuing retreat of Grinnell Glacier. The mean annual temperature in Glacier National Park has increased 1.33°C since 1900. The loss of a glacier results in the loss of summer meltwaters, sharply reducing seasonal water supplies and severely affecting local ecosystems. (credit: USGS, GNP Archives)

Finally, global warming will raise ocean levels due to meltwater from glaciers and the greater volume occupied by warmer water. Shorelines will be inundated,

destroying habitat; a reduction in habitat of 90 percent from 100 km² to 10 km² reduces the number of species supported by about 50 percent.

LINK TO LEARNING

Go to this website (http://openstax.org/l/whats_missing2) for an interactive exploration of endangered and extinct species, their ecosystems, and the causes of their endangerment or extinction.

Conservation of Biodiversity

The threats to biodiversity at the genetic, species, and ecosystem levels have been recognized for some time.

In the

United States, the first national park with land set aside to remain in a wilderness state was Yellowstone Park in 1890. However, attempts to preserve nature for various reasons have occurred for centuries. Today, the main efforts

to preserve biodiversity involve legislative approaches to regulate human and corporate behavior, setting aside protected areas, and habitat restoration.

Changing Human Behavior

Legislation has been enacted to protect species throughout the world. The legislation includes international treaties

as well as national and state laws. The Convention on International Trade in Endangered Species of Wild Fauna and

Flora (CITES) treaty came into force in 1975. The treaty, and the national legislation that supports it, provides a legal

framework for preventing ?listed? species from being transported across nations? borders, thus protecting them from being caught or killed in the first place when the purpose involves international trade. The listed species that

are protected to one degree or another by the treaty number some 33,000. The treaty is limited in its reach because

it only deals with international movement of organisms or their parts. It is also limited by various countries? ability or

willingness to enforce the treaty and supporting legislation. The illegal trade in organisms and their parts is probably

a market in the hundreds of millions of dollars.

Within many countries there are laws that protect endangered species and that regulate hunting and fishing. In the

United States, the Endangered Species Act was enacted in 1973. When an at-risk species is listed by the Act, the

U.S. Fish & Wildlife Service is required by law to develop a management plan to protect the species and bring it back

to sustainable numbers. The Act, and others like it in other countries, is a useful tool, but it suffers because it is often difficult to get a species listed, or to get an effective management plan in place once a species is listed.

Additionally, species may be controversially taken off the list without necessarily having had a change in their situation. More fundamentally, the approach to protecting individual species rather than entire ecosystems (although the management plans commonly involve protection of the individual species? habitat) is both inefficient

and focuses efforts on a few highly visible and often charismatic species, perhaps at the expense of other species

that go unprotected.

The Migratory Bird Treaty Act (MBTA) is an agree

chemical diversity the variety of metabolic typically toxic, but is sequestered by the plant to compounds in an ecosystem defend against herbivores

chytridiomycosis a disease of amphibians caused by species-area relationship the relationship between the fungus *Batrachochytrium dendrobatidis*; area surveyed and number of species encountered; thought to be a major cause of the global amphibian typically measured by incrementally increasing the decline area of a survey and determining the cumulative

ecosystem diversity the variety of ecosystems numbers of species

endemic species a species native to one place tragedy of the commons an economic principle that exotic species (also, invasive species) a species that resources held in common will inevitably be over-

has been introduced to an ecosystem in which it did exploited
not evolve white-nose syndrome a disease of cave-hibernating

extinction the disappearance of a species from Earth; bats in the eastern United States and Canada
local extinction is the disappearance of a species associated with the fungus *Geomyces destructans* from a region

Chapter Summary

21.1 Importance of Biodiversity Crop diversity is a requirement for food security, and it

is being lost. The loss of wild relatives to crops also

Biodiversity exists at multiple levels of organization, threatens breeders' abilities to create new varieties, and is measured in different ways depending on the Ecosystems provide ecosystem services that support goals of those taking the measurements. These include human agriculture: pollination, nutrient cycling, pest numbers of species, genetic diversity, chemical control, and soil development and maintenance. Loss diversity, and ecosystem diversity. The number of of biodiversity threatens these ecosystem services and described species is estimated to be 1.5 million with risks making food production more expensive or about 17,000 new species being described each year. impossible. Wild food sources are mainly aquatic, but Estimates for the total number of eukaryotic species on few are being managed for sustainability. Fisheries? Earth vary but are on the order of 10 million. ability to provide protein to human populations is Biodiversity is negatively correlated with latitude for threatened when extinction occurs.

most taxa, meaning that biodiversity is higher in the
tropics. The mechanism for this pattern is not known 21.2 Threats to Biodiversity
with certainty, but several plausible hypotheses have

The core threats to biodiversity are human population
been advanced.

growth and unsustainable resource use. To date, the
Humans use many compounds that were first most significant causes of extinction are habitat loss, discovered or derived from living organisms as introduction of exotic species, and overharvesting. medicines: secondary plant compounds, animal toxins, Climate change is predicted to be a significant cause of and antibiotics produced by bacteria and fungi. More extinction in the comin

g century. Habitat loss occurs

medicines are expected to be discovered in nature. through deforestation, damming of rivers, and other

Loss of biodiversity will impact the number of activities. Overharvesting is a threat particularly to

pharmaceuticals available to humans. Biodiversity may aquatic species, but the taking of bush meat in the

provide important psychological benefits to humans. humid tropics threatens many species in Asia, Africa,

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and the Americas. Exotic species have been the cause individual countries protecting species and agreements of a number of extinctions and are especially damaging on global warming have had limited success; there is at to islands and lakes. Exotic species? introductions are present no international agreement on targets for increasing because of the increased mobility of human greenhouse gas emissions. In the United States, the populations and growing global trade and Endangered Species Act protects listed species but is transportation. Climate change is forcing range hampered by procedural difficulties and a focus on changes that may lead to extinction. It is also affecting individual species. The Migratory Bird Act is an adaptations to the timing of resource availability that agreement between Canada and the United States to negatively affects species in seasonal environments. protect migratory birds. The non-profit sector is also The impacts of climate change are currently greatest in very active in conservation efforts in a variety of ways. the arctic. Global warming will also raise sea levels,

Conservation preserves are a major tool in biodiversity eliminating some islands and reducing the area of all

protection. Presently, 11 percent of Earth?s land others.

surface is protected in some way. The science of island

21.3 Preserving Biodiversity biogeography has informed the optimal design of

preserves; however, preserves have limitations

Five mass extinctions with losses of more than 50 imposed by political and economic forces. In addition, percent of extant species are observable in the fossil climate change will limit the effectiveness of present record. Recent extinctions are recorded in written preserves in the future. A downside of preserves is that history and are the basis for one method of estimating they may lessen the pressure on human societies to contemporary extinction rates. The other method uses function more sustainably outside the preserves. measures of habitat loss and species-area

relationships. Estimates of contemporary extinction Habitat restoration has the potential to restore rates vary but are as high as 500 times the background ecosystems to previous biodiversity levels before rate, as determined from the fossil record, and are species become extinct. Examples of restoration predicted to rise. include reintroduction of keystone species and removal

of dams on rivers. Zoos have attempted to take a more The

lophase 138 up-regulation 416
Temperate forests 548 ureter 401